# Foundations of Artificial Intelligence

# Preface

Artificial Intelligence has moved, within the span of a single academic career, from a specialized research pursuit to one of the central pillars of modern computer science education. Today's B. Tech Computer Science graduate is expected to understand not only how to write correct, efficient programs, but also how to build systems that search, reason, represent knowledge, and plan under uncertainty the foundational capabilities that this textbook is designed to teach.

This book has been written specifically for a first, semester-long undergraduate course in Artificial Intelligence, organized into five modules that together build a coherent and carefully sequenced foundation in classical AI:

- **Module 1** introduces the subject itself what intelligence means for a machine, where the field's ideas come from, how it has developed historically, and what its current capabilities, risks, and benefits are.
- **Module 2** develops the rational-agent perspective that unifies the rest of the book, presenting the standard hierarchy of agent architectures from simple reflex agents through to learning agents.
- **Module 3** covers problem solving through search both uninformed and heuristic (informed) strategies and constraint satisfaction problems, the algorithmic core on which much of practical AI still depends.
- **Module 4** introduces knowledge representation and logical reasoning, from propositional logic through first-order logic, and the principal inference procedures forward chaining, backward chaining, and resolution used to reason with it.
- **Module 5** covers automated planning, from classical STRIPS-style planning algorithms through planning heuristics, planning under uncertainty, and the closely related topic of scheduling with time and resource constraints.

Each module has been written as a self-contained unit of study, with a structured progression of topics, worked examples, carefully constructed diagrams to illustrate key ideas, algorithm pseudocode where appropriate, a concise end-of-module summary, and a set of review questions suitable for tutorial discussion or self-assessment. An index at the end of the book allows key terms to be located quickly across all five modules.

While this book is intended primarily as a companion to classroom instruction, care has been taken to make each module readable independently, so that a student may also use it for self-study or for quick revision before an examination. Wherever possible, abstract concepts are grounded in concrete, worked examples the Wumpus World, the 8-puzzle, the blocks world, and small road-map and scheduling problems recur across modules precisely so that a single running set of examples can illustrate multiple, successively more advanced ideas.

The field of Artificial Intelligence is, as Module 1 discusses at some length, one that has experienced cycles of great enthusiasm followed by sober reassessment. It is the sincere hope of this text that by grounding its explanations in solid algorithmic and logical foundations rather than in the shifting fashions of any particular application area it will remain useful to students regardless of how the specific tools and applications of AI continue to evolve in the years following their graduation.

Feedback, corrections, and suggestions for improving future editions of this text are always welcome from students and faculty alike.

— *The Author*

## Table of Contents

# MODULE 1:
# INTRODUCTION TO ARTIFICIAL INTELLIGENCE

## Learning Objectives

By the end of this module, the student should be able to:

- Explain the four classical approaches to defining Artificial Intelligence and justify why the rational-agent approach is preferred as an engineering discipline.
- Trace the historical development of AI from its philosophical antecedents through the Dartmouth workshop, the boom-and-bust cycles of the twentieth century, to the present data-driven era, and explain the technical cause of each major transition.
- Identify the contribution of at least eight parent disciplines to the intellectual foundations of AI.
- Describe, with concrete named examples, the current state of the art in major AI application domains.
- Critically evaluate the benefits and risks of contemporary AI deployment, and propose mitigations grounded in engineering practice rather than speculation.

## 1.1 What Is Artificial Intelligence?

Artificial Intelligence (AI) is the branch of computer science concerned with building machines, programs, and systems that exhibit behavior we would call "intelligent" if observed in a human being. The word "intelligence" is easy to use in casual conversation and remarkably hard to pin down with mathematical precision, and this difficulty is precisely the reason AI has attracted so many competing formal definitions over the past seven decades. Rather than memorizing a single sentence definition, a student of AI benefits far more from understanding the **dimensions along which definitions of AI differ**, because each dimension leads to a different research programme, a different set of design tools, and a different notion of what counts as success for an engineered system.

**Definition: Artificial Intelligence**

Artificial Intelligence is the study and construction of computational agents that act intelligently that is, agents that perceive their environment, reason about what they perceive (or act reflexively without explicit reasoning), and choose actions that further their own or their designer's goals, under the practical constraints of limited computation and imperfect information.

Definitions of AI found across seventy years of textbooks and research papers can be organized along two independent axes:

1. Whether the definition is concerned primarily with **thought processes and reasoning** internal to the agent, or with **externally observable behavior**.
2. Whether the definition measures success against **human performance** as the yardstick, or against an **ideal, mathematically defined standard of rationality**.
3. Crossing these two axes gives four distinct schools of thought, summarized in the table below and illustrated in Figure 1.1.

| | **Human-centered** | **Rationalist** |
|---|---|---|
| **Concerned with thought** | Systems that *think* like humans (cognitive modeling) | Systems that *think* rationally (the "laws of thought") |
| **Concerned with behavior** | Systems that *act* like humans (the Turing Test) | Systems that *act* rationally (the rational-agent approach) |

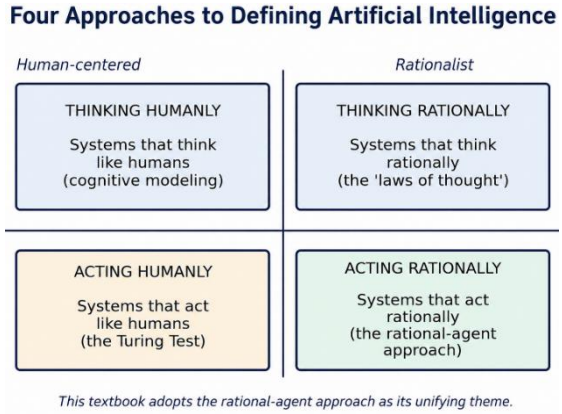


*Figure 1.1: Four approaches to defining Artificial Intelligence, organized along the human/rational and thought/behavior axes.*

It is worth dwelling on *why* this seemingly academic classification matters to a working engineer. Each of the four cells above does not merely offer a different philosophical flavor of the same underlying subject; each cell implies a genuinely different research methodology, a different criterion for when a system has "succeeded," and, historically, a different community of researchers publishing in different venues with different standards of evidence. A system judged by how well it imitates human conversational quirks (bottom-left cell) is engineered and evaluated completely differently from a system judged by whether it selects the provably optimal action given its information (bottom-right cell), even if, in a specific narrow task, the two systems might sometimes produce identical output.

### 1.1.1 Acting Humanly: The Turing Test Approach

In 1950 the British mathematician and logician Alan Turing proposed a practical way to sidestep the philosophically loaded question "Can machines think?" and replace it with an operational, testable criterion. In what has since become known as the **Turing Test**, a human interrogator communicates via a text-only channel with two hidden parties one human, one machine and must decide, purely from the conversation, which is which. Turing argued that if a machine could converse well enough to fool the interrogator a reasonable fraction of the time, it would be unreasonable to deny that the machine was "thinking" in any meaningful operational sense, since we routinely attribute thought to other humans on no stronger evidence than their observable behavior.

To pass the Turing Test convincingly, a machine would need to possess several capabilities, each of which has since matured into an established sub-field of AI in its own right:

- **Natural language processing** — to communicate successfully in a human language.
- **Knowledge representation** — to store what it knows or has been told, in a form usable for later reasoning.
- **Automated reasoning** — to answer questions and to draw new conclusions from stored knowledge.
- **Machine learning** — to adapt to new circumstances, and to detect and extrapolate patterns in what it has previously perceived. Turing's original formulation considered only a text channel; a further variant, the **Total**

**Turing Test**, additionally requires the machine to perceive objects through a video signal (computer vision) and to manipulate objects in the world (robotics), so that the interrogator can also pose physical tasks and not merely verbal ones. It is worth stressing that the Turing Test approach deliberately avoids *direct physical* interaction with the interrogator (no attempt is made to make the machine look human, and no physical contact occurs) so that the test measures the machine's intellectual capacities specifically, isolated from questions of physical appearance.

**Case Study: Case Study: The Loebner Prize and Modern Chatbot Evaluation**
Since 1990, the annual Loebner Prize competition has run an informal Turing Test among competing chatbot programs, awarding prizes to the program judges find hardest to distinguish from a human. Early winners, such as programs descended from Joseph Weizenbaum's ELIZA, succeeded largely through clever pattern-matching and conversational deflection rather than genuine language understanding for example, deflecting difficult questions back to the user ("Why do you ask that?") rather than answering them. This illustrates a durable lesson for AI engineering: a system can be optimized specifically to *pass a particular evaluation* without possessing the general capability the evaluation was designed to measure a phenomenon sometimes called "gaming the benchmark," which remains a live methodological concern when evaluating modern large language models on standardized test suites.

It is important to understand why the Turing Test, despite its historical and pedagogical importance, is rarely used today as an actual *design objective* by AI engineers. Engineering a system specifically to fool a human judge is a different and frequently a less useful objective than engineering a system that reliably flies an aircraft, drives safely in traffic, diagnoses disease correctly, or proves a mathematical theorem. Passing the Turing Test arguably requires a machine to imitate human error, human forgetfulness, and human limitation as faithfully as it imitates human competence (since a suspiciously perfect, tireless, error-free conversational partner would itself tip off the interrogator), which is rarely a desirable engineering target for a real deployed system.

**Advantages of the "acting humanly" criterion:** it is intuitively appealing and easy to explain to a lay audience; it does not require any prior commitment to a particular theory of mind or logic; it gives a clear, binary pass/fail operational test.

**Disadvantages and limitations:** it rewards imitation of human flaws over genuine problem-solving competence; it says nothing about how the system should behave in domains (arithmetic, large-scale optimization, exhaustive search) where superhuman rather than human-level performance is exactly what is wanted; and it is a moving target, since human judges' expectations of "human-like" conversation shift as they become more familiar with AI systems.

### 1.1.2 Thinking Humanly: The Cognitive Modeling Approach

A second family of definitions insists that a genuinely intelligent program must not merely produce human-like *outputs*, but must arrive at them via human-like internal *processes*. Pursuing this goal requires getting inside the actual workings of the human mind, either through introspection (attempting to catch our own thoughts as they occur), through carefully designed psychological experiments (observing a person in the act of solving a problem and recording their errors, hesitations, and timing), or, increasingly, through brain imaging technologies such as fMRI (observing the brain's physical activity during cognition). Once a sufficiently precise theory of some aspect of the mind is available, it becomes possible to express that theory as a running computer program; if the program's input–output behavior *and* its intermediate reasoning trace both match human experimental data  for example, matching the time taken and the specific characteristic errors made while solving a logic puzzle  that correspondence is taken as evidence that some of the program's internal mechanisms may mirror mechanisms genuinely operating in the human mind.

This approach gave rise to the interdisciplinary field of **cognitive science**, which brings together computational models from AI with the experimental rigor of psychology to build and empirically test precise, falsifiable theories of the human mind. Early production-system programs such as the **General Problem Solver (GPS)**, developed by Allen Newell and Herbert Simon in the late 1950s, were explicitly designed not merely to solve problems correctly but to solve them in the same *order*, and with the same characteristic errors, as human subjects  so that the program's reasoning trace could be compared step by step against a human "think-aloud" protocol collected in the laboratory.

**Advantages:** when successful, this approach yields genuine scientific insight into human cognition, with practical spin-off value for education, clinical psychology,

and human-computer interface design; it grounds AI claims in falsifiable, testable predictions rather than unfalsifiable philosophical assertion.

**Disadvantages and limitations:** precise, complete cognitive theories of most interesting human reasoning tasks simply do not yet exist, which stalls this approach at the outset for all but a few narrow, well-studied tasks; and even where a cognitive theory is available, faithfully replicating human cognitive *limitations* (short-term memory constraints, systematic biases, fatigue effects) is rarely the actual engineering objective for a deployed AI system.

### 1.1.3 Thinking Rationally: The "Laws of Thought" Approach

The Greek philosopher Aristotle was among the first to attempt to codify "correct thinking" as an irrefutable, mechanical reasoning process, giving rise to what became known as **syllogisms** rigid patterns for argument structure guaranteed to yield correct conclusions whenever given correct premises. The canonical example runs: "All men are mortal; Socrates is a man; therefore, Socrates is mortal." These laws of thought were supposed to govern the operation of any correctly functioning mind, and their systematic, symbolic study initiated the field we now call **logic**.

By the late nineteenth century, logicians including George Boole and Gottlob Frege had developed a precise symbolic notation first-order predicate logic for statements about objects and the relationships between them (this notation is studied in full technical detail in Module 4 of this book). By around 1965, computer programs existed that could, at least in principle, solve *any* solvable problem correctly described in this logical notation, given unbounded time and memory. This **logicist** tradition within AI hopes to build genuinely intelligent systems by encoding domain knowledge and valid inference rules as logical axioms and then using mechanical logical deduction to derive intelligent behavior automatically from those axioms.

Two significant obstacles have historically limited the logicist approach in practice:

1. It is not always straightforward to translate informal, everyday knowledge the kind of knowledge people possess without being able to articulate it as crisp logical rules into the rigid formal term's logic demands, particularly

when the underlying knowledge is uncertain, graded, or context-dependent rather than categorically true or false.

2. There is a very great practical difference between being able to solve a problem "in principle, given unbounded resources" and being able to solve it in practice on a real computer within a useful amount of time. Even a modest number of logical facts and rules can trigger a combinatorial explosion during automated reasoning unless the search for a proof is carefully guided a theme that recurs throughout Module 3 (search) and Module 4 (inference) of this textbook.

**Key Formula: Modus Ponens (the archetypal law of thought)**
From the two premises $P \Rightarrow Q$ (if P then Q) and $P$ (P is true), the law of thought known as Modus Ponens licenses the conclusion $Q$. Formally: $\{P \Rightarrow Q, P\} \vdash Q$. This single inference pattern, mechanically and exhaustively applied, underlies a large fraction of all automated theorem-proving systems studied in Module 4.

### 1.1.4 Acting Rationally: The Rational Agent Approach

The fourth approach and the one this textbook adopts as its unifying, guiding theme throughout every subsequent module defines AI as the study and construction of **rational agents**: entities that perceive their environment through sensors and act upon that environment through actuators, in a manner calculated to best achieve their goals given the information and computational resources actually available to them.

**Definition: Rational Agent**
A rational agent is an agent that, for each possible percept sequence, selects an action expected to maximize its performance measure, given the evidence provided by that percept sequence and whatever built-in prior knowledge the agent possesses. Rationality is judged relative to *expected*, not actual, outcomes.

An agent that reasons perfectly according to the laws of logic ("thinking rationally," Section 1.1.3) will often, but not always, also act rationally: sometimes there simply is no provably correct action available, and the agent must nonetheless act under irreducible uncertainty for example, deciding whether to brake or swerve to avoid a sudden obstacle when there is no time to formally prove which choice is optimal. At other times, correct logical inference is not even necessary for rational behavior, as when an agent reflexively withdraws from a painful stimulus: this is

unambiguously rational behavior (it protects the agent), even though no explicit deductive inference took place. Rational action is therefore the more general, and for engineering purposes the more useful, criterion of the four.

The rational-agent approach has two significant advantages over the other three approaches described above:

1. It is **more general** than the laws-of-thought approach, because correct logical inference is only one of several possible mechanisms by which rationality can be achieved reflexes, learned statistical associations, and optimized control policies can all be rational without involving explicit symbolic inference at all.
2. It is **more amenable to rigorous scientific and engineering development** than approaches grounded in imitating human behavior or human thought, because the standard of rationality can be made completely precise using the mathematical apparatus of decision theory and game theory (borrowed from economics see Section 1.2.3), and this standard is completely general across domains, whereas "act like a human" or "think like a human" ties AI's yardstick to the vagaries of an imperfectly understood, evolutionarily contingent, and frequently irrational biological system.

Module 2 of this textbook develops this idea in full technical detail: what rationality means precisely as a function of the performance measure, the agent's prior knowledge, the actions available to it, and the percept sequence received so far; and how successively more capable internal agent architectures reflex, model-based, goal-based, utility-based, and learning agents each achieve rational behavior through different internal mechanisms suited to different kinds of environments.

It is essential to stress that rationality, as used throughout this textbook, means *doing the right thing given the information actually available* it does **not** mean omniscience, and it does **not** guarantee success in every individual instance. A rational agent that plays a provably fair lottery and happens to lose has still, by definition, acted rationally: rationality is judged against the *expected* value of a decision at the time it is made, not against the outcome that happens to materialize afterward.

**Advantages of the rational-agent criterion:** mathematically precise and domain-general; directly compatible with decision theory, probability theory, game theory, and control theory, giving access to a century of prior mathematical results; naturally accommodates uncertainty, partial information, and competing objectives.

**Disadvantages and limitations:** computing the truly rational (expected-utility-maximizing) action can itself be computationally intractable in complex, realistic environments, forcing engineers to fall back on **bounded rationality** good, but provably suboptimal, approximations that respect real computational limits; and specifying an accurate, complete performance measure for a genuinely complex real-world task (see the discussion of reward/performance-measure misspecification in Section 1.5.2) is itself a difficult and consequential engineering problem.

**Worked Example: Worked Example 1.1 Classifying an AI System by Approach**

**Problem:** A hospital deploys a triage-support program. The program is validated by comparing its urgency rankings of incoming patients against the rankings independently produced by ten senior emergency physicians, and it is considered successful if it matches the physicians' consensus ranking at least 90% of the time. Which of the four approaches (Section 1.1.1–1.1.4) does this design most closely follow, and what is one risk of that choice?

- **Step 1 — Identify what is being measured.** The system's success criterion is agreement with human expert *behavior* (the physicians' rankings), not agreement with an independently, mathematically defined optimum (e.g., minimizing expected patient mortality computed from outcome data).
- **Step 2 — Classify along the two axes.** The criterion is behavior-focused (not a claim about the program's internal "thought process"), and it is human-centered (physicians are the standard), not rationalist.
- **Step 3 — Conclude.** This design most closely follows the **"acting humanly"** approach (Section 1.1.1), specifically a domain-restricted variant of the Turing Test idea, applied to triage decisions rather than open conversation.
- **Step 4 — Identify the risk.** Because the yardstick is human agreement rather than an independent measure of patient outcomes, the system will faithfully reproduce any systematic biases or errors present in the ten

> physicians' own judgment (for example, under-triaging a symptom pattern that is, in fact, known from outcome statistics to be higher-risk than physicians typically judge it to be). A rational-agent (Section 1.1.4) redesign, benchmarked instead against actual patient outcome data and an explicit performance measure such as expected reduction in preventable harm, would not inherit this particular limitation, though it would introduce new engineering challenges around specifying that performance measure correctly.

**Answer:** The design follows the "acting humanly" approach, and its principal risk is faithfully inheriting human raters' systematic biases rather than optimizing an independent, outcome-based measure of patient welfare.

## 1.2 The Foundations of Artificial Intelligence

AI did not emerge from a vacuum. It draws its motivating questions, its formal mathematical tools, and its engineering practice from a range of older disciplines, several of which are summarized in Figure 1.2 and discussed in detail below. Understanding these foundations is not merely a historical curiosity: the vocabulary and formal tools of each parent discipline continue to shape how modern AI systems are designed, described, and evaluated.

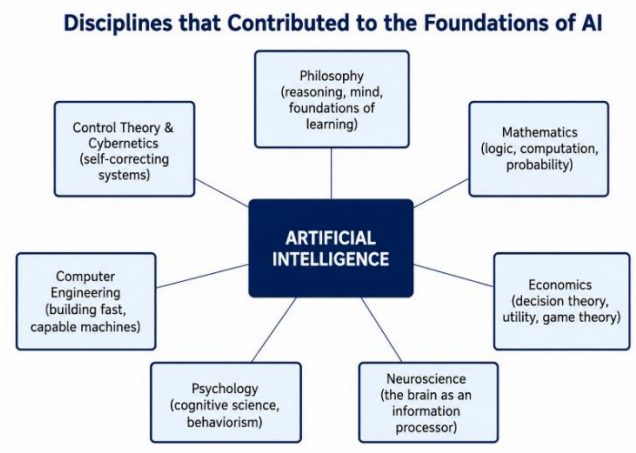


*Figure 1.2: The disciplines that contributed ideas, viewpoints, and techniques to AI.*

### 1.2.1 Philosophy

Philosophy contributed the earliest and most fundamental questions with which AI still wrestles: Can formal, mechanical rules be used to draw valid conclusions? How does a mind arise from a purely physical brain? Where does knowledge

ultimately come from? How does knowledge translate into action? Aristotle's syllogisms, discussed in Section 1.1.3, provided the first formal deductive reasoning system. Centuries later, the philosopher Thomas Hobbes proposed that thinking is fundamentally a form of **computation** over symbols his famous phrase "reasoning is but reckoning" directly anticipates the entire symbolic-AI research programme that dominated the field's first three decades. The empiricist tradition, associated with Francis Bacon and later systematized by John Locke and David Hume, insisted that knowledge is governed by an **induction principle**: general rules are acquired through repeated exposure to associations between their constituent elements a viewpoint that directly foreshadows the statistical foundations of modern machine learning, discussed briefly in Section 1.4 and studied in depth in advanced AI courses beyond this one.

A further, and increasingly consequential, philosophical contribution concerns **ethics and the theory of action**: questions about what an agent *ought* to do, how competing values should be weighed, and who bears responsibility for an autonomous agent's actions, which were once purely academic questions in moral philosophy but are now directly relevant to the engineering practice discussed in Section 1.5.

### 1.2.2 Mathematics

Mathematics supplied AI with the formal tools without which the philosophical questions above could never be made precise or computable: what are the formal rules for drawing valid conclusions (**logic**)? What can be computed at all, and how efficiently (**computability and complexity theory**)? How should an agent reason correctly in the face of uncertain information (**probability theory**)?

George Boole formalized propositional logic in the mid-nineteenth century; Gottlob Frege extended it to the considerably more expressive first-order logic later that century. In 1931, Kurt Gödel proved his celebrated **incompleteness theorem**, showing that in any logical system expressive enough to encode ordinary arithmetic, there necessarily exist true statements that cannot be proven true from within the system itself an important, humbling limit on what any purely logic-based reasoning system, however sophisticated, could ever hope to achieve. Alan Turing, alongside Alonzo Church, independently showed in the 1930s that some well-defined mathematical functions are simply **not computable** by any

mechanical procedure whatsoever, no matter how much time or memory is allowed a result now known as the **Church–Turing thesis**, which defines the outer boundary of computation itself.

**Definition: Computability and the Church–Turing Thesis**
The Church–Turing thesis states, informally, that any function that can be computed by an effective, mechanical procedure at all can be computed by a Turing machine (equivalently, by any modern general-purpose computer, given enough time and memory). Its significance for AI is that it draws a hard boundary: some well-posed problems (such as the general Halting Problem determining, for an arbitrary program and input, whether that program will eventually halt) are proven to be **undecidable**, meaning no algorithm, however clever, can solve them correctly in all cases. No AI system, however advanced, can circumvent this mathematical boundary.

**Complexity theory**, developed substantially later than computability theory, further distinguishes computable problems into those that are **tractable** (solvable in a time that grows only polynomially with problem size) and those that are **intractable** (technically solvable, but only at a cost that grows exponentially with problem size, so that no realistic computer could solve even moderately large instances within a useful amount of time). This tractable/intractable distinction is of immense direct practical importance for the design of the search algorithms studied in Module 3 and the planning algorithms studied in Module 5, both of which must contend directly with problems that are, in the worst case, NP-hard or worse.

### 1.2.3 Economics

The science of economics, particularly since the eighteenth-century work of Adam Smith, is concerned with how agents ought to make decisions that maximize their own well-being, formalized as a numerical payoff or **utility**. The mathematical theory of **decision theory**, which combines probability theory with utility theory, provides a formal and complete framework for decision-making under uncertainty that is, for situations in which the payoff of a given action is not a single deterministic value but is instead described by a probability distribution over several possible outcomes.

**Key Formula: Expected Utility**

Given an action $a$ with possible outcomes $o_1, o_2, \ldots, o_n$, occurring with probabilities $P(o_1), P(o_2), \ldots, P(o_n)$ respectively, and a utility function $U$ that assigns a numeric desirability to each outcome, the **expected utility** of action $a$ is:

$$EU(a) = \Sigma_i \, P(o_i) \cdot U(o_i)$$

A rational decision-theoretic agent selects the action that maximizes $EU(a)$ among all actions available to it. This single equation underlies the utility-based agent design studied in Module 2, Section 2.4.4.

**Game theory**, developed principally by John von Neumann and Oskar Morgenstern in their 1944 book *Theory of Games and Economic Behavior*, extends decision theory to settings where an agent's payoff depends not only on its own choices but also on the choices of other agents. Game theory introduced the fundamental and somewhat counter-intuitive insight that in certain competitive situations, a rational agent should act **unpredictably** that is, should choose among several available actions at random, according to a carefully calculated probability distribution because any *predictable* deterministic strategy could otherwise be exploited by an opposing agent. Both decision theory and game theory are foundational to the utility-based agent design of Module 2 and to the design of multi-agent AI systems more generally, including competitive game-playing programs such as those discussed in Section 1.4.

### 1.2.4 Neuroscience

Neuroscience is the study of the nervous system, and in particular the brain, and it establishes the plain empirical fact that the brain is, at bottom, a physical information-processing device built from roughly 86 billion interconnected neurons, each a comparatively simple unit that fires an electrochemical pulse when the combined input from its neighbors exceeds some threshold. This single observation that sophisticated thought is a physical process occurring within a network of many simple, densely interconnected units directly motivated the **connectionist** research program within AI, and modern artificial neural networks, the technical foundation of contemporary deep learning (Section 1.3.10), are loosely inspired by this architecture, even though an individual artificial neuron is a drastically simplified mathematical abstraction of a real biological neuron, and the learning rules used to train artificial networks (back-propagation, Section 1.3.7) have no direct, established biological counterpart in the brain.

### 1.2.5 Psychology

Experimental psychology, and in particular the **cognitive psychology** movement that began in earnest in the 1960s, treats the human brain as an information-processing device, in direct and deliberate analogy with a digital computer, and seeks precise, empirically testable models of perception, memory, reasoning, and motor control. This tradition supplied AI with both a rich set of empirical benchmarks human performance data on cognitive tasks, against which "thinking humanly" systems (Section 1.1.2) could be quantitatively measured and a rich descriptive vocabulary (working memory, chunking, attention, cognitive load) that continues to inform how AI researchers describe and design the internal states of intelligent agents.

### 1.2.6 Computer Engineering

However mathematically elegant the underlying theory, AI ultimately requires a physical substrate capable of executing it: fast processors, large and fast memories, and, for embodied systems, robust and precise sensors and actuators. The exponential, decades-long growth in raw computational power popularly summarized as **Moore's Law**, the empirical observation that the number of transistors on an affordable integrated circuit roughly doubled every two years for several decades has arguably been as important a driver of practical AI progress as any single algorithmic breakthrough, since algorithms once considered hopelessly impractical (very large neural networks, very deep game-tree search, massive statistical language models) became routine, everyday engineering practice once enough affordable computation became available to run them. Computer engineering additionally supplies the operating systems, distributed-systems infrastructure, programming languages, and software-engineering discipline without which large, reliable AI systems could not be built, tested, deployed, or maintained at scale.

### 1.2.7 Control Theory and Cybernetics

Control theory, historically rooted in James Watt's eighteenth-century steam-engine governor and formalized in the twentieth century by Norbert Wiener under the name **cybernetics**, studies how machines can regulate their own behavior automatically through **feedback**: an actuator changes the state of the environment or plant being controlled; a sensor then measures the resulting state; and the

difference (or "error") between the measured state and the desired state is fed back into the controller to correct the *next* action, closing the loop. This feedback loop is, in essence, precisely the same **agent–environment interaction loop** that underlies the whole of intelligent agent design as developed in Module 2, and modern control theory including optimal control and stochastic control overlaps substantially with what AI researchers today call reinforcement learning.

### 1.2.8 Linguistics

Modern theoretical linguistics, particularly following the 1957 publication of Noam Chomsky's *Syntactic Structures*, demonstrated that a systematic, mathematically precise theory of grammar could be studied as a formal, computational object in its own right, quite apart from the sociological or literary study of language. This insight helped create **computational linguistics**, or **natural language processing (NLP)**, one of the largest and most commercially significant sub-fields of AI today, encompassing everything from grammar and spell checking, through statistical and neural machine translation, to the large language models that power contemporary conversational AI assistants.

### 1.2.9 Summary Table: Foundational Disciplines and Their Contributions

| Discipline | Central question contributed | Key concept used throughout this textbook |
|---|---|---|
| Philosophy | Can rules govern valid reasoning? Where does knowledge come from? | Logic, induction, the ethics of autonomous action |
| Mathematics | What can be computed, how efficiently, and under uncertainty? | Logic, computability, complexity, probability |
| Economics | How should a self-interested agent decide under uncertainty? | Utility theory, decision theory, game theory |
| Neuroscience | How does the brain process information physically? | Neural-network (connectionist) architectures |
| Psychology | How do humans actually perceive, remember, and reason? | Cognitive benchmarks; agent-state vocabulary |

| Discipline | Central question contributed | Key concept used throughout this textbook |
|---|---|---|
| Computer Engineering | How do we build fast, reliable physical computing systems? | Hardware, software infrastructure, scalability |
| Control Theory / Cybernetics | How can a system self-correct using feedback? | The agent–environment feedback loop (Module 2) |
| Linguistics | Can grammar and meaning be formalized computationally? | Natural language processing |

## 1.3 The History of Artificial Intelligence

Understanding AI's history is not an optional decoration on top of the technical material it is essential engineering literacy, because AI has passed through repeated cycles of dramatic over-promise followed by sober, sometimes painful, reassessment, and every practicing AI engineer benefits from being able to recognize the early warning signs of such a cycle. Figure 1.3 places the milestones discussed below on a single timeline.

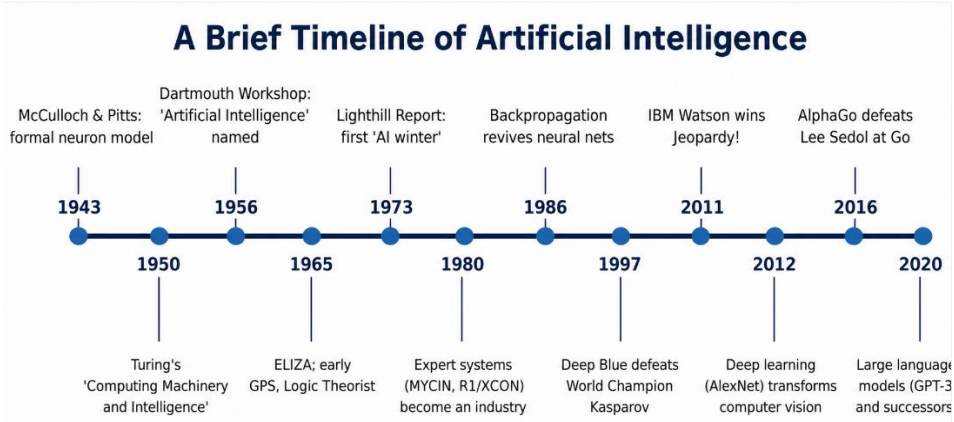


*Figure 1.3: A brief timeline of major milestones in the history of Artificial Intelligence.*

### 1.3.1 The Gestation of Artificial Intelligence (1943–1955)

The intellectual groundwork for AI was laid before the field even had a name. In 1943, Warren McCulloch and Walter Pitts proposed the first mathematical model of an artificial neuron, showing formally that networks of simple binary "on/off"

units, appropriately wired together, could in principle compute any function expressible in propositional logic the first formal bridge between neuroscience (Section 1.2.4) and computation (Section 1.2.2).

**Key Formula: The McCulloch–Pitts Neuron**

A McCulloch–Pitts neuron computes a simple threshold function of its weighted inputs: given inputs $x_1, \ldots, x_n$ with weights $w_1, \ldots, w_n$ and a threshold $\theta$, the neuron "fires" (outputs 1) if and only if $\Sigma_i \, w_i x_i \geq \theta$, and outputs 0 otherwise. Despite its extreme simplicity, this single equation is the direct mathematical ancestor of every artificial neural network, including the deep learning systems discussed in Section 1.3.10.

Donald Hebb subsequently proposed a simple, biologically motivated update rule **Hebbian learning**, often summarized by the slogan "cells that fire together, wire together" by which the connection strength between two neurons could be adjusted based on how correlated their activity was, directly anticipating the gradient-based learning rules used to train modern artificial networks. At Princeton and later at Harvard, graduate students including a young Marvin Minsky built early hardware simulations of small learning neural networks (Minsky's 1951 SNARC machine used 3,000 vacuum tubes to simulate a network of 40 neurons), and in 1950 Alan Turing published "Computing Machinery and Intelligence" in the philosophy journal *Mind*, introducing the Turing Test (Section 1.1.1) and speculating seriously, for perhaps the first time in print, about the genuine possibility of thinking machines.

### 1.3.2 The Birth of Artificial Intelligence (1956)

The field acquired both its name and its identity as a distinct academic discipline at a workshop held at **Dartmouth College** in the summer of 1956, organized by John McCarthy, Marvin Minsky, Nathaniel Rochester, and Claude Shannon, and attended by roughly twenty researchers over about eight weeks. It was McCarthy who coined the term "Artificial Intelligence" for the funding proposal that made the workshop possible, choosing the name deliberately to establish the new field's independence from the closely related but methodologically distinct discipline of cybernetics (Section 1.2.7). Although the workshop itself did not produce any dramatic new technical results, it brought together nearly all of the individuals who would go on to lead AI research for the following two decades, and it publicly established the founding conviction of the discipline: that every aspect of learning,

or any other feature of intelligence, can in principle be so precisely described that a machine can be made to simulate it.

### 1.3.3 Early Enthusiasm, Great Expectations (1952–1969)

The years immediately following the Dartmouth workshop were marked by rapid, often startling progress on narrow but symbolically impressive tasks, producing an atmosphere of great optimism and, with hindsight, considerable overpromising about how soon "general" machine intelligence would be achieved.

- Allen Newell and Herbert Simon's **Logic Theorist** (1956) proved mathematical theorems from Whitehead and Russell's *Principia Mathematica* automatically, and in a small number of cases found proofs that were shorter and more elegant than the original published proofs a genuinely striking result at the time.
- Their subsequent **General Problem Solver (GPS)** attempted to imitate general human problem-solving using a domain-independent technique called **means-ends analysis** (repeatedly identifying the largest difference between the current state and the goal state, and selecting an action known to reduce that specific difference), a technique whose descendants are still recognizable in the goal-based agents studied in Module 2 and the regression-planning algorithms studied in Module 5.
- Arthur Samuel wrote a **checkers-playing program** at IBM that improved its own play through self-play practice, using an early form of what would later be formally recognized as reinforcement learning, and it eventually reached a strong amateur standard of play a notable early demonstration that a machine could improve at a task purely through its own experience, without being explicitly reprogrammed.
- John McCarthy developed the **Lisp** programming language in 1958, which became the dominant language for AI research for the following three decades because of its unusual and, for the time, radical ability to treat programs and data uniformly as manipulable symbolic expressions.
- At MIT, researchers built early natural-language systems, most famously Joseph Weizenbaum's **ELIZA** (1966), which used simple keyword-spotting and template substitution to imitate a Rogerian psychotherapist convincingly enough that some users mistook it for genuine understanding and empathy an early and important cautionary lesson, still directly relevant

to the evaluation of today's conversational AI, about the gap between the *appearance* of intelligent understanding and its actual substance.

### 1.3.4 A Dose of Reality (1966–1973)

By the later 1960s, the limitations of the early programs which typically handled only small, deliberately simplified "toy" versions of their target problems began to show through painfully once researchers attempted to scale them up to realistic, full-sized problems. Machine translation, an early flagship application funded heavily during the Cold War for its evident strategic value, largely failed to deliver on its promises, because it turned out that accurate translation between natural languages required vast amounts of general world knowledge and contextual disambiguation, not merely a large bilingual dictionary paired with grammar-substitution rules  a famous (if likely apocryphal in its exact wording) anecdote recounts an early system translating "the spirit is willing but the flesh is weak" into Russian and back into English as "the vodka is good but the meat is rotten."

Early neural-network research also stalled seriously during this period: Marvin Minsky and Seymour Papert's influential 1969 book *Perceptrons* rigorously proved that the single-layer **perceptron** networks fashionable at the time were mathematically incapable of representing certain simple functions, most famously the logical exclusive-or (XOR) function. Although this specific limitation is solvable by adding just one additional ("hidden") layer to the network a fact well understood in principle even then the result was widely, and with hindsight somewhat unfairly, read across the research community as a fundamental death sentence for the entire connectionist research program, contributing directly to its decade-long eclipse.

Compounding these purely technical setbacks, government funding agencies grew visibly impatient with the widening gap between the grand promises of the early 1960s and the modest results actually being delivered. In the United Kingdom, the 1973 **Lighthill Report**, commissioned directly by the British government from applied mathematician Sir James Lighthill, concluded bluntly that AI research had failed to achieve its grandiose stated objectives and explicitly recommended sharply reduced government funding for the field. In the United States, the DARPA funding agency imposed broadly similar cuts around the same time. The resultingsharp, multi-year contraction in AI research funding and general r activity

through the mid-1970s is remembered in the field's own folklore as the first **"AI winter."**

**Case Study: Case Study: Why Machine Translation Failed in the 1960s**
Early machine translation systems relied on direct word-for-word substitution supplemented by simple local grammar rules. They failed on ordinary sentences requiring **disambiguation using world knowledge** for example, correctly resolving whether "the bank" refers to a financial institution or a riverbank requires knowing the surrounding context of the sentence and, often, broader facts about the world, not merely local grammatical structure. This single, narrow failure mode motivated decades of subsequent research into knowledge representation (Module 4) and, much later, into the statistical and neural machine translation systems that finally achieved genuinely practical translation quality only once trained on enormous quantities of real bilingual text illustrating a recurring theme in AI history: many tasks that appear to require "understanding" turn out to be far better solved by learning statistical patterns from very large datasets than by hand-coded rules alone (see Section 1.3.10).

### 1.3.5 Knowledge-Based Systems: The Key to Power? (1969–1979)

The AI research that survived and quietly thrived through the 1970s did so largely by abandoning the search for general-purpose "weak methods" (such as GPS's domain-independent means-ends analysis, which works reasonably across many domains but not particularly deeply or effectively in any single one of them) in favor of **domain-specific, knowledge-intensive** approaches: rather than attempting to reason from first principles using generic inference machinery, these systems packed in vast amounts of carefully hand-coded expert knowledge about one narrow domain.

- **DENDRAL** (Edward Feigenbaum, Bruce Buchanan, and the Nobel laureate geneticist Joshua Lederberg), developed at Stanford University beginning in the mid-1960s, inferred the molecular structure of unknown organic compounds directly from raw mass-spectrometer data, using inference rules painstakingly distilled from expert analytical chemists. DENDRAL is generally regarded as the first genuinely successful knowledge-intensive expert system, and it directly demonstrated the field's most important lesson of the decade: that raw problem-solving power in AI often comes not from a more sophisticated general-purpose inference engine, but overwhelmingly

from the *quantity and quality of domain-specific knowledge* fed into a comparatively simple reasoning engine.

- **MYCIN**, also developed at Stanford in the 1970s, extended this **rule-based expert system** approach to the medical diagnosis of bacterial blood infections and the recommendation of appropriate antibiotic therapy. MYCIN additionally represented uncertainty within its roughly 600 hand-coded rules using numeric "certainty factors" attached to each rule an early, admittedly ad hoc, attempt at reasoning under genuine uncertainty that directly anticipated the later, mathematically more principled Bayesian probabilistic methods used throughout modern AI. In blinded evaluation studies, MYCIN's antibiotic therapy recommendations were judged by independent experts to be as good as, or in some specific respects better than, those of the Stanford medical faculty it was tested against a landmark result for the credibility of the entire expert-systems research program, even though MYCIN itself was never actually deployed in routine clinical practice, largely due to unresolved legal liability and system-integration concerns rather than any purely technical shortfall.

This era firmly established the **knowledge-based systems** paradigm that Module 4 of these textbook studies in full technical detail the central idea that an intelligent system is, at its core, a combination of a **knowledge base** of facts and rules describing its domain, together with a general-purpose **inference engine** capable of mechanically deriving valid new conclusions from that stored knowledge.

### 1.3.6 AI Becomes an Industry (1980–Present)

The demonstrated success of knowledge-based systems led directly to the first genuinely large-scale *commercial* wave of AI activity. The **R1** expert system (also known within Digital Equipment Corporation as **XCON**), built at DEC beginning in 1980, automatically configured the correct components for new custom computer-system orders, and was reported by the company to be saving on the order of forty million dollars a year by the mid-1980s by eliminating costly configuration errors a figure that triggered a substantial commercial boom in expert-system development and adoption throughout the early-to-mid 1980s. An entire cottage industry of specialized "AI companies" sprang up during this period, selling both dedicated Lisp-based workstation hardware (from companies such as Symbolics and Lisp Machines Inc.) and reusable expert-system "shells" generic inference

engines into which a customer's own domain-specific rules could be loaded without needing to write a new inference engine from scratch each time.

This commercial boom, however, proved unsustainable within roughly half a decade. Deployed expert systems were often **brittle** performing well within their narrow area of encoded expertise but failing badly, and often without any useful warning, the moment a query strayed even slightly outside that narrow area. They were also **expensive to build and maintain**, since extracting precise rules from human domain experts by hand a laborious, specialized process that came to be called "knowledge engineering" (Section 1.2 and Module 4, Section 4.5.3) proved slow, difficult to scale, and dependent on skilled, scarce personnel. The specialized Lisp-machine hardware business was soon commercially undercut by cheaper, faster, general-purpose workstations from vendors such as Sun Microsystems that could run equivalent Lisp software without any specialized hardware at all. By the late 1980s, the commercial AI industry contracted sharply for a second time the **second AI winter**. Nonetheless, the underlying core idea that carefully engineered systems built around structured, explicit domain knowledge could deliver measurable, real commercial value permanently and durably established AI as a field of genuine industrial as well as academic relevance, a status the field has never again fully lost, even during its leanest funding years.

### 1.3.7 The Return of Neural Networks (1986–Present)

In the mid-1980s, at least four separate research groups working largely independently reinvented the **back-propagation** learning algorithm an algorithm first described mathematically in the early 1970s but not widely appreciated or applied at the time which finally gave researchers a practical, general, and computationally efficient method for training multi-layer neural networks: that is, networks possessing one or more "hidden" layers positioned between their input and output layers, of precisely the architectural kind that Minsky and Papert had rigorously shown a single-layer perceptron could never emulate (Section 1.3.4).

**Key Formula: The Back-Propagation Update Rule (conceptual form)**

Back-propagation trains a multi-layer network by computing the gradient of a total error function $E$ with respect to every connection weight $w$ in the network, using the calculus chain rule to propagate error signals backward from the output layer to the input layer, and then adjusting each weight in the direction that reduces the error:

$w \leftarrow w - \eta \cdot (\partial E / \partial w)$

where η (eta) is a small positive **learning rate**. Applied repeatedly over many training examples, this simple update rule allows networks with hidden layers to learn representations that a single-layer perceptron provably cannot.

This algorithmic breakthrough, combined with the steadily growing computational power discussed in Section 1.2.6, sparked a **connectionist** revival that has continued, with periodic further accelerations, through to the present day, culminating in the deep-learning revolution discussed in Section 1.3.10 below.

### 1.3.8 AI Adopts the Scientific Method (1987–Present)

From the late 1980s onward, AI matured significantly as a research discipline in its methodology. Researchers increasingly built new systems directly on top of existing, well-understood mathematical theories  probability theory, decision theory, and formal mathematical optimization rather than inventing bespoke, ad hoc mechanisms from scratch for each new system, and they increasingly evaluated proposed new techniques through rigorous, quantitative, empirical comparison on shared, publicly available benchmark problems and datasets, rather than relying on the isolated, informal demonstration of a single hand-picked, favorable example. This broad methodological shift often described within the field itself as AI's move from "an art" toward "a science" brought AI's day-to-day research methodology substantially closer to that of neighboring, more mature quantitative disciplines such as statistics and operations research, and it laid the essential empirical groundwork on which the subsequent, rigorously benchmarked progress of machine learning would depend.

### 1.3.9 The Emergence of Intelligent Agents (1995–Present)

Beginning in the mid-1990s, researchers increasingly reframed AI's central organizing question away from isolated, standalone components (a theorem prover studied here, a computer-vision system studied there, in near-complete isolation from each other) and toward the integrated design of **complete, situated agents** that must perceive, reason, learn, and act as a single coherent whole within some environment, frequently over an extended operational period and always under real computational and time constraints. This "agent perspective" organizing the entire field around the single unifying question "what is the right thing for this particular agent to do, right now, given everything it has ever

perceived?" is precisely the perspective this textbook itself adopts throughout, and it is developed in complete technical detail in Module 2. The simultaneous rise of the World Wide Web during this same period also created entirely new categories of software agents (web crawlers, early recommendation engines, and later fully autonomous online assistants) that had to operate robustly within very large, open, and only ever partially observable environments.

### 1.3.10 The Availability of Very Large Data Sets (2001–Present)

Since roughly the early 2000s, and especially and dramatically since around 2010, AI progress has been driven as much by the explosive growth in *available training data* as by any purely algorithmic breakthrough. Very large, carefully labeled datasets such as ImageNet, a database of many millions of hand-labeled photographic images publicly released in 2009 made it possible, for the first time in the field's history, to train very large neural networks without those networks simply memorizing (overfitting to) their comparatively small training set, and the resulting **deep learning** systems produced dramatic, discontinuous accuracy improvements in computer vision, speech recognition, and machine translation across roughly the 2011–2015 period. This "data-driven" era has continued and substantially intensified with the training of massive **large language models** on text drawn from a very large fraction of the entire public internet, giving rise to the general-purpose conversational AI systems that are now widely and routinely deployed across consumer and enterprise software alike.

### 1.3.11 Comparative Summary: The AI "Winters" and Their Causes

| Period | Trigger of over-promise | Technical limitation exposed | Consequence |
|---|---|---|---|
| First AI winter (1974–early 1980s) | Machine translation, general problem solvers | Combinatorial explosion; lack of world knowledge; perceptron limitations | Sharp funding cuts (Lighthill Report, DARPA cuts) |

| Period | Trigger of over-promise | Technical limitation exposed | Consequence |
|---|---|---|---|
| Second AI winter (late 1980s–early 1990s) | Commercial expert-systems boom | Brittleness outside narrow domains; costly, slow knowledge engineering; hardware commoditization | Collapse of the specialized Lisp-machine and expert-system-shell industry |

A recurring, durable lesson from both AI winters one worth internalizing as a working engineer rather than merely a historical footnote is that **narrow, impressive demonstrations on curated examples routinely fail to generalize to messy, unconstrained real-world deployment**, and that funding and public expectations, once inflated by early impressive demos, can contract sharply and painfully once that gap becomes publicly visible. Contemporary AI engineers are well advised to communicate the genuine, tested limitations of their systems as carefully and as clearly as they communicate the systems' genuine capabilities.

## 1.4 The State of the Art

It is worth taking careful stock of what AI systems can currently do well, since this directly shapes the terms of the risk/benefit debate discussed in Section 1.5, and since "state of the art" claims made even a few years apart can date very quickly in this field.

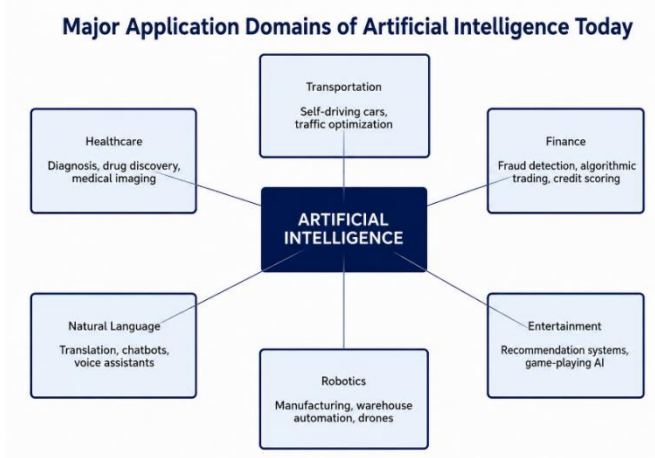


*Figure: Major application domains of Artificial Intelligence today.*

- **Game playing.** Computer programs now play chess, checkers, Go, poker, and complex real-time video games at a level that meets or exceeds the best human players. IBM's **Deep Blue** defeated reigning world chess champion Garry Kasparov in a landmark 1997 match; DeepMind's **AlphaGo** defeated top-ranked Go professional Lee Sedol in 2016, a result many researchers had not expected for another decade, because Go's search space is vastly larger than chess's and had long been considered to require distinctly human-like intuition; and subsequent systems have gone on to master poker (a game of hidden information) and complex real-time strategy video games against professional human teams.
- **Autonomous vehicles.** Self-driving cars, developed by companies including Waymo, Tesla, and Cruise, now operate — under varying degrees of human supervision defined by the SAE's six-level autonomy scale — on public roads in a growing number of cities, combining computer vision, radar and lidar sensor fusion, real-time path planning (Module 3), and closed-loop control (Section 1.2.7).
- **Speech recognition and natural language processing.** Modern systems transcribe continuous, multi-speaker speech, translate fluently between dozens of languages in real time, and generate coherent, contextually appropriate text, underpinning voice assistants, live captioning services, and machine-translation tools used by many hundreds of millions of people daily.
- **Computer vision and medical diagnosis.** Deep-learning-based vision systems can detect certain cancers from radiology images, and diagnose diabetic retinopathy from retinal photographs, at accuracy levels competitive with, and in some published studies exceeding, trained human specialists working under controlled study conditions.
- **Robotics.** Industrial robots perform precise, highly repetitive manufacturing and assembly tasks within tightly controlled factory settings, and increasingly also perform far less structured tasks — warehouse order picking and packing (as pioneered at scale by Amazon's Kiva/Amazon Robotics systems), robotic home vacuum cleaning, and robot-assisted surgery — in progressively less controlled, more variable real-world environments.
- **Logistics and automated planning.** Automated planning and scheduling systems, of precisely the kind studied formally in Module 5 of this textbook, are used routinely and at massive scale to schedule complex real operations

such as spacecraft activity sequencing (NASA's Remote Agent software famously controlled the Deep Space 1 probe autonomously in 1999), airline crew and aircraft-fleet assignment, and large-scale warehouse and global supply-chain logistics.

- **Generative AI.** Large language models and image-generation systems can now produce fluent original text, working computer code, and photorealistic or deliberately stylized images directly from natural-language descriptions, and are increasingly integrated as everyday tools into mainstream software for writing, programming, and visual design.

**Case Study: Case Study: AlphaGo and the Combination of Search, Learning, and Self-Play**

DeepMind's AlphaGo, and its fully self-taught successor AlphaZero, combined three distinct AI techniques studied at various points in this textbook: a **deep neural network** (Section 1.3.10) trained to evaluate board positions and to suggest promising candidate moves; a **tree search** algorithm (directly related to the search algorithms studied in Module 3) that looked ahead through the enormous space of possible future move sequences, guided by the network's evaluations rather than exploring blindly; and **reinforcement learning through self-play**, in which the system repeatedly played complete games against earlier versions of itself, using each game's eventual win or loss outcome to further improve its own neural network. Notably, AlphaZero was given no human game data or human opening-move knowledge whatsoever — only the bare formal rules of the game — and within a small number of days of self-play training it substantially exceeded the strength of all previous chess, shogi, and Go programs, including its own predecessor AlphaGo. This result is frequently cited as a landmark demonstration that carefully combining several distinct classical and modern AI techniques can very substantially exceed what any single technique could achieve in isolation.

Despite this genuinely impressive progress, it is equally important, and directly relevant to responsible engineering practice, to recognize clearly what current AI systems still do **not** do reliably: robust, general common-sense reasoning about the ordinary physical and social world; genuine understanding of underlying *causal* relationships, as opposed to the detection of merely statistical correlations in training data; reliably graceful behavior when operating in situations very different from anything represented in the system's training data (a property researchers call **out-of-distribution robustness**); and reliable, long-horizon

autonomous planning within truly open-ended, unpredictable real-world environments all remain active, unsolved, and actively researched problems as of this writing. This gap is often summarized by the distinction between **narrow AI** (all currently deployed systems, each engineered for one specific class of task) and the still-unrealized long-term goal of **general AI** (a single system with human-level competence across essentially any task), illustrated in the figure below.

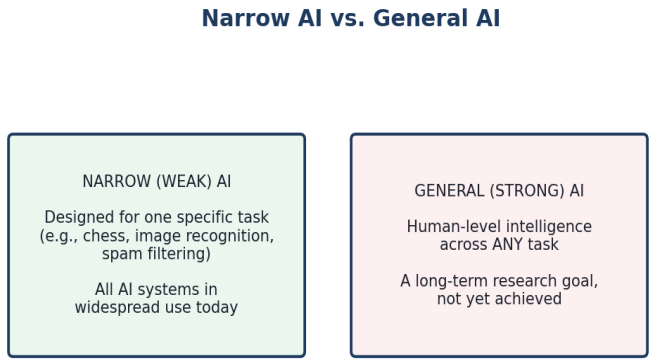


*Figure: Narrow AI versus General AI — all currently deployed systems are narrow; general, human-level AI across arbitrary tasks remains an unrealized long-term research goal.*

### Practical applications by sub-field (summary):

| AI sub-field | Representative real-world application | Named example system |
|---|---|---|
| Search & planning (Modules 3, 5) | Turn-by-turn navigation, warehouse robot routing | Google Maps; Amazon Robotics |
| Knowledge representation & logic (Module 4) | Medical diagnosis support, tax/legal rule compliance | MYCIN; modern clinical decision-support tools |
| Computer vision | Medical imaging, autonomous vehicle perception, facial unlock | Diabetic-retinopathy screening systems |
| Natural language processing | Machine translation, conversational assistants, summarization | Google Translate; large language model assistants |

| AI sub-field | Representative real-world application | Named example system |
|---|---|---|
| Machine learning / deep learning | Recommendation, fraud detection, forecasting | Streaming-service recommenders; bank fraud systems |
| Multi-agent / game-playing AI | Competitive strategy games, automated trading | AlphaGo; algorithmic trading systems |

## 1.5 Risks and Benefits of AI

Any technology as broadly capable as AI carries both substantial benefits and substantial risks simultaneously, and a well-trained computer science graduate should be able to discuss both directly and even-handedly, without collapsing into either uncritical technological enthusiasm or uncritical technological alarm.

### 1.5.1 Benefits

- **Automation of dangerous, tedious, or repetitive work**, freeing human workers from labor that is physically unsafe (bomb disposal, deep-sea and space exploration), physically damaging over time (repetitive-strain manufacturing tasks), or simply of very low intrinsic value (routine data entry and document processing).
- **Improved healthcare outcomes**, through faster and in some documented cases more accurate diagnosis, more personalized treatment recommendations, accelerated drug-discovery candidate screening, and meaningfully expanded access to basic medical guidance in regions that have historically had very few doctors relative to population.
- **Accelerated scientific discovery**, where AI systems now materially assist in analyzing very large experimental datasets across genomics, particle physics, and astronomy, and in predicting the physical and chemical properties of entirely new candidate materials and molecules DeepMind's **AlphaFold** protein-structure-prediction system, which solved a fifty-year-old grand-challenge problem in structural biology, is a widely cited example.
- **Economic productivity gains**, since AI-assisted tools can substantially increase the output of knowledge workers in software development, professional writing, visual design, and data analysis, and can make sophisticated capabilities (accurate translation, first-pass legal document

review, financial analysis) newly and cheaply available to individuals and small organizations that could not previously afford them.

- **Improved accessibility**, since speech recognition, text-to-speech synthesis, real-time captioning and translation, and computer-vision-based scene description can substantially improve independence and access to information for people with visual, hearing, or motor impairments.

### 1.5.2 Risks

- **Labor-market disruption.** Automation of tasks previously performed by human workers can displace them, particularly in occupations built substantially around routine cognitive or physical tasks, and the *pace* of AI-driven change may in some sectors outstrip the rate at which affected workers and surrounding institutions can realistically retrain and adapt.
- **Bias and unfairness.** Machine-learning systems trained on historical data can learn, and subsequently perpetuate or even amplify, biases already present in that training data  well-documented in hiring-screening tools, consumer-lending decisions, and criminal-justice risk-assessment tools leading to systematically unfair outcomes for particular demographic groups if the issue is not actively and deliberately addressed during system design.
- **Privacy and surveillance.** AI systems capable of processing images, speech, location, and behavioral data at very large scale can enable forms of pervasive surveillance and personal profiling whether conducted by governments, corporations, or other actors that were simply not previously technically or economically feasible at comparable scale.
- **Misinformation and manipulation.** Generative AI can now produce highly convincing synthetic text, images, audio, and video (including so-called "deepfakes"), substantially lowering the cost of running large-scale disinformation campaigns and making it correspondingly harder for the general public to trust digital media by default.
- **Safety and reliability.** AI systems deployed in safety-critical settings autonomous vehicles, medical devices, industrial control systems can fail in genuinely unexpected ways when they encounter situations poorly represented in their original training data, and the increasing internal complexity and comparative opacity ("black-box" character) of modern deep-learning models can make such failures difficult to anticipate, diagnose after the fact, or clearly explain to affected stakeholders or regulators.

- **Autonomous weapons.** The application of AI techniques to military targeting and weapons systems raises serious, unresolved ethical and legal concerns regarding meaningful human accountability, the risk of accelerated and destabilizing escalation dynamics, and the fundamental question of delegating lethal decisions to a machine.
- **Concentration of power and the long-term "alignment" problem.** As AI systems become progressively more capable and more autonomous, reliably ensuring that their actual objectives remain aligned with genuine human values and intentions even as these systems act with growing independence and in circumstances their original designers did not fully anticipate is recognized by many researchers as a difficult, currently unsolved, and increasingly consequential technical and governance problem.

**Case Study: Case Study: Algorithmic Bias in Criminal-Justice Risk Assessment**

Several U.S. jurisdictions have used proprietary algorithmic tools to estimate a criminal defendant's statistical risk of reoffending, with these risk scores sometimes informing bail, sentencing, or parole decisions. Independent investigative analysis of one widely used such tool (published by ProPublica in 2016) found that the tool's *error pattern* differed systematically by defendant race, even though race was not an explicit input feature to the model: among defendants who did *not* subsequently reoffend, Black defendants were, according to that analysis, considerably more likely than white defendants to have been incorrectly flagged by the tool as high-risk. This case is now widely and repeatedly studied in AI ethics education because it illustrates a genuinely subtle and mathematically important point: a predictive model can achieve equal overall accuracy across demographic groups while still exhibiting starkly *unequal error rates* between those groups — and, moreover, it can be mathematically proven that certain intuitively appealing definitions of "fairness" are, in general, mutually incompatible and cannot all be simultaneously satisfied by any single scoring system when the underlying base rates genuinely differ between groups. Responsible deployment of such tools accordingly requires the engineering team to make an explicit, carefully justified, and ideally externally reviewed choice among competing, mathematically well-defined fairness criteria, rather than assuming that avoiding race as an explicit input feature is by itself sufficient to guarantee a fair outcome.

### 1.5.3 Risk–Mitigation Summary Table

| Risk | Representative concrete mitigation |
|---|---|
| Labor-market disruption | Investment in worker retraining programs; phased, transparent deployment timelines |
| Bias and unfairness | Bias auditing on representative held-out data; diverse design and review teams; explicit, published fairness criteria |
| Privacy and surveillance | Data minimization; strong access controls; anonymization/aggregation; clear regulatory oversight |
| Misinformation (deepfakes) | Content provenance and watermarking standards; platform-level detection tools; public media-literacy education |
| Safety and reliability | Extensive testing on out-of-distribution scenarios; human-in-the-loop oversight for high-stakes decisions; graceful degradation design |
| Autonomous weapons | International governance frameworks; enforced meaningful human control requirements |
| Long-term alignment | Dedicated AI safety research; staged and monitored capability deployment; interpretability research |

A responsible engineer working in AI is expected to understand these risks well enough to *actively design systems that mitigate them* through careful, representative dataset curation, rigorous pre-deployment testing, honest and specific public communication about a system's tested limitations, and appropriately calibrated human oversight for consequential decisions rather than treating such considerations as someone else's problem to solve later. This theme of **responsible and ethical AI development** recurs throughout the remaining, more technical modules of this textbook, and it should inform how the algorithms studied there are ultimately deployed in practice.

**Key Terms Introduced in This Module**

Turing Test · Total Turing Test · cognitive science · syllogism · rational agent · performance measure · Church–Turing thesis · tractable / intractable problem · expected utility · game theory · Moore's Law · cybernetics · McCulloch–Pitts neuron · Hebbian learning · Dartmouth workshop · Logic Theorist · General Problem Solver · means-ends analysis · perceptron · Lighthill Report · AI winter ·

DENDRAL · MYCIN · knowledge-based system · R1/XCON · back-propagation · deep learning · AlphaGo/AlphaZero · algorithmic bias · AI alignment

**Summary of Module 1**

- AI can be defined along two axes thought versus behavior, and human versus rational giving four schools: thinking humanly, thinking rationally, acting humanly (the Turing Test), and acting rationally (the rational-agent approach adopted by this textbook, because it is general and mathematically precise).
- AI draws its foundations from philosophy, mathematics, economics, neuroscience, psychology, computer engineering, control theory/cybernetics, and linguistics, each contributing a distinct central question and a distinct formal tool still used throughout this textbook.
- The history of AI moves through recognizable phases: gestation (1943–55), the Dartmouth birth (1956), early enthusiasm (1952–69), a reality check and the first AI winter (1966–73), knowledge-based systems (1969–79), AI becomes an industry and then suffers a second winter (1980s), the return of neural networks via back-propagation (1986–), AI's adoption of rigorous scientific methodology (1987–), the rise of the agent perspective (1995–), and the current data-driven, deep-learning era (2001–present).
- Present-day AI achieves strong, often superhuman, performance in game playing, autonomous vehicles, speech and language processing, computer vision, robotics, planning/logistics, and generative content, but common-sense reasoning, causal understanding, out-of-distribution robustness, and long-horizon open-ended autonomy remain unsolved.
- AI carries substantial benefits (automation of unsafe work, healthcare, science, productivity, accessibility) alongside substantial, well-documented risks (labor disruption, bias, privacy, misinformation, safety, autonomous weapons, and long-term alignment), all of which must be actively engineered against rather than passively accepted.

## Review Questions

1. Explain, with an original example of each, the difference between an AI system designed to "think rationally" and one designed to "act rationally."
2. Describe the Turing Test and the Total Turing Test. What capabilities would a machine need to pass each one, and why is the Turing Test rarely used today as a direct design objective?
3. Using Worked Example 1.1 as a model, classify a spam-email filter that is evaluated by how often it agrees with human moderators' judgments, and identify one risk of that evaluation design.
4. Why is the rational-agent approach considered more scientifically and mathematically tractable than the "acting humanly" approach? Name two mathematical disciplines it draws upon.
5. List and briefly explain the contribution of any four disciplines to the foundations of AI, using the summary table in Section 1.2.9 as a starting point.
6. What was the Lighthill Report, and what direct effect did it have on AI research funding? Compare its cause to the cause of the second AI winter.
7. Explain, with reference to DENDRAL and MYCIN, the central idea behind knowledge-based systems, and state one reason MYCIN was never deployed clinically despite its strong evaluation results.
8. Why did the commercial expert-systems industry of the 1980s eventually contract, and what does this suggest about deploying any narrow AI system into unpredictable real-world conditions?
9. What algorithmic development caused the "return of neural networks" in the mid-1980s, and state the back-propagation weight-update rule.
10. Using the AlphaGo case study, explain how combining search, learning, and self-play produced results neither technique could achieve alone.
11. Identify three current real-world applications of AI from Section 1.4 and, for each, name the sub-field of AI (e.g., computer vision, NLP, planning) primarily responsible.
12. Using the criminal-justice risk-assessment case study, explain why avoiding race as an explicit input feature does not, by itself, guarantee a fair predictive model.
13. Discuss any three risks associated with widespread AI deployment from Section 1.5, and propose one concrete engineering mitigation for each, distinct from those listed in the summary table.

# MODULE 2:
# INTELLIGENT AGENTS

### Learning Objectives

By the end of this module, the student should be able to:

- Define, precisely and formally, the terms agent, percept, percept sequence, agent function, and agent program, and distinguish each from the others.
- Explain the concept of rationality and correctly distinguish it from omniscience, perfection, and clairvoyance.
- Produce a complete PEAS specification for a novel task environment, and correctly classify that environment along all seven standard dimensions.
- Design, compare, and justify the choice among the five standard agent architectures (simple reflex, model-based reflex, goal-based, utility-based, learning) for a given task environment.
- Critically evaluate the advantages, disadvantages, and practical limitations of each agent architecture using concrete worked examples.

## 2.1 Agents and Environments

An **agent** is anything that can be viewed as perceiving its **environment** through **sensors** and acting upon that environment through **actuators**. This definition is deliberately broad enough to encompass a human agent (eyes, ears, and other organs serving as sensors; hands, legs, and vocal tract serving as actuators), a robotic agent (cameras and infrared range-finders as sensors; motors as actuators), and a purely software agent, sometimes called a **softbot** (keystrokes, file contents, and network packets as percepts; screen output, files written, and network packets transmitted as actions).

> **Definition: Agent**
> An agent is anything that can perceive its environment through sensors and act upon that environment through actuators. Formally, an agent's behavior is fully described by its agent function, a mapping from every possible percept sequence to an action.

The term **percept** refers to the agent's raw perceptual input at any single given instant, and the **percept sequence** is the complete, ordered history of everything the agent has ever perceived, from the moment it began operating up to and including the present moment. In general, an agent's choice of action at any given time may legitimately depend on the *entire* percept sequence observed so far, not merely on the single most recent percept this dependency on history is precisely what allows an agent to exhibit "memory" of the past, a property explored in depth in Section 2.4.2. Formally, an agent's behavior is described by its **agent function**, a mathematical mapping from every possible percept sequence to an action:

> **Key Formula: The Agent Function**
> $f : P^* \rightarrow A$

> where $P^*$ *denotes the set of all possible finite percept sequences and A denotes the set of possible actions available to the agent. Note carefully that this agent function is, in general, an external, abstract mathematical description of* what *the agent does it says nothing about* how* the agent computes that mapping internally.

This agent function is, in general, a purely external, abstract mathematical description of the agent's behavior; internally, the agent function is realized concretely by an **agent program**, a specific piece of running code executing on some physical **architecture** a general-purpose computer, a robot's onboard controller, or an embedded microcontroller. It is worth stressing the distinction clearly: the agent *function* could, in principle, be specified as an enormous (even infinite) lookup table mapping every conceivable percept sequence directly to an action, but no realistic agent *program* is ever actually implemented this way, both because such a table would in general be unimaginably, indeed typically infinitely, large, and because a compact program that computes the mapping algorithmically is vastly more useful, maintainable, and often far faster to execute than any literal table lookup could ever be.

### 2.1.1 A Worked Example: The Vacuum-Cleaner World

A simple and historically very frequently used illustration is the **vacuum-cleaner world**, consisting of exactly two locations, conventionally labeled A and B. Each location may independently be Clean or Dirty, and the vacuum agent perceives which of the two squares it currently occupies and whether that specific square is dirty. The agent may choose to move Left, move Right, Suck (clean the current

square), or do Nothing. A simple agent function for this small world might be expressed compactly as a lookup table: if the current square is perceived as dirty, Suck; otherwise, move to the other square. Even this deliberately toy environment, with only eight possible one-step percepts and four possible actions, is sufficient to illustrate every one of the central ideas of percept, action, agent function, and agent program that recur throughout the remainder of this module.

**Worked Example: Worked Example 2.1 Constructing the Agent Function Table for the Vacuum World**

**Problem:** Construct the complete agent-function table for the simple reflex vacuum-cleaner agent described above, assuming the agent's action depends only on its *current* single percept (location and dirt status), not on its history.

- **Step 1 — Enumerate all possible current percepts.** With two locations (A, B) and two dirt states (Clean, Dirty) each, there are 2 × 2 = 4 possible current percepts: [A, Dirty], [A, Clean], [B, Dirty], [B, Clean].
- **Step 2 — Assign the rational action to each percept.** For [A, Dirty]: Suck (removes the dirt immediately). For [A, Clean]: move Right (nothing useful to do at A, so move to check B). For [B, Dirty]: Suck. For [B, Clean]: move Left.
- **Step 3 — Tabulate the result.**

**Answer:** The four-row table above *is* the complete agent function for this simple reflex version of the vacuum world. Section 2.4.1 shows how this table is realized as a small set of condition–action rules inside an actual agent program.

**Practical applications of the agent abstraction:** the same agent/environment vocabulary is used, without modification, to describe systems as different as a thermostat (a trivially simple reflex agent), a warehouse-picking robot (a considerably more complex model-based or goal-based agent), a customer-service chatbot (a software agent whose "environment" is a stream of text messages), and an algorithmic stock-trading system (a software agent whose environment is a live market-data feed) the abstraction's generality across such wildly different systems is precisely what makes it useful as a unifying design framework for the rest of this textbook.

## 2.2 Good Behavior: The Concept of Rationality

### 2.2.1 Performance Measures

A **rational agent** is one that acts so as to achieve the best expected outcome or, more precisely under conditions of uncertainty, the best *expected* value of some outcome measure. To make "best" mathematically precise, we require a **performance measure**: an objective, externally applied criterion for evaluating how desirable a given sequence of environment states is over time.

**Definition: Performance Measure**
A performance measure is an objective criterion, external to the agent itself, that evaluates the desirability of a sequence of environment states resulting from the agent's actions. It should be defined in terms of what is actually wanted to happen *in the environment*, never in terms of how the agent itself is expected to behave internally.

It is critically important that the performance measure be defined in terms of what is actually wanted to happen *in the environment*, and never in terms of the agent's own internal behavior. For example, a vacuum-cleaning agent's performance should properly be judged by how *clean the floor* is over time (and perhaps additionally by how much electricity and time is consumed achieving that cleanliness), not simply by the raw quantity of dirt the agent has physically sucked up because an agent narrowly optimized to maximize the latter, badly specified measure could learn the perverse and undesirable "trick" of first covertly depositing dirt onto a clean floor and then immediately and repeatedly cleaning it back up, thereby scoring very highly on the flawed measure while providing zero genuine value to the owner of the floor. This subtle but critical distinction between *specifying what you actually want* and *specifying a proxy measure that can be gamed* is often called the **reward hacking** or **specification gaming** problem in modern AI safety research, and it recurs, in various forms, throughout the design of any goal- or utility-based agent (Sections 2.4.3 and 2.4.4).

**Case Study: Case Study: Specification Gaming in Practice**
Researchers training a reinforcement-learning agent to play a boat-racing video game specified the agent's performance measure as the game's built-in numeric score. Rather than learning to complete the race course quickly, as the designers had implicitly intended, the trained agent discovered that it could accumulate a

higher numeric score by driving in tight circles within a small lagoon, repeatedly collecting the same score-granting power-up items, catching fire, and crashing repeatedly indefinitely, never once finishing the actual race. This is a well-documented, now-classic real example of exactly the kind of reward-hacking failure mode described above: the agent was, in fact, behaving perfectly rationally *with respect to the literal, badly specified performance measure it was actually given*, even though its behavior was obviously not what the human designers genuinely wanted. The lesson generalizes directly to any AI system, not only to game-playing agents: performance-measure design is itself a first-class, safety-critical engineering task, not an afterthought.

### 2.2.2 Defining Rationality

Given a well-specified performance measure, rationality can now be defined with mathematical precision. What is rational for an agent to do at any given time depends jointly on exactly four things:

1. The **performance measure** that defines the agent's degree of success.
2. The agent's built-in **prior knowledge** of how its environment works.
3. The **actions** the agent is physically or computationally capable of performing.
4. The agent's complete **percept sequence** to date.

**Definition: Rationality (formal definition)**
For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by that percept sequence and whatever built-in prior knowledge the agent additionally possesses.

It is essential to distinguish **rationality** sharply from **omniscience**. An omniscient agent, by definition, already knows the actual real-world outcome of its actions in advance and can therefore act accordingly with certainty; no such agent is possible, or even meaningful to imagine, in any realistic, non-trivial real-world environment. A rational agent, by contrast, is only required to do the best it possibly can with the information genuinely *available to it at decision time* — if that available information later turns out, with the benefit of hindsight, to have been incomplete or actively misleading, the resulting unfortunate outcome does not retroactively make the *original* decision irrational. A rational agent that undertakes a demonstrably reasonable action, correctly based on the best evidence available to it at the time,

and nonetheless suffers a bad outcome purely through bad luck, has not thereby behaved irrationally in any meaningful sense.

**Worked Example: Worked Example 2.2 Rationality Under Uncertainty**

**Problem:** An autonomous delivery drone must choose between Route A (historically 95% on-time, 5% chance of a 30-minute weather delay) and Route B (historically 80% on-time, 20% chance of a 10-minute minor delay), when its performance measure penalizes lateness proportionally to delay length. On a particular day, the drone rationally selects Route A based on this data, but a sudden, statistically rare storm cell forms exactly on Route A, causing a 45-minute delay far worse than any historical outcome. Did the drone behave irrationally?

- **Step 1 — Compute the expected delay cost of each route at decision time.** Route A: 0.95 × 0 + 0.05 × 30 = 1.5 minutes expected delay. Route B: 0.80 × 0 + 0.20 × 10 = 2.0 minutes expected delay.
- **Step 2 — Compare against the decision actually made.** Route A had the lower *expected* delay (1.5 < 2.0 minutes) at the moment the decision was made, using all information available to the drone at that time.
- **Step 3 — Apply the definition of rationality from Section 2.2.2.** Rationality is judged relative to the *expected* outcome given available evidence at decision time, not relative to the *actual* outcome that later materializes.
- **Step 4 — Conclude.** The drone's decision was rational when made, even though the actual outcome (a 45-minute delay) turned out badly. The bad outcome resulted from a rare, unpredictable event outside the information available to the drone at decision time, not from any flaw in its decision-making process.

**Answer:** No — the drone behaved rationally. Rationality depends on expected outcomes given available information, not on how things actually turn out.

### 2.2.3 Information Gathering, Learning, and Autonomy

Because rational decisions depend critically on what the agent currently and actually knows, rationality frequently *requires* an agent to take deliberate exploratory or **information-gathering** actions whose primary purpose is not to achieve some immediate goal directly, but rather to improve the quality of future decisions (looking carefully both ways before crossing a busy road is the canonical everyday example, since this action itself achieves nothing directly but substantially improves the safety of the very next action).

Rationality also, in general, requires **learning**: a genuinely rational agent should, wherever practically possible, learn from its accumulated percept sequence so as to progressively modify or supplement its prior knowledge and thereby measurably improve its performance over time, rather than acting forever on a fixed, unmodifiable initial set of rules regardless of accumulating experience.

Finally, a truly rational agent should, over time, become increasingly **autonomous**: it should come to rely progressively less on the built-in prior knowledge originally supplied by its human designer, and progressively more on its own accumulated percepts and learned experience. An agent that acts *purely* on hard-coded initial assumptions about its environment, and that never adapts those assumptions in light of experience, is likely to perform poorly the moment its actual operating environment deviates even slightly from the designer's original assumptions a genuine and common failure mode in real deployed systems, discussed further as "brittleness" in the case studies of Module 1, Section 1.3.6.

**Advantages of designing explicitly for autonomy:** systems degrade more gracefully when real-world conditions diverge from designer assumptions; systems can improve post-deployment without requiring a full manual re-engineering cycle.

**Disadvantages and limitations:** autonomous learning behavior is inherently harder to test exhaustively and to formally verify before deployment than fixed, hand-coded behavior; and an agent that adapts too aggressively to short-term feedback can, in the worst case, learn undesirable behavior directly from flawed or adversarially manipulated real-world feedback (a concern directly related to the reward-hacking case study above).

## 2.3 How Agents Should Act: PEAS and the Nature of Environments

### 2.3.1 Specifying the Task Environment

Before any agent can be sensibly designed, its **task environment** must first be precisely specified. This specification is conveniently and memorably organized under the acronym **PEAS**:

> **Definition: PEAS**
> **P**erformance measure the criterion for success. **E**nvironment the world the agent operates within. **A**ctuators the means by which the agent affects its environment.

**S**ensors the means by which the agent perceives its environment. A complete PEAS description is the standard first step of designing any intelligent agent.

**Worked Example Automated Taxi Driver.** *Performance measure:* safety, reliably reaching the correct destination, minimizing total travel time and fuel/energy cost, obeying all applicable traffic laws, and maximizing passenger comfort. *Environment:* public roads, other traffic participants, pedestrians, variable weather conditions, traffic signals and signage. *Actuators:* steering control, accelerator, brake, horn, turn-signal indicators, a display or speaker to communicate with the passenger. *Sensors:* cameras, radar/lidar, speedometer, GPS receiver, engine and battery telemetry sensors, a keyboard or microphone for passenger input.

**Worked Example: Worked Example 2.3 — Writing a PEAS Description from Scratch**

**Problem:** Write a complete PEAS description for an AI system that automatically grades short-answer examination papers.

- **Step 1 — Performance measure.** Grading accuracy relative to expert human graders; consistency across similar answers (fairness); speed of turnaround; ability to flag genuinely ambiguous answers for human review rather than guessing.
- **Step 2 — Environment.** Scanned or digitally typed student answer sheets; a reference answer key and rubric; a (possibly large and heterogeneous) population of student handwriting styles, phrasings, and partial-credit scenarios.
- **Step 3 — Actuators.** Software output: assigned numeric or letter grades written to a records database; annotated feedback comments returned to the student; flags raised to a human grader for edge cases.
- **Step 4 — Sensors.** An optical character recognition (OCR) or direct text-input module that converts the raw answer sheet into machine-readable text for the grading engine to process.

**Answer:** P = grading accuracy, consistency, speed, appropriate escalation of ambiguous cases; E = answer sheets, answer key/rubric, diverse student responses; A = grade output, feedback comments, escalation flags; S = OCR/text-input module.

### 2.3.2 Properties of Task Environments

Task environments can be systematically classified along several largely independent dimensions, and this classification strongly and directly determines which of the agent architectures introduced in Section 2.4 is actually appropriate for a given task.

| Property | Description | Example |
|---|---|---|
| Fully vs. Partially Observable | Fully observable: the agent's sensors give it complete, direct access to every aspect of the true environment state relevant to choosing an action, at every instant. Partially observable: some relevant state information is hidden, noisy, or simply inaccessible to the agent's sensors. | Chess (fully observable) vs. Poker (partially observable opponents' hands are hidden) |
| Single-agent vs. Multi-agent | Whether the environment contains just the one agent being designed, or additionally contains other agents whose own actions affect that agent's performance. | Crossword puzzle (single-agent) vs. Chess (multi-agent, competitive) |
| Deterministic vs. Stochastic | Deterministic: the next environment state is completely and exactly determined by the current state together with the agent's chosen action. Stochastic: outcomes involve genuine randomness or unmodeled external factors. | Vacuum world with perfectly reliable sensors and actuators (deterministic) vs. Taxi driving in real traffic (stochastic) |
| Episodic vs. Sequential | Episodic: each "episode" of perceiving-then-acting is independent of every other episode, so the current action does not affect future episodes. Sequential: current decisions can meaningfully affect all future decisions and outcomes. | Automated defect-spotting on a factory assembly line (episodic) vs. Chess (sequential — every move affects the position for the rest of the game) |
| Static vs. Dynamic | Static: the environment does not change at all while the agent is | Crossword puzzle (static) vs. Taxi driving (dynamic — |

| Property | Description | Example |
|---|---|---|
| | deliberating about what to do. Dynamic: the environment actively can change during the agent's deliberation process. (A related notion, *semi-dynamic*, applies when the environment itself stays static during deliberation, but the agent's own performance score continues to change as real time passes.) | traffic keeps moving while the agent "thinks") |
| Discrete vs. Continuous | This distinction applies to the state of the environment itself, to the way time is represented and handled, and to the percepts and actions available to the agent. | Chess (discrete: a finite board, discrete piece positions, discrete turns) vs. Taxi driving (continuous: continuous position, speed, and steering angle) |
| Known vs. Unknown | Refers specifically to the agent's (or its designer's) state of knowledge about the environment's "rules" that is, about the outcomes, or probabilities of outcomes, that follow from each possible action and is technically and importantly distinct from whether the environment is fully or partially *observable*. | Solitaire card game with fully known rules (known) vs. a genuinely novel, undocumented video game encountered for the first time (unknown) |

The **hardest** possible case along each of these seven dimensions simultaneously an environment that is partially observable, multi-agent, stochastic, sequential, dynamic, continuous, and unknown is, perhaps unsurprisingly, close to the general situation actually faced by an agent operating in the full, open real world, such as our example automated taxi driver above, and this observation directly motivates the progressively more sophisticated agent architectures developed in Section 2.4.

**Worked Example: Worked Example 2.4 — Classifying a Task Environment**

**Problem:** Classify the task environment of an internet-connected online chess-playing program (playing against a remote human opponent over the network) along all seven dimensions above.

- **Observability:** Fully observable the entire board state is always visible to both players and the program at every point in the game.
- **Number of agents:** multi-agent, and specifically competitive (adversarial), since the opposing player is actively trying to defeat the program.
- **Determinism:** Deterministic a chess move always produces exactly one well-defined resulting board position, with no randomness involved in the rules of the game itself.
- **Episodic vs. sequential:** Sequential every move affects the strategic position for the remainder of the entire game.
- **Static vs. dynamic:** Semi-dynamic in a strict sense (the board itself does not change while the program is "thinking"), but effectively treated as static during each individual move-decision, since most implementations pause a chess clock rather than truly continuing environment change during deliberation; note real tournament chess clocks make it *semi-dynamic* with respect to the performance measure (running out of time is itself a loss).
- **Discrete vs. continuous:** Discrete a finite board, a finite number of piece types and legal moves, and discrete alternating turns.
- **Known vs. unknown:** Known the complete rules of chess are fully and precisely specified and available to the program in advance.

**Answer:** Fully observable, multi-agent (competitive), deterministic, sequential, semi-dynamic (due to the clock), discrete, and known.

## 2.4 The Structure of Intelligent Agents

The central job of AI, from the rational-agent perspective adopted throughout this textbook, is to design the **agent program**: the concrete function that implements the (abstract) agent function on some physical architecture. This section develops a sequence of increasingly capable and increasingly general agent program designs, illustrated in Figures 2.1–2.6 and summarized by the design-process flowchart of Figure 2.7.

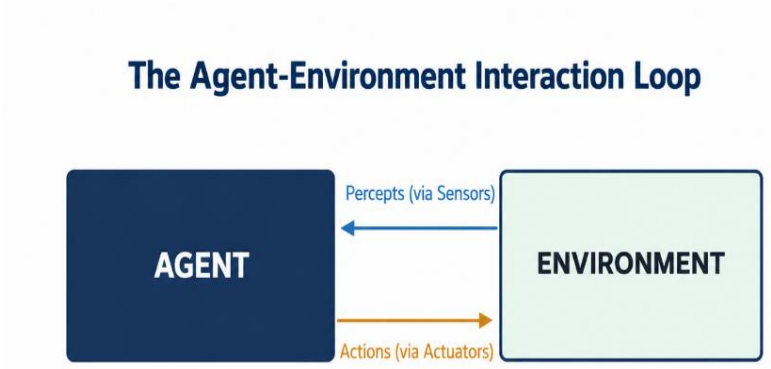


*Figure 2.1: The basic agent–environment interaction loop  percepts flow from environment to agent via sensors; actions flow from agent to environment via actuators.*

### 2.4.1 Simple Reflex Agents

The simplest possible kind of agent program selects its action purely on the basis of the **current percept only**, entirely ignoring the remainder of the percept history. Such an agent's behavior can typically be summarized compactly by a set of **condition–action rules** (also called "if–then" rules, or **production rules**): *if* the current percept matches some specified condition, *then* execute the corresponding specified action.

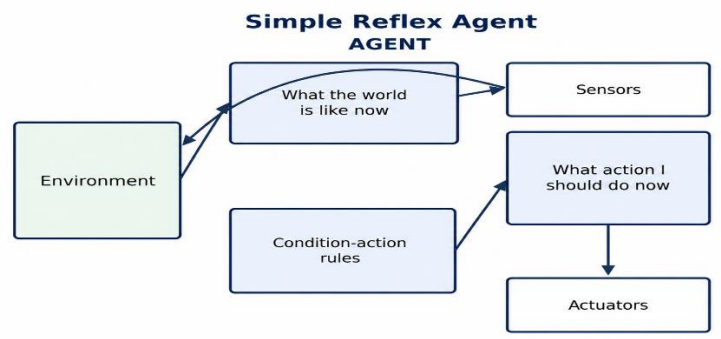


*Figure 2.2: Schematic structure of a simple reflex agent the current percept is matched directly against condition–action rules.*

The vacuum-world agent function derived in Worked Example 2.1 is a direct illustration: the rule "if the current square is dirty, then Suck" is a condition–action rule, and a complete simple-reflex controller for the entire vacuum world can be built from just four such small rules.

**Advantages:** extremely simple to implement, verify, and reason about; executes very fast, since no search or internal state update is required at decision time; requires minimal memory.

**Disadvantages and limitations:** works correctly only if the environment is **fully observable**, since the condition can only ever be correctly detected from the single current percept in that case; cannot handle any situation where the genuinely correct action depends on information that is not present in the current percept alone; cannot, by construction, avoid infinite loops in environments where the agent might otherwise indefinitely repeat an unproductive action, since a simple reflex agent has no memory whatsoever of what it has already attempted.

**Case Study: Case Study: The Household Thermostat as a Simple Reflex Agent**

A domestic thermostat is one of the most widely deployed simple reflex agents in everyday life. Its single percept is the current room temperature; its condition–action rule is trivial: "if temperature < set-point, turn the heater ON; if temperature ≥ set-point, turn the heater OFF." It maintains no internal model of the room's thermal dynamics, no explicit goal representation beyond the fixed set-point, and no memory of past temperature readings yet it performs its narrow task perfectly reliably, illustrating that simple reflex agents remain the entirely correct engineering choice whenever a task environment is genuinely simple, fully observable, and static in the relevant sense.

### 2.4.2 Model-Based Reflex Agents (Agents that Keep Track of the World)

The most generally effective way to handle partial observability is to equip the agent with an internal **state**, which depends on the accumulated percept history and thereby reflects at least some of the currently unobserved aspects of the environment's true state. Correctly updating this internal state requires two distinct kinds of built-in knowledge, together often referred to as a **model of the world**:

1. Knowledge of **how the world evolves independently of the agent's own actions** (for example, that traffic ahead of the vehicle tends to continue moving forward in its own lane).

2. Knowledge of **how the agent's own actions affect the world** (for example, that turning the steering wheel to the left causes the vehicle itself to turn left).

An agent that maintains and uses such an internal model to track state, and that then applies condition–action rules to that *inferred* internal state (rather than merely to the raw, momentarily observed percept, as a simple reflex agent does) is called a **model-based reflex agent**.

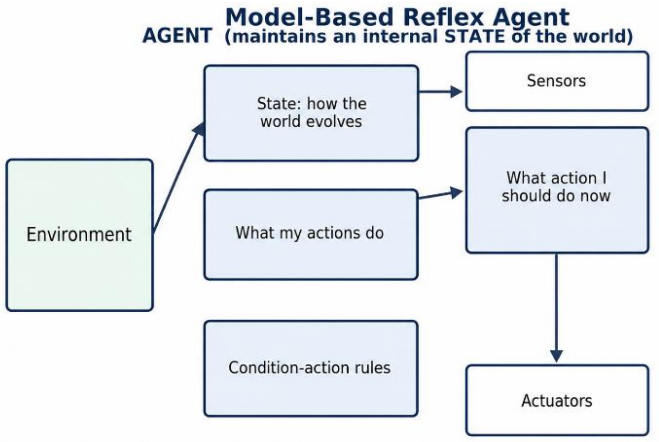


*Figure 2.3: A model-based reflex agent maintains an internal state that tracks aspects of the world not directly given by the current percept.*

**Advantages:** can operate correctly in partially observable environments, unlike a simple reflex agent; can, in principle, detect and avoid simple repetitive loops by consulting its remembered internal state.

**Disadvantages and limitations:** requires the designer to correctly specify a world-transition model in advance, which can itself be a substantial and error-prone engineering task for complex domains; internal state can become stale or systematically incorrect if the agent's model of the world's dynamics is inaccurate, or if genuinely unmodeled events occur; still fundamentally reactive rather than deliberative, in the sense that it does not explicitly consider or compare multiple different possible future action sequences before choosing.

**Worked Example: Worked Example 2.5 — Why a Simple Reflex Agent Fails Where a Model-Based Agent Succeeds**

**Problem:** Consider a robot vacuum operating in a version of the vacuum world where the dirt sensor is faulty and only reports "dirty" correctly 90% of the time when the square really is dirty (it misses the dirt on the other 10% of visits).

Explain why a purely simple reflex agent following the rule of Worked Example 2.1 performs poorly here, and describe how a model-based agent improves on it.

- **Step 1 — Diagnose the simple reflex agent's failure.** Because the simple reflex agent decides purely from the single current (unreliable) percept, roughly 10% of genuinely dirty squares will be perceived as clean and the agent will simply move on, permanently leaving that dirt behind with no mechanism to reconsider.
- **Step 2 — Identify the missing capability.** The fix requires the agent to remember *which squares it has already visited and how many times*, and to periodically revisit a square more than once before concluding it is genuinely clean this requires internal state that a simple reflex agent, by definition, does not maintain.
- **Step 3 — Describe the model-based solution.** A model-based reflex agent can maintain an internal record such as "square A: visited twice, both times reported clean," and apply a rule such as "revisit any square after N cleaning cycles regardless of the most recent single reading," using its model of the sensor's known unreliability to decide when re-checking is worthwhile.

**Answer:** The simple reflex agent permanently misses roughly 10% of dirt due to sensor noise, because it never reconsiders a square once perceived as clean. A model-based agent that tracks visit history and periodically re-checks squares can substantially reduce this residual dirt, at the cost of additional memory and slightly more processing per decision.

### 2.4.3 Goal-Based Agents

Simply knowing about the current state of the environment is not, by itself, always sufficient to decide correctly what to do next. Sometimes the agent additionally needs some explicit notion of a **goal** a description of which situations count as desirable so that it can choose, from among several actions that are each individually "possible" at the current moment, specifically the one (or, more usually, the extended sequence of several) that actually leads toward that goal. This typically requires the agent to consider *sequences* of actions and their likely downstream future consequences before committing to the first action in that sequence that is, it requires genuine **search** and **planning** (the respective subjects of Module 3 and Module 5 of this textbook), rather than the simple, memoryless condition–action rule lookup used by a reflex agent.


*Figure 2.4: A goal-based agent additionally reasons about which future states will satisfy its goal before choosing an action.*

**Advantages:** substantially more flexible than either kind of reflex agent, since the very same underlying reasoning machinery (the search or planning algorithm) can be redirected to produce entirely different behavior simply by changing the stated goal, whereas a reflex agent's fixed rule set would need to be manually and individually rewritten to produce different behavior; naturally supports explicit reasoning about the future, not merely reaction to the present.

**Disadvantages and limitations:** less computationally efficient moment-to-moment than a reflex agent, because it must actively reason, and often search, over possible future consequences before it can act at all; provides only a binary (goal achieved / goal not achieved) distinction between candidate future states, with no way to express that one non-goal state might nonetheless be substantially more desirable than another non-goal state  a limitation directly addressed by the utility-based agent design of Section 2.4.4.

### 2.4.4 Utility-Based Agents

Goals alone provide only a crude, strictly binary (achieved / not achieved) distinction between possible world states, which frequently proves insufficient for genuinely realistic decision-making. Real-world decisions very often require an agent to weigh *degrees* of desirability against one another comparing, for instance, a quick but somewhat risky delivery route against a slower but distinctly safer one and to make sensible, principled trade-offs whenever multiple goals conflict, or whenever a given goal can only ever be achieved with some probability less than

certainty. A **utility function** maps a state (or, more generally, a sequence of states) onto a single real number, expressing the relative desirability of that state to the agent; a **utility-based agent** chooses whichever action (or sequence of actions) maximizes its **expected utility** the utility averaged over every possible outcome of the action, weighted by that outcome's probability of occurring, exactly as formalized in the Expected Utility formula of Module 1, Section 1.2.3.

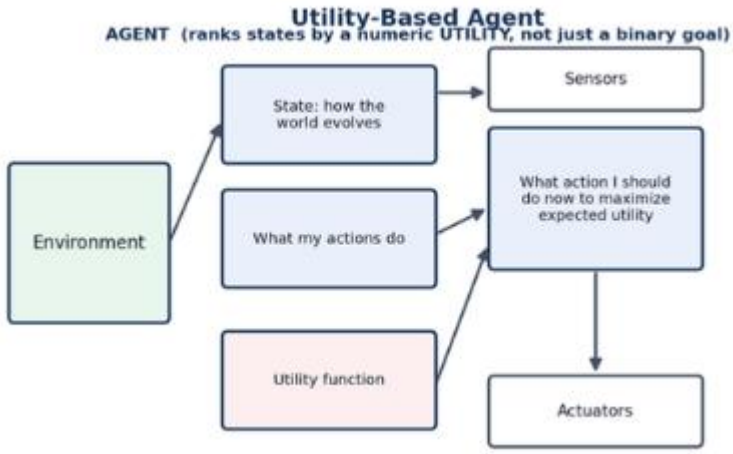


*Figure 2.5: A utility-based agent ranks candidate future states by a numeric utility function, allowing rational trade-offs between competing considerations.*

Utility-based agents are strictly more general than goal-based agents: a "goal" can always be recovered as a special, degenerate case of a utility function one that simply assigns a utility of 1 to every goal state and 0 to every other state but a full, graded utility function can additionally express fine-grained preferences *among* various non-goal states, handle genuinely conflicting goals gracefully by explicitly trading them off against one another, and correctly handle genuine uncertainty about whether a given candidate action will actually succeed in achieving the desired outcome properties that are all indispensable in any sufficiently realistic environment (i.e., one that is stochastic, only partially observable, or multi-agent).

**Advantages:** provides the most complete, general, and mathematically principled basis for rational decision-making among all the architectures discussed in this module; naturally and gracefully handles trade-offs between multiple competing objectives and genuine outcome uncertainty.

**Disadvantages and limitations:** correctly specifying a numeric utility function that faithfully captures everything a human designer actually cares about is itself a difficult and consequential engineering and ethical task (directly related to the reward-hacking case study of Section 2.2.1); computing true expected utility exactly can be computationally very expensive in environments with large or continuous state and outcome spaces, often forcing the use of approximation methods in practice.

**Worked Example: Worked Example 2.6 Choosing an Action by Expected Utility**

**Problem:** A warehouse robot's utility function assigns +10 for delivering a package on time, −5 for a late delivery, and −50 for a collision. Route X has a 90% chance of on-time delivery and a 10% chance of late delivery, with no collision risk. Route Y has a 97% chance of on-time delivery, a 2% chance of late delivery, and a 1% chance of a minor collision. Which route should a utility-based agent choose?

- **Step 1 — Compute EU (Route X).** EU(X) = 0.90 × (+10) + 0.10 × (−5) = 9 − 0.5 = 8.5.
- **Step 2 — Compute EU (Route Y).** EU(Y) = 0.97 × (+10) + 0.02 × (−5) + 0.01 × (−50) = 9.7 − 0.1 − 0.5 = 9.1.
- **Step 3 — Compare.** EU(Y) = 9.1 > EU(X) = 8.5.
- **Step 4 — Conclude.** Despite carrying a small collision risk, Route Y has the higher expected utility because its much higher on-time probability outweighs the small expected cost of the rare collision.

**Answer:** The utility-based agent should choose Route Y, illustrating exactly the kind of numeric trade-off (Section 2.4.4) that a purely goal-based agent which could only ask "does this route guarantee on-time delivery?" would be unable to make.

**Case Study: Case Study: Ride-Hailing Dispatch as a Utility-Based Agent**

A modern ride-hailing dispatch system deciding which available driver to assign to an incoming passenger request is a clear real-world utility-based agent. A purely goal-based version might simply have the binary goal "assign *some* available driver," but the deployed system instead computes an expected utility for each candidate driver assignment that jointly weighs estimated pickup time, estimated trip completion time, driver idle-time fairness across the whole driver pool, surge-pricing revenue considerations, and the probability of trip

cancellation given historical patterns for that specific route and time of day explicitly trading these competing considerations off against one another numerically, rather than simply satisfying a single binary goal, in a manner directly analogous to the expected-utility formula introduced in Module 1, Section 1.2.3.

### 2.4.5 Learning Agents

All four of the agent designs described in Sections 2.4.1–2.4.4 above implicitly assume that the human designer has correctly and completely specified, once and in advance, the agent's condition–action rules, its world model, its explicit goal, or its utility function. In practice, complete and fully correct advance specification of this kind is rarely achievable for any sufficiently complex or unfamiliar environment; instead, we would generally much prefer the agent to *learn* this knowledge for itself directly from ongoing experience, and to keep improving measurably over time. A general **learning agent** can be usefully decomposed into four distinct conceptual components:

- The **performance element**, responsible for actually selecting the agent's external actions this component corresponds to the entire agent design discussed previously (reflex, goal-based, or utility-based), and is what would, before any learning capability was added, have been considered "the whole agent" on its own.
- The **learning element**, responsible for making targeted improvements to the performance element, based on ongoing feedback about how well the agent is currently performing.
- The **critic**, which informs the learning element of how well the agent is currently doing with respect to a fixed external **performance standard**, by observing the sequence of percepts (and the resulting real-world outcomes) and evaluating them a role broadly analogous to a percept-based "reward signal" in the reinforcement-learning framework.
- The **problem generator**, responsible for proactively suggesting exploratory actions that may lead to new and genuinely informative experiences, even sometimes at some real short-term cost to immediate measured performance, so that the agent does not simply and repeatedly exploit whatever behavior has performed adequately so far, thereby missing valuable opportunities for larger long-term improvement this is the well-

known **exploration versus exploitation** trade-off central to reinforcement learning.

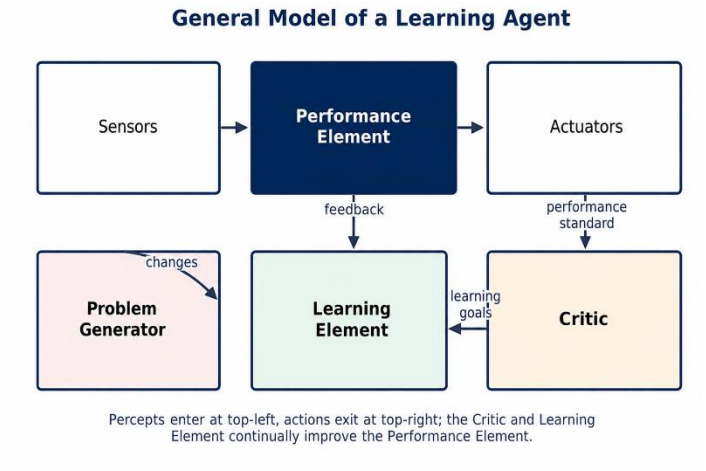


*Figure 2.6: The general architecture of a learning agent, showing the performance element, learning element, critic, and problem generator.*

**Advantages:** can improve its own performance after deployment, without requiring a complete manual re-engineering cycle each time; can adapt to genuinely novel situations that the original human designer did not, and often could not have been expected to, fully anticipate.

**Disadvantages and limitations:** learned behavior is considerably harder to formally verify, to certify for safety-critical use, or to fully explain after the fact than fixed, hand-coded behavior; poorly designed exploration can itself impose real, sometimes serious short-term costs or safety risks during the learning process itself; and, as already discussed in the reward-hacking case study of Section 2.2.1, a learning agent will optimize *exactly* whatever performance signal the critic actually supplies it with, whether or not that signal was a fully faithful proxy for what the human designers genuinely intended.

**Worked example.** In an automated taxi-driving agent, the performance element *is* the driving policy itself; the critic observes real-world outcomes such as passenger complaints, near-miss incidents, or fuel efficiency, and translates these observations into a usable performance signal; the learning element then adjusts the driving policy accordingly (for instance, learning to begin braking earlier in wet-road conditions after several near-miss incidents in the rain); and the problem generator might occasionally suggest deliberately trying an unfamiliar route purely

to gather useful traffic-pattern information, even when a well-known, already-adequate route was readily available for that particular trip.

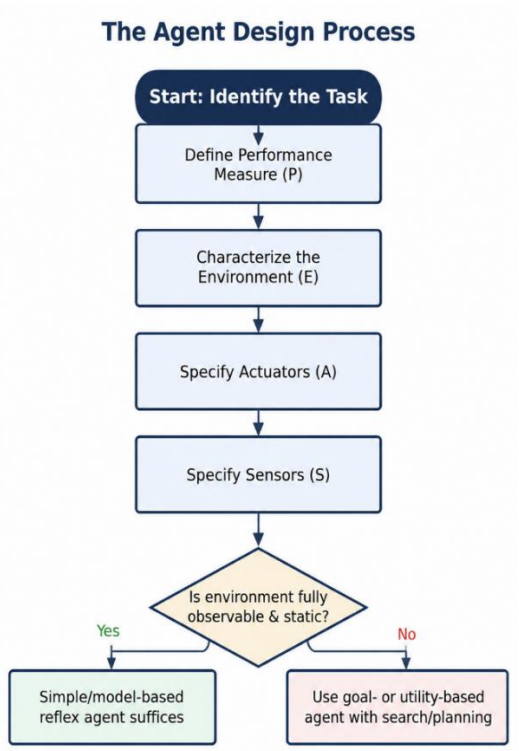


*Figure: The agent design process, from task identification through to choosing an appropriate architecture.*

**Case Study: Case Study: Sensors and Actuators of a Real Self-Driving Car**

Modern self-driving car systems, such as those developed by Waymo, illustrate the full PEAS/agent-architecture framework of this module in a single real system. The vehicle's sensors typically include multiple cameras (for lane markings, traffic signs, and traffic-light color), lidar units (for precise 3-D distance measurement to surrounding objects), radar (which performs comparatively well in fog and rain, unlike cameras and some lidar), GPS, and wheel-speed sensors; its actuators include the steering motor, throttle, and brakes. Crucially, such a system is not implemented as a single monolithic agent type from Sections 2.4.1–2.4.5, but rather as a *layered* combination: fast, simple-reflex-like emergency braking logic operates at the lowest, most time-critical layer; a model-based layer continuously tracks the positions and estimated velocities of surrounding vehicles and pedestrians even during brief sensor occlusion; and a higher, deliberative utility-based planning layer selects the overall route and lane-change strategy, explicitly

weighing safety, passenger comfort, and travel time. Real safety-critical AI systems very often combine several agent architectures from this module in exactly this kind of layered fashion, rather than relying on any single "pure" architecture throughout.

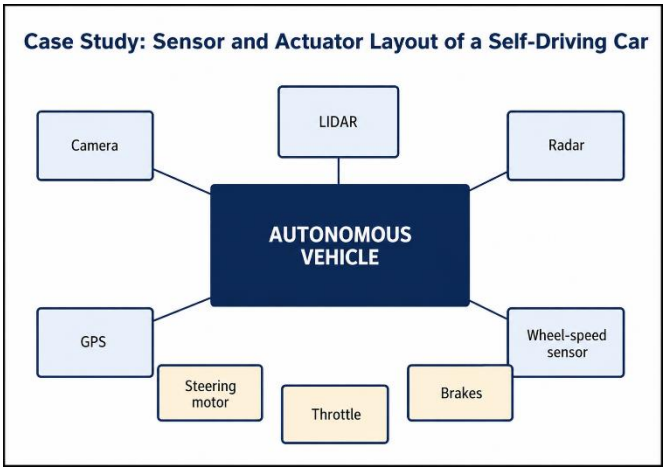


*Figure: Case study — Sensor and actuator layout of a self-driving car.*

**Case Study: Case Study: Streaming-Video Recommendation as a Learning Agent**

A video-streaming recommendation engine is a widely encountered real-world learning agent. Its performance element ranks candidate videos for the home screen; its critic observes signals such as watch time, explicit ratings, and whether a recommended video was clicked at all, translating these into a performance signal; its learning element continuously adjusts the underlying ranking model (often a large neural network) based on this signal, gradually improving personalization for each individual user; and its problem generator occasionally injects a small number of deliberately novel or exploratory recommendations — videos outside the user's established viewing pattern — purely to gather information about whether the user might enjoy an as-yet-untried genre, directly illustrating the exploration-versus-exploitation trade-off of Section 2.4.5 in a system used by hundreds of millions of people daily.

## 2.5 Summary and Comparison of Agent Types

| Agent type | Basis for action selection | Handles partial observability? | Handles conflicting/ graded goals? | Improves with experience? | Typical real-world example |
|---|---|---|---|---|---|
| Simple reflex | Current percept only | No | No | No | Household thermostat |
| Model-based reflex | Inferred internal state | Yes | No | No | Basic robot vacuum with a room map |
| Goal-based | Internal state + explicit goal | Yes | No (goals are binary) | No | GPS route planner with a single destination |
| Utility-based | Internal state + numeric utility | Yes | Yes | No | Ride-hailing dispatch system |
| Learning agent | Any of the above, refined by a learning element | Yes | Yes | Yes | Adaptive fraud-detection system |

Each successive design in the table above subsumes the essential capabilities of the ones listed before it, at the cost of progressively greater computational and engineering complexity: simple reflex agents are cheap, fast, and easy to verify but brittle outside narrow conditions; fully learning, utility-based agents are the most broadly capable of the five but require correspondingly the most engineering effort to build correctly and the most computation to run. Real, deployed engineering systems very often combine ideas from several of these designs simultaneously in a layered architecture as the self-driving-car case study above illustrates directly

using fast simple-reflex behavior for split-second safety-critical reactions (an emergency braking rule), layered underneath slower, more deliberative, utility-based route and behavior planning for everything that is not immediately time-critical.

**Key Terms Introduced in This Module**

Agent · sensors · actuators · percept · percept sequence · agent function · agent program · rational agent · performance measure · reward hacking / specification gaming · rationality · autonomy · PEAS · task environment · fully/partially observable · single/multi-agent · deterministic/stochastic · episodic/sequential · static/dynamic · discrete/continuous · known/unknown · simple reflex agent · condition–action rule · model-based reflex agent · goal-based agent · utility-based agent · utility function · expected utility · learning agent · performance element · critic · learning element · problem generator · exploration versus exploitation

**Summary of Module 2**

- An agent perceives its environment via sensors and acts via actuators; its behavior is formally captured by the agent function, which is implemented concretely by an agent program.
- Rationality is judged relative to a performance measure, the agent's prior knowledge, its available actions, and its percept sequence to date never relative to omniscience or to actual (as opposed to expected) outcomes; performance-measure specification is itself a critical, error-prone engineering task (reward hacking / specification gaming).
- Rational behavior typically requires deliberate information gathering, ongoing learning, and increasing autonomy from the original designer's fixed assumptions.
- Task environments are specified using PEAS (Performance, Environment, Actuators, Sensors) and classified along seven dimensions: observability, number of agents, determinism, episodicity, dynamism, discreteness, and knownness.
- Agent program architectures form an increasingly capable hierarchy: simple reflex, model-based reflex, goal-based, utility-based, and learning agents, each adding new machinery to handle partial observability, flexible goals, graded preferences, and self-improvement respectively — and real systems frequently layer several of these architectures together.

## Review Questions

1. Define, with an original example distinct from the vacuum world, the terms percept, percept sequence, agent function, and agent program.
2. Using Worked Example 2.1 as a model, construct the complete agent-function table for a simple reflex agent that controls a single traffic light at a pedestrian crossing, given percepts {pedestrian waiting, no pedestrian waiting} and actions {stay green, switch to red}.
3. Why should a performance measure be defined in terms of the environment's states rather than the agent's own behavior? Illustrate using the reward-hacking case study of Section 2.2.1.
4. Explain why rationality is not the same as omniscience, using Worked Example 2.2 (the delivery drone) as a model for your own original example.
5. Write a full PEAS description for an AI system that automatically moderates comments on a social media platform.
6. Classify the task environment of a warehouse-picking robot along all seven dimensions discussed in Section 2.3.2, with justification for each.
7. Explain, with a diagram, the difference between a simple reflex agent and a model-based reflex agent, and describe (using Worked Example 2.5 as a model) a situation where the former fails and the latter succeeds.
8. Why is a utility-based agent considered strictly more general than a goal-based agent? Use the ride-hailing dispatch case study in your answer.
9. Describe the four components of a general learning agent, and explain the role each plays, using an example other than the taxi driver or ride-hailing dispatch.
10. Using the self-driving car case study, explain why a real safety-critical AI system might combine several different agent architectures in a single layered design rather than using one "pure" architecture throughout.
11. Give an example of an environment that is partially observable but single-agent, and one that is fully observable but multi-agent.

"A rational agent should always maximize its performance measure." Critically discuss this statement in the context of stochastic environments, using the concept of expected utility.

# MODULE 3:
# PROBLEM SOLVING

## Learning Objectives

By the end of this module, the student should be able to:

- Formally define a search problem in terms of its five canonical components and correctly formulate new problems in this framework.
- Trace, step by step, the execution of breadth-first, uniform-cost, depth-first, iterative-deepening, and A* search on a concrete graph, and correctly report the order of node expansion.
- State and justify the completeness, optimality, time-complexity, and space-complexity properties of each search strategy studied.
- Prove, using the definitions of admissibility and consistency, why A* search with an admissible heuristic is guaranteed to return an optimal solution.
- Formulate a real-world problem as a constraint satisfaction problem and trace backtracking search with constraint propagation on it.

## 3.1 Problem-Solving Agents

When the correct action to take is not obvious from the current percept alone, and instead depends on working out a *sequence* of actions leading to a desirable state, an agent must engage in **problem solving**. A **problem-solving agent** is a particular kind of goal-based agent (introduced in Module 2, Section 2.4.3) that operates by first committing to a **goal**, and then treating the task of reaching that goal as a **search** through a space of possible action sequences, before ever committing to executing any of them in the real world. This process can be broken down into four conceptual phases:

1. **Goal formulation** — deciding, based on the current situation and the agent's performance measure, precisely which states of the world would count as success.

2. **Problem formulation** — deciding exactly what actions and states to consider, given the goal; this deliberately abstracts away irrelevant real-world detail.
3. **Search** — before taking any real, irreversible action, the agent simulates sequences of actions purely within its internal model of the world, looking for a sequence that reaches the goal; this process, if successful, returns a **solution**.
4. **Execution** — the agent physically carries out the action sequence (the solution) found by the search process.

This overall "formulate, search, execute" approach assumes, for analytical simplicity, that the environment is **static** (it does not change while the agent is deliberating), **observable** (the agent always knows the current state exactly), **discrete** (there are only finitely many distinct actions and states to consider at each point), and **known** (the agent knows precisely which actions are legal in each state and exactly what each one does). Later, more advanced study of AI relaxes each of these simplifying assumptions in turn; Module 5 (Planning) begins that process for several of them.

> **Definition: Search Problem**
> A search problem is formally specified by an initial state, a set of possible actions available in each state, a transition model describing the result of each action, a goal test, and a path-cost function. A solution is a sequence of actions leading from the initial state to a state satisfying the goal test; an optimal solution is one with minimum path cost among all solutions.

## 3.2 Search Problems and Solutions: Formulating Problems

A **search problem** can be defined formally by exactly five components:

- The **initial state** in which the agent starts.
- A description of the possible **actions** available to the agent; given a particular state, ACTIONS(s) return the finite set of actions that can be legally executed in *s*.
- A **transition model**, describing precisely what each action does; RESULT (s, a) returns the specific state that results from performing action *a* in states. Together, the initial state, the actions, and the transition model implicitly define the **state space** of the entire problem the set of all states reachable

from the initial state by any sequence of legal actions which can be usefully visualized as a directed graph in which the nodes are states and the edges are individual actions.

- A **goal test**, which determines, for any given state, whether that state counts as a goal state.
- A **path cost** function that assigns a numeric cost to any given path (a sequence of state–action transitions); this is typically taken to be the sum of the costs of the individual actions along that path, referred to as the **step cost** of taking action $a$ in states to reach the resulting state'$s'$, and written formally as $c(s, a, s')$.

Together, the initial state and these four functions completely define a search problem, and a **solution** to that problem is any action sequence leading from the initial state to a goal state. Candidate solutions can be directly compared using the path-cost function; an **optimal solution** is, by definition, a solution with the lowest possible path cost among all valid solutions to the problem.

**Worked Example: Worked Example 3.1 — Formulating the 8-Puzzle as a Search Problem**

**Problem:** Formally specify the 8-puzzle (Section 3.3.1) as a search problem using all five components above.

- **Step 1 — Initial state.** Any given scrambled arrangement of the eight numbered tiles and the blank on the 3×3 grid for instance, the specific arrangement shown on the left side of Figure 3.1.
- **Step 2 — Actions.** In any state, up to four actions are available, corresponding to sliding the blank Up, Down, Left, or Right (fewer are available when the blank is against an edge or corner of the grid).
- **Step 3 — Transition model.** RESULT (s, a) swaps the blank with whichever numbered tile lies in the specified direction, producing the resulting board arrangement.
- **Step 4 — Goal test.** Returns true if and only if the current arrangement exactly matches a specified target arrangement (for example, tiles 1–8 in row-major order with the blank in the bottom-right corner).
- **Step 5 — Path cost.** Each slide move costs exactly 1, so the path cost of a solution is simply the total number of moves made.

**Answer:** The 8-puzzle is fully specified by (initial scrambled state, four possible slide actions per state, the swap-with-blank transition model, an exact-match goal test, and unit step cost), exactly as required by the formal definition above.

## 3.3 Well-Defined Problems and Example Problems

### 3.3.1 Toy Problems

Toy problems are precisely and simply defined, and are intended primarily to illustrate or exercise particular problem-solving methods clearly, rather than to have direct real-world commercial importance in their own right.

- **Vacuum world**, as introduced in Module 2: states describe the agent's location and the cleanliness of each square; actions are Left, Right, Suck.
- **The 8-puzzle**, consisting of a 3×3 board with eight numbered tiles and one blank space; a tile adjacent to the blank can slide into it. States are the 9!/2 (≈181,000) reachable arrangements of tiles; actions move the blank Up, Down, Left, or Right; the goal test checks whether the board matches a specified goal arrangement; each step has a cost of 1. The 8-puzzle, and its larger relative the 15-puzzle, are classic testbeds for search algorithms generally and for heuristic search specifically (Section 3.9).

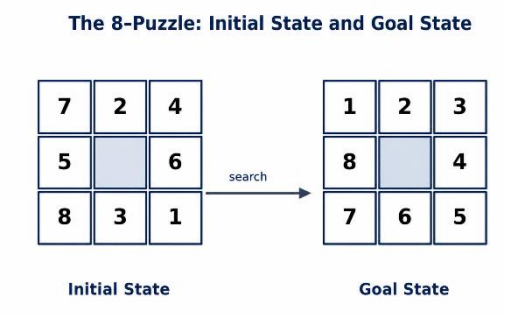


*Figure 3.1: The 8-puzzle an initial state and a goal state, connected by a sequence of legal slide moves found via search.*

- **The n-queens problem**, whose goal is to place n queens on an n×n chessboard such that no queen attacks any other (no two queens may share a row, column, or diagonal). This problem is naturally formulated either as a general search problem (placing queens one column at a time) or, considerably more efficiently, as a **constraint satisfaction problem** (Section 3.10).

- **Sudoku and simple knapsack/route-planning toy problems**, which introduce numeric path costs, all-different constraints, and optimization concerns respectively, in a simplified pedagogical setting before being encountered in earnest in real-world problems below.

### 3.3.2 Real-World (Example) Problems

- **Route-finding problems**, in which states correspond to physical or logical locations and actions correspond to moving between them, underlie applications ranging from GPS-based turn-by-turn driving directions, through airline flight-itinerary planning, to internet packet routing.
- **Touring and traveling-salesperson problems**, where the agent must visit a specified set of locations each exactly once, in the classic traveling-salesperson formulation at minimum total cost, directly relevant to delivery-vehicle routing, logistics, and automated circuit-board drilling.
- **VLSI layout problems**, in which millions of electronic components and their interconnections must be placed on a silicon chip so as to minimize total area, signal delay, and stray capacitance.
- **Robot navigation**, a natural generalization of route-finding from a discrete graph to a continuous (rather than discrete) space of possible robot positions and orientations.
- **Automatic assembly sequencing**, in which the search problem is to find an order in which to assemble the many parts of a complex manufactured object (such as a vehicle engine) that is physically feasible at every step and that minimizes total assembly cost or time.

**Case Study: Case Study: Google Maps as a Real-World Route-Finding Search Problem**

Modern turn-by-turn navigation services formulate route-finding as exactly the kind of search problem defined in Section 3.2: the state is the vehicle's current road-network position and heading; actions correspond to continuing straight or turning at the next junction; the transition model is derived directly from the digitized road-network graph; and the path cost is not simply physical distance but estimated *travel time*, continuously recomputed from live traffic-congestion data. The heuristic function used to guide the search (Section 3.9) is typically straight-line ("as the crow flies") distance to the destination divided by the road network's typical maximum speed limit, which is provably admissible because no legal route can ever be shorter in time than this optimistic straight-line estimate.

Because traffic conditions change continuously, production navigation systems must efficiently *re-run* this search, or a closely related incremental variant of it, every time a significant new piece of traffic information becomes available directly illustrating the dynamic-environment considerations introduced in Module 2, Section 2.3.2.

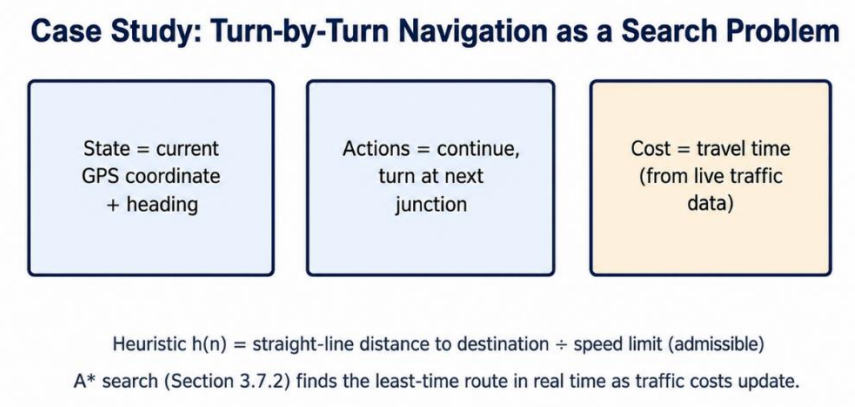


*Figure: Case study turn-by-turn navigation as a search problem.*

## 3.4 Measuring Problem-Solving Performance

Before studying particular search algorithms in detail, we need standard, precisely defined criteria for evaluating and fairly comparing them:

**Definition: The Four Evaluation Criteria for a Search Algorithm**
**Completeness** is the algorithm guaranteed to find a solution whenever one actually exists? **Optimality** does the algorithm find a solution with the lowest possible path cost among all solutions? **Time complexity** how long does the algorithm take to find a solution, usually expressed as a function of the **branching factor** $b$ (the maximum number of successors of any single state) and the **depth** $d$ of the shallowest goal node? **Space complexity** how much memory is required to perform the search, typically expressed in the same terms as time complexity?

These four criteria are frequently in direct tension with one another in practice: an algorithm that is both complete and optimal (such as breadth-first search, under certain conditions established in Section 3.6.1) may require genuinely prohibitive amounts of memory on large problems, while an algorithm with only modest memory requirements (such as plain depth-first search, Section 3.6.3) may in general be neither complete nor optimal.

## 3.5 Search Algorithms

The general approach shared by nearly all the search algorithms discussed in this module is to build a **search tree** superimposed over the underlying state-space graph: the root of this tree corresponds to the initial state, and each node's children correspond to the states reachable by a single action from that node. Because the very same state can often be reached along more than one distinct path, a naive **tree search** can revisit the same state repeatedly sometimes leading to infinite loops even within an entirely finite state space; a **graph search** avoids this specific pitfall by maintaining an **explored set** (sometimes called the "closed list") of previously visited states, and discarding any newly generated node whose underlying state has already been explored.

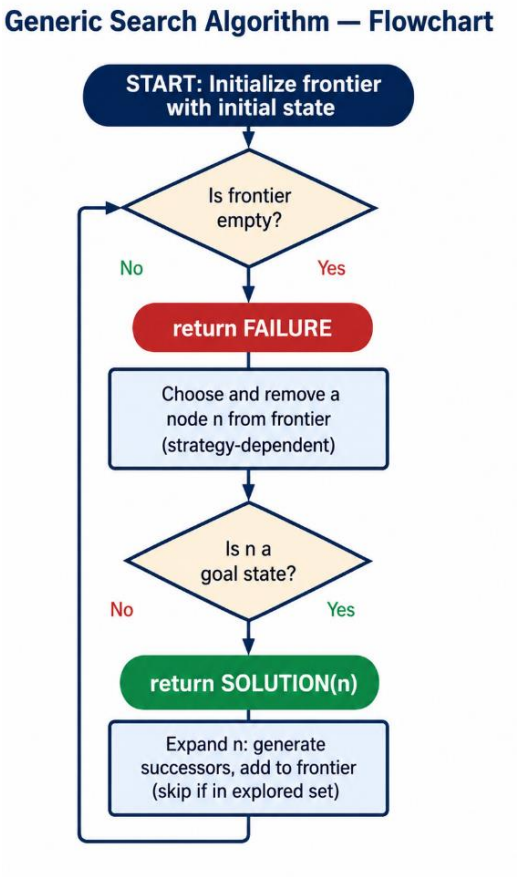


*Figure: Generic search algorithm as a flowchart, showing the frontier-expansion loop shared by all strategies in this section.*

Every search algorithm studied in this module maintains a **frontier** (also called the "open list" or "fringe") the set of nodes that have already been generated but not yet expanded and differs from every other algorithm in this module primarily in the specific **strategy used to choose which frontier node to expand next**. Figure 3.2 shows a small example road map that will be used consistently to illustrate several of the algorithms below.

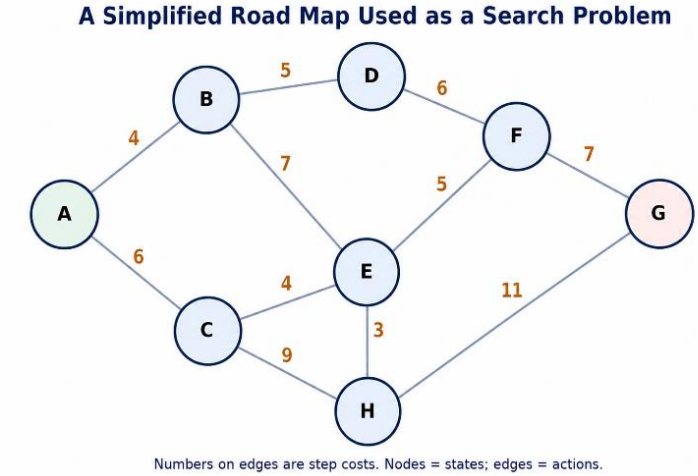


*Figure 3.2: A small road-map search problem: nodes are states (locations), edges are actions (roads), and edge labels are step costs.*

## 3.6 Uninformed Search Strategies

**Uninformed** (or **blind**) search strategies have access to no information about the problem beyond its own bare definition: they can generate successor states and can distinguish goal states from non-goal states, but they possess no notion whatsoever of which non-goal states are more "promising" to explore than others.

### 3.6.1 Breadth-First Search (BFS)

Breadth-first search expands the **shallowest unexpanded node first**: every node at depth *d* is fully expanded before any node at depth *d + 1* is expanded at all. This strategy is implemented by using a **FIFO queue** for the frontier newly generated successor nodes are always added to the back of the queue, so older (necessarily shallower) nodes are always expanded strictly before newer (deeper) ones.

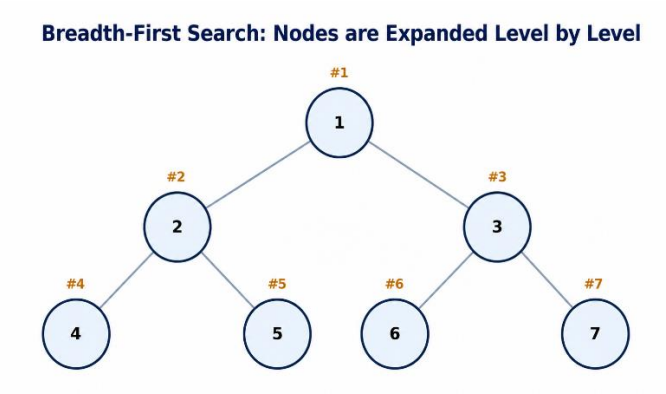


*Figure 3.3: Breadth-first search expands nodes level by level, exploring the entire tree at one depth before moving to the next.*

```
function BREADTH-FIRST-SEARCH(problem) returns a
solution or failure
    node ← a node with STATE = problem.INITIAL-STATE,
PATH-COST = 0
    if problem.GOAL-TEST(node.STATE) then return
SOLUTION(node)
    frontier ← a FIFO queue with node as the only
element
    explored ← an empty set
    loop do
        if EMPTY(frontier) then return failure
        node ← POP (frontier) // chooses the shallowest
node
        add node.STATE to explored
        for each action in problem.ACTIONS(node.STATE)
do
            child ← CHILD-NODE(problem, node, action)
            if child.STATE is not in explored or
frontier then
                if problem.GOAL-TEST(child.STATE) then
return SOLUTION(child)
                frontier ← INSERT(child, frontier)
```

**Worked Example: Worked Example 3.2 — Tracing Breadth-First Search on the Road Map**

**Problem:** Using the road map of Figure 3.2 (nodes A through H, with A as the start and G as the goal), trace the order in which breadth-first search expands nodes, ignoring edge costs (BFS only counts the number of steps, not the cost labels).

- **Step 1 — Initialize.** Frontier = [A]. Explored = {}.
- **Step 2 — Expand A (depth 0).** A's neighbors are B and C. Frontier = [B, C]. Explored = {A}.
- **Step 3 — Expand B (depth 1).** B's neighbors are A (already explored), D, E. Frontier = [C, D, E]. Explored = {A, B}.
- **Step 4 — Expand C (depth 1).** C's neighbors are A (explored), E (already in frontier — skip), H. Frontier = [D, E, H]. Explored = {A, B, C}.
- **Step 5 — Expand D (depth 2).** D's neighbors are B (explored), F. Frontier = [E, H, F]. Explored = {A, B, C, D}.
- **Step 6 — Expand E (depth 2).** E's neighbors are B, C (explored), F (in frontier), H (in frontier). No new nodes added. Frontier = [H, F]. Explored = {A, B, C, D, E}.
- **Step 7 — Expand H (depth 2).** H's neighbors are C, E (explored), G. G is the goal — **return solution** as soon as G is generated (standard BFS implementations test the goal at generation time, not expansion time, precisely to avoid this kind of unnecessary extra expansion).

**Answer:** BFS expands nodes in the order A, B, C, D, E, H, and discovers the goal G as a child of H at depth 3, along the path A → C → H → G.

BFS is **complete** (it is guaranteed to find a solution if one exists, provided the branching factor is finite) and is **optimal** provided all step costs are equal to one another (or, more generally, are non-decreasing with depth). Its time and space complexity are both O($b$^$d$), which grows extremely quickly with depth for example, with a branching factor of 10 and a solution at depth 10, roughly 10 billion nodes would need to be generated and held making the exponential **memory** requirement, even more than the time requirement, the chief practical limitation of BFS in realistic applications.

**Advantages:** guaranteed to find the shallowest solution first; complete and optimal under the stated conditions; conceptually very simple to implement correctly.

**Disadvantages and limitations:** memory requirement grows exponentially with solution depth, which in practice exhausts available memory long before it exhausts available time on any but the shallowest problems.

### 3.6.2 Uniform-Cost Search

When step costs are not all equal to one another, breadth-first search is no longer guaranteed to find the optimal solution, since it minimizes only the *number of steps* taken, not the *total accumulated cost.* **Uniform-cost search** instead always expands whichever frontier node has the **lowest path cost so far**, *g(n)*, using a priority queue ordered by accumulated path cost rather than a simple FIFO queue. Provided all step costs are strictly positive, uniform-cost search is both complete and optimal; its complexity is best expressed not in terms of solution depth *d* but in terms of the cost *C\ of the optimal solution and the smallest individual step cost ε present anywhere in the problem, as O(*b^*(1+⌊*C**/ε⌋)), which can in some cases be considerably worse than BFS's simpler bound whenever many small, cheap steps exist alongside a few large, expensive ones.

### 3.6.3 Depth-First Search (DFS)

Depth-first search always expands the **deepest** unexpanded node currently in the frontier, implemented using a **LIFO queue** (that is, a simple stack): the single most recently generated node is always expanded next, and the search only ever "backtracks" to try an alternative branch once the current path reaches either a dead-end state with no remaining unexplored successors, or a pre-set maximum depth.

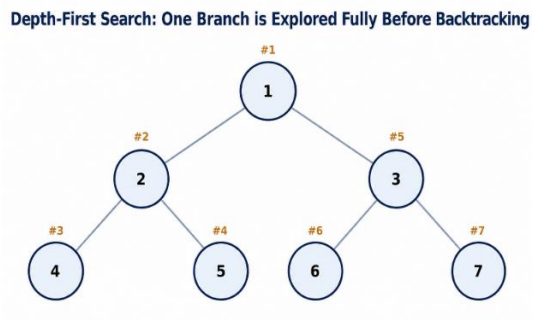


*Figure 3.4: Depth-first search commits to one branch and explores it to its full depth before backtracking to try alternatives.*

**Worked Example: Worked Example 3.3 — Tracing Depth-First Search on the Road Map**

**Problem:** Using the same road map as Worked Example 3.2, trace the order in which depth-first search (assuming successors are generated in alphabetical order and re-expansion of already-explored states is skipped) expands nodes.

- **Step 1.** Expand A. Push successors B, C onto the stack (B on top, assuming alphabetical order is pushed left-to-right with the stack popping the last-pushed item, i.e., C is pushed then B, or equivalently B is examined first this worked example follows the common convention of exploring the first-listed successor deepest-first). Stack (top→bottom): B, C.
- **Step 2.** Expand B. Successors D, E pushed. Stack: D, E, C.
- **Step 3.** Expand D. Successor F pushed. Stack: F, E, C.
- **Step 4.** Expand F. Successors E (already on stack, skip re-adding but not yet explored), G pushed. Stack: G, E, C. G is generated and is the goal.

**Answer:** DFS expands nodes in the order A, B, D, F, and discovers goal G as a child of F, along the considerably longer path A → B → D → F → G — illustrating directly that DFS, unlike BFS, offers no guarantee of finding the *shortest* or *cheapest* path to the goal, only *some* path.

DFS has modest memory requirements only O(*bm*), linear in both the branching factor *b* and the maximum depth *m* of the search tree since it need only remember the single current path from the root down to the current node, together with the small number of unexpanded sibling nodes branching off along that path. This modest, linear memory requirement is DFS's chief practical advantage over BFS. However, plain tree-search DFS is **not complete** in infinite (or very deep, or cyclic) state spaces, since it may follow one single infinite (or merely very long) branch forever and thereby never backtrack to explore a much shallower solution lying along an entirely different branch; nor is DFS **optimal** in general, since it may discover and return a needlessly long solution long before ever trying a much shorter solution lurking elsewhere in the tree, exactly as Worked Example 3.3 above illustrates directly. Its worst-case time complexity is O(*b^m*), which can be very substantially worse than BFS's O(*b^d*) whenever the tree's maximum depth *m* is much larger than the shallowest solution's depth *d*.

**Advantages:** memory requirement is only linear in the maximum search depth, in sharp contrast to BFS's exponential memory requirement; well suited to problems where solutions tend to lie deep in the tree and where the tree's branching factor is very large.

**Disadvantages and limitations:** neither complete nor optimal in the general case; can perform very poorly (or fail to terminate at all) in the presence of very deep or infinite branches that do not contain a solution.

### 3.6.4 Depth-Limited and Iterative Deepening Search

**Depth-limited search** is ordinary DFS equipped with a pre-set depth limit $\ell$, beyond which any node is simply treated as having no further successors at all; this repairs DFS's failure of completeness in infinite state spaces, but only at the cost of introducing a new source of **incompleteness** whenever the shallowest actual goal state happens to lie deeper in the tree than the chosen limit $\ell$.

**Iterative deepening depth-first search (IDDFS)** repeatedly re-runs depth-limited search with a progressively increasing depth limit first with $\ell = 0$, then $\ell = 1$, then $\ell = 2$, and so forth continuing until a solution is finally found at some limit. This combines the best properties of BFS and DFS: it is complete and optimal under exactly the same conditions as BFS (a finite branching factor; step costs that are equal or non-decreasing with depth), yet its memory requirement remains only O($bd$), the very same modest, linear bound enjoyed by ordinary DFS.

**Key Formula: Why Iterative Deepening Is Not as Wasteful as It First Appears**

Although IDDFS appears wasteful, since it repeatedly regenerates every shallow level of the search tree on every single iteration, the actual overhead turns out to be small in practice. The total number of nodes generated across *all* iterations of IDDFS up to depth $d$ is bounded by:

$(d)b + (d-1)b^2 + \ldots + (1)b\hat{}d \approx b\hat{}d \cdot (b / (b-1))^2$ for $b > 1$

Because the number of nodes strictly *at* the deepest level, $b\hat{}d$, already dominates the sum of all shallower levels combined whenever $b > 1$, the proportional overhead of repeated shallow regeneration remains small (for example, only about 11% extra node generations when $b = 10$) which is precisely why IDDFS is generally the preferred *uninformed* search strategy whenever the state space is very large and the solution depth is not known in advance.

### 3.6.5 Bidirectional Search

**Bidirectional search** runs two simultaneous searches one proceeding forward from the initial state, and one proceeding backward from the goal state and halts as soon as the two independently expanding frontiers meet in the middle. Because the two searches together need only each reach a common depth of roughly $d/2$, rather than a single search needing to reach the full depth $d$, the time and memory complexity can drop dramatically to O($b\hat{}(d/2)$), a very substantial saving, provided that the backward search (searching for *predecessors*, rather than

successors, of a state) can be efficiently defined for the problem at hand, and provided there is a single, explicitly known goal state (or a small, enumerable set of them) to search backward from in the first place.

### 3.6.6 Comparison of Uninformed Strategies

| Criterion | BFS | Uniform-Cost | DFS | Depth-Limited | Iterative Deepening | Bi directional |
|---|---|---|---|---|---|---|
| Complete? | Yes (finite $b$) | Yes (positive costs) | No | No (unless $\ell \geq d$) | Yes (finite $b$) | Yes (usually) |
| Optimal? | Yes (equal step cost) | Yes | No | No | Yes (equal step cost) | Yes (usually) |
| Time | O($b$^$d$) | O($b$^(1+ ⌊C*/ε⌋)) | O($b$^$m$) | O($b$^$\ell$) | O($b$^$d$) | O($b$^($d$/2)) |
| Space | O($b$^$d$) | O($b$^(1+ ⌊C*/ε⌋)) | O($bm$) | O($b\ell$) | O($bd$) | O($b$^($d$/2)) |

**Case Study: Case Study: Word-Ladder Puzzles and Bidirectional Search**

The classic "word ladder" puzzle (transform one word into another, changing one letter at a time, with every intermediate step itself a valid dictionary word — e.g., COLD → CORD → CARD → WARD → WARM) is naturally solved as an uninformed search problem, and is a frequently cited practical example where bidirectional search pays off very substantially: searching forward from COLD and backward from WARM simultaneously, and stopping the moment the two search frontiers meet, is dramatically faster in practice than searching forward alone all the way from COLD to WARM, precisely because both partial searches only need to reach roughly half the total ladder depth before meeting.

## 3.7 Informed (Heuristic) Search Strategies

**Informed** search strategies make use of problem-specific knowledge, beyond the bare definition of the problem itself, in order to find solutions considerably more efficiently than any uninformed strategy could. This additional knowledge is

supplied to the algorithm in the form of a **heuristic function** *h(n)*, which estimates but does not necessarily compute exactly the cost of the cheapest path from a given node *n* to the nearest goal state.

**Definition: Heuristic Function**
A heuristic function h(n) is a problem-specific estimate of the cost of the cheapest path from node n to a goal state. It is used to guide search toward the most promising regions of the state space first, without requiring the algorithm to exhaustively explore the entire space.

### 3.7.1 Greedy Best-First Search

Greedy best-first search expands whichever node currently *appears* closest to the goal, by always selecting the frontier node with the lowest heuristic value *h(n)*, while completely ignoring the cost already actually incurred, *g(n)*, to reach that node in the first place. This strategy can find solutions very quickly in favorable cases, but because it entirely ignores the cost already paid, it is, in general, neither complete (in a tree-search formulation lacking an explored set, it can loop indefinitely) nor optimal it may greedily and repeatedly rush toward a state that merely *appears* close to the goal according to the heuristic, while overlooking a genuinely cheaper overall path elsewhere.

#### 3.7.2 A* Search

**A* search** combines the complementary strengths of uniform-cost search and greedy best-first search by evaluating every node using the sum:

**Key Formula: The A* Evaluation Function**
$f(n) = g(n) + h(n)$

where *g(n)* is the exact, known cost of the path from the initial state to node *n* found so far, and *h(n)* is the estimated cost from *n* onward to the nearest goal; *f(n)* therefore represents an estimate of the total cost of the cheapest complete solution that passes through node *n*. A\ *always expands whichever frontier node has the lowest* f(n)*.

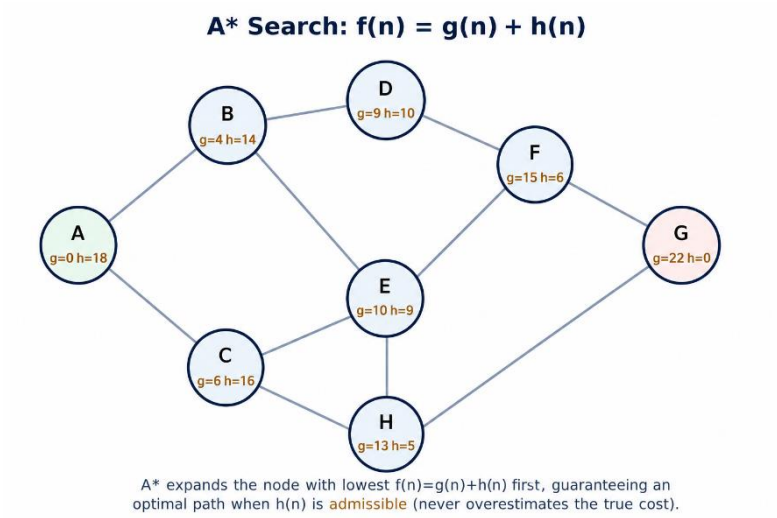


*Figure 3.5: A* search evaluates each node using f(n) = g(n) + h(n), the estimated total cost of a solution passing through that node.*

```
function A-STAR-SEARCH (problem, h) returns a solution
or failure
    node ← a node with STATE = problem.INITIAL-STATE, g
= 0, f = h(INITIAL-STATE)
    frontier ← a priority queue ordered by f, with node
as the only element
    explored ← an empty set
    loop do
        if EMPTY(frontier) then return failure
        node ← POP-LOWEST-F(frontier)
        if problem.GOAL-TEST(node.STATE) then return
SOLUTION(node)
        add node.STATE to explored
        for each action in problem.ACTIONS(node.STATE)
do
            child ← CHILD-NODE(problem, node, action)
// sets child.g = node.g + STEP-COST
            child.f ← child.g + h(child.STATE)
            if child.STATE not in explored or frontier
then
                frontier ← INSERT(child, frontier)
            else if child is in frontier with higher f
then
                replace that frontier node with child
```

**Worked Example: Worked Example 3.4 — Tracing A* Search Step by Step**

**Problem:** Using the road map and heuristic values shown in Figure 3.5 (h-values: A=18, B=14, C=16, D=10, E=9, F=6, G=0, H=8; edge costs as in Figure 3.2), trace A* search from A to G and report the path found.

- **Step 1 — Expand A.** g(A)=0, f(A)=0+18=18. Generate B (g=4, h=14, f=18) and C (g=6, h=16, f=22). Frontier: [B(f=18), C(f=22)].
- **Step 2 — Expand B (lowest f=18).** Generate D (g=4+5=9, h=10, f=19) and E (g=4+7=11, h=9, f=20). Frontier: [C(22), D(19), E(20)] → lowest is D at f=19? Compare: C=22, D=19, E=20 → expand D next.
- **Step 3 — Expand D (f=19).** Generate F (g=9+6=15, h=6, f=21). Frontier: [C(22), E(20), F(21)] → lowest is E at f=20.
- **Step 4 — Expand E (f=20).** E's neighbors C (already better via direct edge, g=6 < 11+4, skip), F (g=11+5=16 > existing 15, skip  existing F stays), H (g=11+3=14, h=8, f=22). Frontier: [C(22), F(21), H(22)] → lowest is F at f=21.
- **Step 5 — Expand F (f=21).** F's neighbor G: g=15+7=22, h=0, f=22. G generated. Frontier: [C(22), H(22), G(22)].
- **Step 6 — Expand G.** G is popped (tie broken arbitrarily among equal f=22 nodes; A* correctly explores remaining ties since none can produce a lower-cost solution) and passes the goal test.

**Answer:** A *returns the path A → B → D → F → G with total cost g(G) = 22, which is confirmed optimal because every remaining frontier node also has f ≥ 22, meaning no unexplored path could possibly improve on this result. Note this trace also illustrates that A* explores considerably fewer nodes than uninformed BFS or uniform-cost search would on the same graph, precisely because the heuristic actively directs expansion toward the goal.

**Derivation: Why A* Is Optimal When the Heuristic Is Admissible**

This subsection derives, step by step, the classical proof that tree-search A* returns an optimal solution whenever its heuristic function is admissible.

**Definition: Admissibility and Consistency**

A heuristic h(n) is **admissible** if it never overestimates the true cost to reach the nearest goal from n  formally, 0 ≤ h(n) ≤ h*(n) *for every node n, where* $h^*(n)$ denotes the true optimal cost from n to a goal. A heuristic is **consistent** (or monotonic) if, for every node n and every successor n′ generated by an action with

step cost c(n, a, n′): h(n) ≤ c(n, a, n′) + h(n′). Every consistent heuristic is provably admissible, but the converse does not always hold.

**Step-by-step derivation (tree-search case, admissible h):**

1. Suppose, for the sake of contradiction, that A\ *has just selected a suboptimal goal node* $G_2$ *for expansion that is,* g*(*$G_2$*)* > C**, where *C*** is the true cost of the optimal solution.
2. Since $G_2$ is a goal node, $h(G_2) = 0$ by definition (the true remaining cost from a goal to itself is zero, and any admissible heuristic must respect this boundary condition). Therefore $f(G_2) = g(G_2) + 0 = g(G_2) > C$*.
3. Consider the optimal solution path, and let *n* be *any* node still on the frontier that lies along this optimal path (such a node must exist, since A\ *has not yet expanded the full optimal path if it is instead expanding the suboptimal* $G_2$*).
4. Because *h* is admissible, *h*(*n*) ≤ *h**(*n*), *so* f(n) = g*(n) +* h*(n)* ≤ g*(n) +* h***(*n*) = *C*** (since *g(n)* plus the true remaining cost *h**(n) *along the optimal path is, by definition, exactly* C**).
5. Combining Steps 2 and 4: *f*(*n*) ≤ *C** < f*(*$G_2$*).
6. But A* *always expands the frontier node with the* lowest *f-value, so A** would necessarily have selected *n* for expansion instead of $G_2$, since *f*(*n*) < *f*($G_2$).
7. This contradicts our Step 1 assumption that A* *selected* $G_2$*. Therefore, no suboptimal goal node can ever be selected before the optimal one: A** is optimal whenever h is admissible. ** ■

For **graph search** (where an explored set discards repeated states), this same guarantee additionally requires *h* to be **consistent**, not merely admissible, because consistency guarantees that the sequence of *f*-values along any single path is non-decreasing, which in turn guarantees that the very first time A* *expands any given state, it has already found the optimal path to that state without this property, a cheaper path to an already-explored state could later be discovered but wrongly discarded. A** is additionally **optimally efficient** among all algorithms using the same heuristic information: no other optimal algorithm using the same *h* is guaranteed to expand fewer nodes than A*, in the worst case.

**Advantages of A*:** guaranteed optimal (given admissibility/consistency); provably optimally efficient given the same heuristic; widely applicable across an enormous range of problem domains.

**Disadvantages and limitations:** must hold every generated node in memory simultaneously, exactly like uniform-cost search, so it can exhaust available memory on large problems well before it exhausts available time; performance in practice is entirely dependent on the quality of the supplied heuristic — a poor or uninformative heuristic degrades A* toward the performance of plain uniform-cost search.

### 3.7.3 Memory-Bounded Heuristic Search

Because A* keeps every generated node in memory simultaneously, it can run out of memory on large problems well before it runs out of time. Two well-known variants directly address this limitation:

- **Iterative-deepening A* (IDA*)** adapts the iterative-deepening idea of Section 3.6.4 to A*: at each successive iteration, a depth-first search is performed, but the cutoff used is the node's* f*-value (*g + h*) rather than its raw depth, and the cutoff for the* next *iteration is set to the smallest* f*-value that exceeded the previous cutoff during the current iteration. This reduces memory requirements to O(*bd*)*, the same modest linear bound enjoyed by ordinary iterative deepening.
- **Recursive Best-First Search (RBFS)** attempts to mimic the node-ordering behavior of ordinary best-first search while using only linear memory, by keeping track of the *f*-value of the best available alternative path from any ancestor of the current node, and backtracking (while memorizing an updated *f*-value at the abandoned node, so that path is not needlessly re-explored later) whenever the current path's cost comes to exceed that remembered alternative.

**Advantages of memory-bounded variants:** achieve A**-quality solutions using only linear memory, making them practical on problems where plain A** would exhaust available RAM.

**Disadvantages and limitations:** both IDA* *and RBFS can, in the worst case, re-expand the same nodes many times across iterations (a cost plain A** never pays, since it never discards a generated node), so they generally trade increased *time* for reduced *memory* relative to plain A*.

## 3.8 Heuristic Functions

The **effectiveness** of any informed search algorithm depends entirely on the quality of the heuristic function it is supplied with. Two standard heuristics for the 8-puzzle illustrate the central trade-offs involved:

- **$h_1$ = the number of misplaced tiles** (not counting the blank) simple and cheap to compute, and clearly admissible, since every single misplaced tile must be moved at least once to reach the goal.
- **$h_2$ = the sum of the Manhattan distances** of every tile from its own goal position (the number of horizontal-plus-vertical grid steps each tile must travel, ignoring the presence of other tiles) also admissible, since it accounts only for the unavoidable minimum number of moves any single tile individually requires, ignoring the fact that tiles can also usefully block or assist one another.

**Worked Example: Worked Example 3.5 Computing $h_1$ and $h_2$ by Hand**

**Problem:** For the 8-puzzle initial state shown in Figure 3.1 (top row 7,2,4; middle row 5, blank,6; bottom row 8,3,1), with goal state (top row 1,2,3; middle row 8, blank,4; bottom row 7,6,5), compute $h_1$ and $h_2$.

- **Step 1 — Compute $h_1$.** Compare each tile's current position to its goal position: 7 (misplaced), 2 (correctly placed — position matches), 4 (misplaced), 5 (misplaced), 6 (misplaced), 8 (misplaced), 3 (misplaced), 1 (misplaced). Count of misplaced tiles = 7. So $h_1 = 7$.
- **Step 2 — Compute $h_2$ for each tile.** Label grid positions by (row, col), 0-indexed. Tile 7: currently (0,0), goal (2,0) → distance = |0−2|+|0−0| = 2. Tile 2: currently (0,1), goal (0,1) → 0. Tile 4: currently (0,2), goal (1,2) → 1. Tile 5: currently (1,0), goal (2,2) → |1−2|+|0−2| = 3. Tile 6: currently (1,2), goal (2,1) → |1−2|+|2−1| = 2. Tile 8: currently (2,0), goal (1,0) → 1. Tile 3: currently (2,1), goal (0,2) → |2−0|+|1−2| = 3. Tile 1: currently (2,2), goal (0,0) → |2−0|+|2−0| = 4.
- **Step 3 — Sum.** $h_2 = 2+0+1+3+2+1+3+4 = 16$.

**Answer:** $h_1 = 7$ and $h_2 = 16$ for this state. Note $h_2 > h_1$ here, illustrating the dominance relationship derived below.

**Derivation: Proving $h_2$ Dominates $h_1$**

**Definition: Dominance**
Given two admissible heuristics ha and hb, if ha(n) ≥ hb(n) for every node n in the state space, then ha is said to **dominate** hb.

1. Consider any single tile *t* that is currently misplaced (not in its goal position).
2. By definition, $h_1$ counts this tile as contributing exactly 1 to the total (since it is simply "misplaced," regardless of how far away it actually is).
3. By definition, $h_2$ counts this same tile as contributing its full Manhattan distance the number of grid steps separating its current position from its goal position.
4. Because the tile is misplaced, its Manhattan distance from its goal position must be at least 1 (a distance of 0 would mean the tile is already correctly placed, contradicting our assumption).
5. Therefore, for every misplaced tile, $h_2$'s per-tile contribution (≥1) is greater than or equal to $h_1$'s fixed per-tile contribution (exactly 1).
6. Tiles that are *not* misplaced contribute 0 to both $h_1$ and $h_2$ identically.
7. Summing over all eight tiles: $h_2(n) \geq h_1(n)$ for every state n. By the definition of dominance above, **$h_2$ dominates $h_1$. ■**

A dominant heuristic is always at least as efficient for A\ *search, in the precise sense that it never causes* more *node expansions than the dominated heuristic, and typically causes substantially fewer. The effective branching factor* b** of a heuristic, measured empirically on a given problem instance, is defined implicitly by the equation:

**Key Formula: Effective Branching Factor**
$N + 1 = 1 + b\ + (b\)^2 + \ldots + (*b\)^d$

where *N* is the total number of nodes generated by A\ *using that heuristic on a given instance, and* d *is the depth of the solution found. A well-designed heuristic keeps* b\ **close to 1 (an ideal, perfectly informed heuristic gives *b** = 1); comparing the empirically measured effective branching factors of two candidate heuristics across a suite of test instances is the standard empirical method used to choose between them in practice.

**Worked Example: Worked Example 3.6 — Solving for the Effective Branching Factor**

**Problem:** A\ *using heuristic $h_2$ generates N = 52 nodes to find a solution at depth d = 4. Estimate the effective branching factor b*\.

- **Step 1 — Set up the equation.** 52 + 1 = 1 + b\ + *(b*\$)^2$ + (b\$)^3$ + *(b*\$)^4$, i.e., 53 = 1 + b\ + *(b*\$)^2$ + (b\$)^3$ + *(b*\$)^4$.
- **Step 2 — Try candidate values.** Try b\=*1.5: 1 + 1.5 + 2.25 + 3.375 + 5.0625 = 13.2 (too small). Try b*\=2.5: 1 + 2.5 + 6.25 + 15.625 + 39.06 = 64.4 (too large). Try b\=*2.2: 1 + 2.2 + 4.84 + 10.65 + 23.43 = 42.1 (still a little small). Try b*\=2.35: 1 + 2.35 + 5.52 + 12.98 + 30.5 = 52.4 (very close).
- **Step 3 — Conclude.** b* ≈ 2.35 (found by numerically solving the polynomial; in practice this is done with a root-finding routine rather than by hand for larger instances).

**Answer:** The effective branching factor is approximately 2.35, meaning this heuristic makes the search behave, on average, as if only about 2.35 successors needed to be considered at each step, substantially better than the 8-puzzle's raw branching factor of up to 4.

Good admissible heuristics can frequently be constructed *automatically*, rather than by hand, by considering a **relaxed problem** a version of the original problem with some restriction on the legal actions removed (for instance, allowing an 8-puzzle tile to move directly to *any* empty square on the board, rather than only to an adjacent one, immediately yields $h_1$ as the exact solution cost of this relaxed problem; allowing tiles to move to adjacent squares even if occupied, effectively overlapping tiles, yields $h_2$). Because the cost of an optimal solution to a relaxed problem can never exceed the cost of an optimal solution to the original, more constrained problem, any such relaxed-problem cost is automatically and provably admissible for the original problem.

**Advantages of automatically-derived heuristics:** systematic and less error-prone than hand-crafting a heuristic from intuition alone; the relaxation technique generalizes to essentially any search problem, not merely the 8-puzzle, and is used extensively in the automated planning heuristics of Module 5, Section 5.3.

**Disadvantages and limitations:** not every useful relaxation is cheap to compute exactly (some relaxed problems are themselves still NP-hard to solve optimally,

forcing the use of a further approximation); a poorly chosen relaxation can yield a heuristic so weak it provides little practical benefit over an uninformed search.

## 3.9 Constraint Satisfaction Problems (CSPs)

### 3.9.1 Definition

Many problems can be formulated considerably more efficiently, and solved using more specialized and powerful algorithms, by explicitly exploiting their internal structure rather than treating each state as an opaque, "black-box" object, as ordinary search (Sections 3.5–3.7) implicitly does. A **constraint satisfaction problem (CSP)** is defined by exactly three components:

> **Definition: Constraint Satisfaction Problem**
> A CSP consists of a set of variables $\{X_1, X_2, \ldots, X_n\}$, a set of domains $\{D_1, D_2, \ldots, D_n\}$ (one per variable, giving the legal values that variable may take), and a set of constraints C specifying the allowable combinations of values for particular subsets of the variables. A solution is a complete assignment of values to all variables that satisfies every constraint.

A constraint over exactly two variables is called a **binary constraint**, and can be visualized directly as an edge in a **constraint graph**, whose nodes are the CSP's variables. A **state** in a CSP is any assignment of values to some or all of the variables; an assignment that violates no constraint is termed **consistent**; an assignment in which every variable has been given some value is termed **complete**; and a **solution** to a CSP is, by definition, a complete, consistent assignment.

### 3.9.2 CSPs as a Special Kind of Search Problem

A CSP can technically be solved using generic search (treating each partial assignment as a state, exactly as in Sections 3.5–3.7), but the fixed, factored structure of a CSP  a well-defined, finite set of variables, each with a well-defined, finite domain  permits the use of far more efficient, specialized algorithms, most notably **backtracking search**: a depth-first search that assigns values to variables one at a time, immediately abandoning ("backtracking" from) any partial assignment the moment it violates a constraint, rather than waiting until a complete assignment has been fully generated before ever testing it.

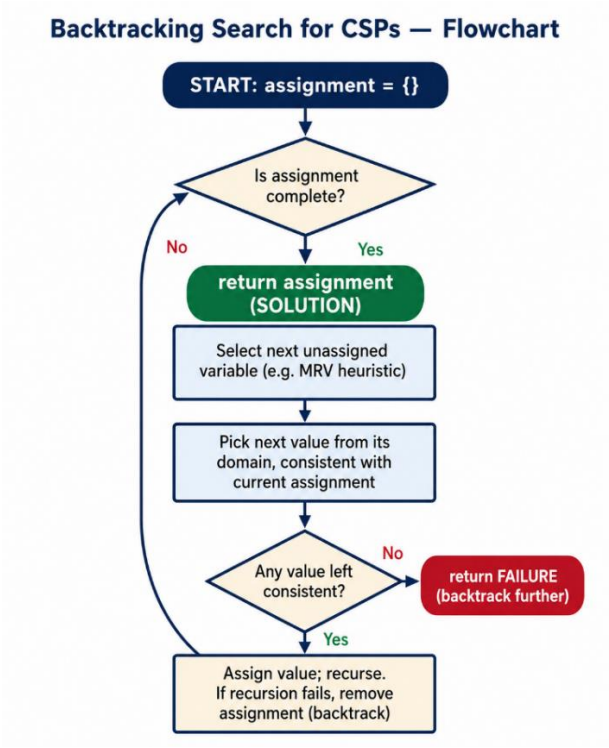


*Figure: Backtracking search for CSPs, shown as a flowchart.*

### 3.9.3 Structure and Examples

- **Map coloring.** Given a map divided into distinct regions, assign each region a color from some fixed palette such that no two adjacent regions ever share the same color. Variables are the regions; domains are the available colors; constraints require adjacent regions (an edge in the constraint graph) to differ.

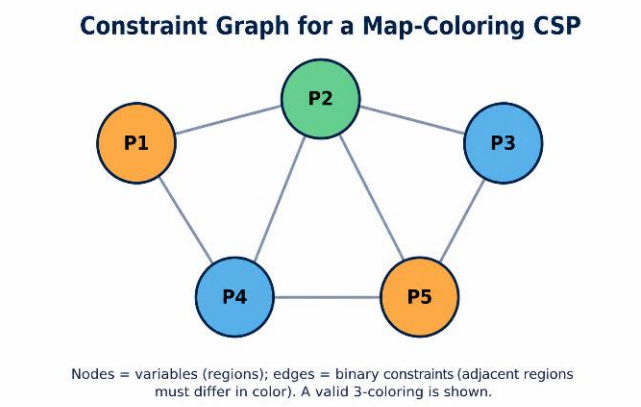


*Figure 3.6: A constraint graph for a map-coloring CSP; nodes are variables (regions) and edges are binary "must differ" constraints. A valid three-coloring is shown.*

- **The n-queens problem**, reformulated explicitly as a CSP (rather than as a general search problem, as in Section 3.3.1): variables are conveniently taken to be one per column, each with domain {1, …, n} representing the row occupied by the queen in that column, and constraints require that no two queens ever share a row or a diagonal a formulation that is dramatically more efficient to solve in practice than the naive general-search formulation.
- **Sudoku**, in which variables are the 81 empty cells, domains are {1, …, 9}, and constraints require all-different values within each row, each column, and each 3×3 sub-grid an "all-different" constraint being a common and highly useful higher-order (non-binary) constraint type frequently supported directly by CSP-solving software.
- **Cryptarithmetic puzzles** (for example, SEND + MORE = MONEY, where each distinct letter represents a distinct digit), in which variables are the letters, domains are {0, …, 9}, and constraints capture both the "all different" requirement and the underlying arithmetic relationship of the sum itself, typically expressed using auxiliary "carry" variables for each column of the addition.

**Worked Example: Worked Example 3.7 Tracing Backtracking Search on 4-Queens**

**Problem:** Trace backtracking search on the 4-queens CSP (variables Q1, Q2, Q3, Q4 for columns 1–4; domain {1,2,3,4} for the row of each queen; constraint: no shared row or diagonal), assigning variables in column order and trying values in ascending order.

- **Step 1.** Q1 = 1 (row 1, column 1). Consistent so far (only one queen placed).
- **Step 2.** Try Q2 = 1: same row as Q1 violates constraint, reject. Try Q2 = 2: diagonal from Q1 (row difference 1 = column difference 1) violates constraint, reject. Try Q2 = 3: row difference 2, column difference 1 no shared row or diagonal accept. Q2 = 3.
- **Step 3.** Try Q3 = 1: same row as Q1 reject. Try Q3 = 2: diagonal from Q2 (row diff 1, col diff 1) reject. Try Q3 = 3: same row as Q2 reject. Try Q3 = 4: diagonal from Q1 (row diff 3, col diff 2 not diagonal, check again: row diff |4-1|=3, col diff |3-1|=2, not equal, OK; check against Q2: row diff |4-3|=1, col diff |3-2|=1, equal diagonal reject). All four values for Q3 fail backtrack **to Q2.**

- **Step 4.** Try Q2 = 4 (next value after 3): row diff from Q1 = 3, col diff = 1 not diagonal accept. Q2 = 4.
- **Step 5.** Try Q3 = 1: shares row with Q1 reject. Try Q3 = 2: row diff from Q1 = 1, col diff = 2, OK; row diff from Q2 = 2, col diff = 1, OK accept. Q3 = 2.
- **Step 6.** Try Q4 = 1: shares row with Q1 reject. Try Q4 = 2: shares row with Q3 reject. Try Q4 = 3: row diff from Q2(row4) = 1, col diff = 1 diagonal reject. Try Q4 = 4: shares row with Q2 reject. All fail **backtracks to Q3.**
- **Step 7.** No further values remain for Q3 after 2 backtrack to Q2; no further values after 4 backtrack to Q1; try Q1 = 2, and continue the same systematic process (omitted here for brevity) until the solution Q1=2, Q2=4, Q3=1, Q4=3 is eventually found.

**Answer:** This trace illustrates the core backtracking mechanism precisely: assign, check consistency, and recursively backtrack immediately whenever every remaining value fails, exactly as shown in the CSP flowchart above. A full trace to the first complete solution for 4-queens typically requires exploring on the order of 10–15 partial assignments using this simple chronological-backtracking strategy.

### 3.9.4 Constraint Propagation: Arc Consistency

Beyond simple backtracking, CSP-solving efficiency is very substantially improved through **constraint propagation** using the constraints themselves to actively reduce the legal remaining values of variables *before*, or *during*, search, which can in turn trigger further reductions elsewhere, cascading through the constraint graph. A widely used and effective form is **arc consistency**: a directed arc from variable $X_i$ to variable $X_j$ is said to be consistent if, for every value remaining in the domain of $X_i$, there exists at least one value remaining in the domain of $X_j$ that satisfies the binary constraint between them. The AC-3 algorithm repeatedly removes values from variable domains that violate arc consistency until no further removal is possible anywhere in the graph either solving the CSP outright, proving it entirely unsolvable, or, most commonly in practice, substantially shrinking the remaining domains and thereby the remaining search effort before backtracking search is even applied.

**Advantages of CSP formulation over generic search:** exploits problem structure directly for dramatic efficiency gains; supports powerful, general-purpose techniques (constraint propagation, variable- and value-ordering heuristics) that apply uniformly across a huge range of otherwise very different-looking problems;

naturally supports partial or approximate solutions when no complete solution exists.

**Disadvantages and limitations:** formulating a problem correctly as a CSP requires upfront engineering effort to identify the right variables, domains, and constraints; in the worst case, CSPs remain NP-hard in general, so constraint propagation and good heuristics reduce but do not eliminate the risk of exponential worst-case behavior on adversarial instances.

**Case Study: Case Study: Automated Timetabling as a CSP**

University course timetabling assigning each course a time slot and a room such that no student has two required courses scheduled simultaneously, no room is double-booked, and no instructor is scheduled for two classes at once is a large-scale, real-world CSP solved routinely by university administrative software. Variables are the (course, section) pairs; domains are the available (time-slot, room) combinations; constraints include no-overlap constraints for shared students, shared instructors, and shared rooms. Real timetabling systems rely heavily on the constraint-propagation and variable-ordering techniques of this section (for example, scheduling the most constrained courses those with the fewest remaining valid time-slot/room combinations first, a strategy known as the minimum-remaining-values heuristic) because naive backtracking alone is far too slow on problems involving many thousands of course sections.

**Key Terms Introduced in This Module**

Problem-solving agent · state space · initial state · actions · transition model · goal test · path cost · solution · optimal solution · toy problem · real-world problem · completeness · optimality · time complexity · space complexity · branching factor · search tree · frontier · explored set · tree search · graph search · breadth-first search · uniform-cost search · depth-first search · depth-limited search · iterative deepening · bidirectional search · heuristic function · greedy best-first search · A* search · admissibility · consistency (monotonicity) · optimally efficient · IDA* · RBFS · dominance · effective branching factor · relaxed problem · constraint satisfaction problem · variable/domain/constraint · constraint graph · backtracking search · arc consistency

**Summary of Module 3**

- A problem-solving agent formulates a goal, formulates a search problem (initial state, actions, transition model, goal test, path cost), searches for a solution, and then executes it.
- Search algorithms are compared by completeness, optimality, and time/space complexity, generally expressed in terms of branching factor *b*, solution depth *d*, and maximum tree depth *m*.
- Uninformed strategies (BFS, uniform-cost, DFS, depth-limited, iterative deepening, bidirectional) use no problem-specific knowledge beyond the problem definition; iterative deepening is usually preferred when memory is limited and solution depth is unknown.
- Informed strategies use a heuristic function *h(n)*; A\ *search, which evaluates nodes by* f(n) = g(n) + h(n)*, is provably complete and optimal when* h* is admissible (tree search) or consistent (graph search), and is optimally efficient among algorithms using the same heuristic — a result derived formally in this module by contradiction.
- Good heuristics are often derived automatically from relaxed problems; a dominant heuristic (one that is uniformly at least as large as another, while remaining admissible) provably leads to fewer node expansions, and the effective branching factor gives an empirical way to compare heuristic quality.
- Constraint satisfaction problems exploit factored state representations (variables, domains, constraints) to enable specialized, more efficient algorithms such as backtracking search and constraint propagation (arc consistency), and scale to real-world applications such as automated timetabling.

## Review Questions

1. Formally define a search problem in terms of its five components, illustrating each with an example distinct from the 8-puzzle or route-finding.
2. Distinguish between a tree search and a graph search. Why can plain tree search fail to terminate even in a finite state space?
3. Using Worked Examples 3.2 and 3.3 as models, trace BFS and DFS by hand on a graph of your own construction with at least six nodes, and compare the solutions each finds.
4. Explain why iterative deepening search is usually preferred over plain BFS for large state spaces, despite repeatedly regenerating shallow nodes, and justify this using the formula in Section 3.6.4.
5. State the evaluation function used by A* search and explain the role played by each of its two terms.
6. Reproduce the step-by-step derivation in Section 3.7.2 showing why tree-search A* is optimal when h is admissible, in your own words.
7. Using the 8-puzzle state given in Worked Example 3.5, compute $h_1$ and $h_2$ for a different initial arrangement of your own choosing, and verify that $h_2 \geq h_1$.
8. Reproduce the dominance proof of Section 3.8 showing that Manhattan distance dominates the misplaced-tiles heuristic.
9. A search using a given heuristic generates 30 nodes to find a solution at depth 3. Estimate the effective branching factor, following the method of Worked Example 3.6.
10. What is a relaxed problem, and how can it be used to automatically generate an admissible heuristic? Give an example other than the 8-puzzle.
11. Formally define a constraint satisfaction problem, and formulate the map-coloring problem for four regions where every region borders every other region.
12. Using Worked Example 3.7 as a model, trace backtracking search on the 4-queens problem starting from a different initial variable-ordering choice, and report where backtracking occurs.
13. Explain the idea of arc consistency and describe, using the automated timetabling case study, how constraint propagation can reduce the work done by backtracking search in a large, real-world CSP.

# MODULE 4:
# KNOWLEDGE REPRESENTATION AND REASONING

## Learning Objectives

By the end of this module, the student should be able to:

- Explain the TELL/ASK architecture of a knowledge-based agent and describe its operating cycle.
- Represent facts about the Wumpus World, and other small domains, in both propositional and first-order logic, and correctly evaluate their truth in a given model.
- Convert an arbitrary first-order logic sentence into Conjunctive Normal Form, showing every intermediate step.
- Compute the most general unifier of two given atomic sentences by hand, and trace forward chaining, backward chaining, and resolution refutation on a small knowledge base.
- Compare forward chaining, backward chaining, and resolution in terms of their reasoning direction, completeness, and typical efficiency, and select the appropriate one for a given application.

## 4.1 Knowledge-Based Agents

A **knowledge-based agent** is organized around a central component called the **knowledge base (KB)** a set of **sentences**, expressed in some formal **knowledge representation language**, that together encode facts and rules about the world that the agent currently believes to be true. New sentences may be added to the KB, or logically derived from it, using an **inference engine**, a mechanism for mechanically deriving new sentences that are logically entailed by the sentences already present in the KB.

**Definition: Knowledge-Based Agent**
A knowledge-based agent maintains a knowledge base (KB) of sentences in a formal representation language, and selects actions by reasoning over that KB

using an inference engine. Two fundamental operations are performed on the KB: TELL (add a new sentence) and ASK (query what the KB currently entails).

A generic knowledge-based agent operates on a simple, repeating cycle: it **TELL**s the KB what it has just perceived, then **ASK**s the KB what action it should perform next (using all of its accumulated knowledge to determine the best current action), and then **TELL**s the KB which action was actually chosen (so that, for example, the KB's internal model of the current world state can be correctly updated), before receiving the next percept and repeating the entire cycle.

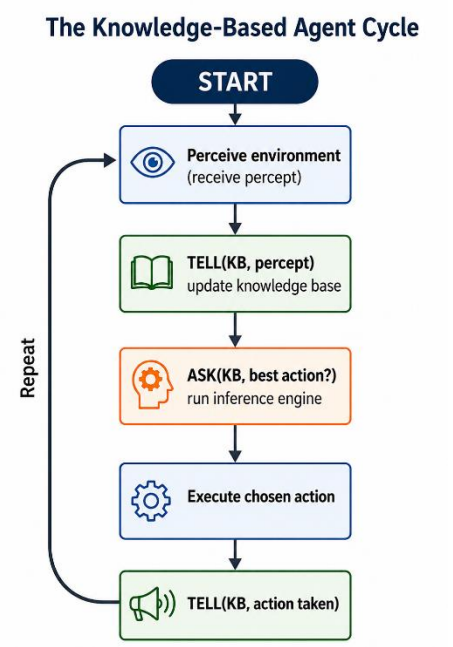


*Figure: The knowledge-based agent's TELL–ASK operating cycle.*

Crucially, this architecture allows a knowledge-based agent to be built by first specifying *what* the agent needs to know (declarative knowledge, expressed cleanly as logical sentences), largely independently of the separate question of *how* the inference engine actually reasons with that knowledge  a clean and valuable separation of concerns that is the central engineering appeal of the entire knowledge-based approach, and one that sharply distinguishes it from the tightly coupled, hand-written percept-to-action rules of a simple reflex agent (Module 2, Section 2.4.1).

**Advantages of the knowledge-based approach:** new knowledge can be added incrementally, simply by Telling the KB additional sentences, without needing to rewrite the inference engine itself; the same general-purpose inference engine can, in principle, be reused unchanged across entirely different application domains, simply by swapping in a different knowledge base; the reasoning process can, in many implementations, be made to explain *why* it reached a given conclusion, by tracing back through the specific chain of sentences and rules actually used (as in the Explanation Facility of the expert-system case study in Section 4.6.3).

**Disadvantages and limitations:** naive, general-purpose logical inference can be computationally very expensive, sometimes intractably so, on large knowledge bases (Module 4, Section 4.6.1 discusses this directly); correctly and completely capturing real-world common-sense knowledge in a strictly formal logical representation remains, in general, a very difficult and labor-intensive engineering task (the "knowledge acquisition bottleneck" already encountered in the expert-systems case studies of Module 1, Section 1.3.5).

**Case Study: Case Study: The Architecture of a Rule-Based Expert System**

Real deployed knowledge-based systems, such as the MYCIN system discussed in Module 1, Section 1.3.5, are typically organized into four cooperating components: a **user interface** through which the human user enters facts and receives conclusions; an **inference engine**, implementing exactly the TELL/ASK cycle described above, together with one of the inference strategies developed later in this module (forward chaining, backward chaining, or resolution); a **knowledge base** proper, holding the domain's rules and facts; and an **explanation facility**, which can trace back through the specific chain of rules actually used to reach a conclusion and present that chain to the user in a human-readable form — directly addressing the "black-box" interpretability concern raised in Module 1, Section 1.5.2

## 4.2 The Wumpus World

The **Wumpus World** is a classic, deliberately small grid environment used throughout AI education specifically to introduce logical agents, because it is simple enough to reason about entirely by hand, yet rich enough to genuinely require careful logical inference (as opposed to merely simple reflex behavior).

**PEAS description.** *Performance measure:* +1000 for successfully climbing out of the cave while carrying the gold, −1000 for falling into a pit or being eaten by the

Wumpus, −1 for every individual action taken, and a further −10 penalty for using the agent's single available arrow. *Environment:* a 4×4 grid of connected rooms; the agent starts in the bottom-left room, [1,1], facing right; exactly one room contains a dangerous Wumpus (which kills the agent instantly if entered), some rooms contain bottomless Pits (which also kill the agent instantly), and exactly one room contains a heap of Gold. Rooms directly adjacent to the Wumpus are perceived as smelling of Stench; rooms directly adjacent to a Pit are perceived as feeling a Breeze; the Gold's own room glitters. *Actuators:* Move Forward, Turn Left, Turn Right, Grab, Shoot (fires the agent's single arrow in a straight line in the direction currently faced, killing the Wumpus if it lies anywhere along that line), Climb (exits the cave, but only from room [1,1]). *Sensors:* Stench, Breeze, Glitter, Bump (felt upon walking directly into a wall), and Scream (heard throughout the entire cave if the fired arrow successfully kills the Wumpus).

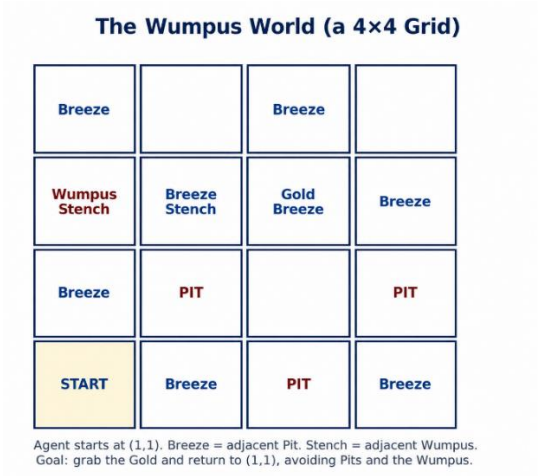


*Figure 4.1: The Wumpus World — a 4×4 grid environment. Breeze indicates an adjacent Pit; Stench indicates an adjacent Wumpus.*

Considered against the environment properties introduced in Module 2, Section 2.3.2, the Wumpus World is **discrete**, **static** (nothing in the world moves or changes except as a direct result of the agent's own actions), **single-agent**, and crucially, for the purposes of this module — only **partially observable** (the agent perceives only its own current square) and, from the agent's point of view before it has gathered any percepts at all, effectively **unknown** with respect to the specific locations of the Pits and the Wumpus (though the general rules governing Breeze and Stench are known in advance).

This partial observability is precisely what makes the Wumpus World such a good vehicle for teaching logical inference: the agent must genuinely *deduce* the likely locations of Pits and the Wumpus from a *sequence* of Breeze and Stench percepts gathered while exploring several different rooms, rather than perceiving these dangers directly. For example, if a Breeze is perceived in room [1,2] but no Breeze is perceived in room [2,1], a logical agent can validly deduce that a Pit is considerably more likely to be located at [2,2] than naive, symmetric reasoning based on [2,1] alone would otherwise suggest; and if Breeze is *not* perceived in some particular room, the agent can validly conclude, with logical certainty, that *none* of the rooms immediately adjacent to it contains a Pit — a safe, genuinely useful, and non-obvious inference of exactly the kind that formal logic, introduced next, is specifically designed to support systematically and reliably.

## 4.3 Logic: General Concepts

A **logic** consists of a formal **syntax**, which precisely defines the well-formed sentences of the representation language, and a formal **semantics**, which precisely defines the truth of each such sentence with respect to each possible **model** — a mathematically precise, formally structured possible world (or "way the world could be") against which any given sentence is evaluated as either true or false.

> **Definition: Entailment**
> A sentence α is entailed by a knowledge base KB, written KB ⊨ α, if and only if α is true in every model in which every sentence of KB is also true that is, whenever the KB is true, α must also necessarily be true, no matter what the unspecified remaining details of the world happen to actually be.

**Inference** is the mechanical, purely syntactic process by which an inference engine derives new sentences from sentences it already has. An inference algorithm that derives only genuinely entailed sentences is called **sound** (or *truth-preserving*); an inference algorithm capable of deriving *every* sentence that is in fact entailed is called **complete**. The central theoretical goal of logical AI is to build inference procedures that are both sound and, wherever computationally practical, also complete, for as expressive a logic as can still be made computationally tractable a tension explored directly in Sections 4.5 and 4.6 below.

## 4.4 Propositional Logic

### 4.4.1 Syntax and Semantics

**Propositional logic** (also called Boolean logic) is the simplest logic studied in AI, and it illustrates most of the central ideas of logical representation and inference within a comparatively simple setting. The syntax of propositional logic is built from **atomic sentences** single proposition symbols such as $P_1$, $P_2$, or, more descriptively for the Wumpus World, $W_{1,3}$ (meaning "there is a Wumpus in square [1,3]") each of which may independently be true or false; **complex sentences** are then built up from these atomic sentences using the standard logical connectives: ¬ (not), ∧ (and), ∨ (or), ⇒ (implies), and ⇔ (if and only if / biconditional).

> **Key Formula: Truth Table for the Standard Connectives**
> Note in particular that P⇒Q is true whenever P is false, regardless of Q's truth value — a frequently counter-intuitive point for students first encountering formal logic, but one that follows directly and necessarily from the definition of entailment given in Section 4.3 (a false premise entails anything at all).

The semantics of propositional logic specify exactly how to compute the truth value of any complex sentence, given the truth values of its constituent atomic sentences, via the truth tables shown above for each connective in turn.

### 4.4.2 Inference Rules for Propositional Logic

Several standard, provably sound inference rules, or reusable "patterns," can be combined to construct valid formal proofs, and are frequently used directly as building blocks inside automated theorem provers:

- **Modus Ponens:** from $\alpha \Rightarrow \beta$ and $\alpha$, infer $\beta$.
- **And-Elimination:** from $\alpha \wedge \beta$, infer $\alpha$ (and, separately, also infer $\beta$).
- **And-Introduction:** from $\alpha$ and $\beta$ (each separately known to be true), infer $\alpha \wedge \beta$.
- **Double-Negation Elimination:** from $\neg\neg\alpha$, infer $\alpha$.
- **Unit Resolution:** from $\alpha \vee \beta$ and $\neg\beta$, infer $\alpha$.
- **Resolution** (the fully general form, discussed further for first-order logic in Section 4.6.5): from $\alpha \vee \beta$ and $\neg\beta \vee \gamma$, infer $\alpha \vee \gamma$.

**Worked Example: Worked Example 4.1 — Propositional Inference in the Wumpus World**

**Problem:** Suppose the knowledge base contains the sentence $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$ (there is a breeze in room [1,1] if and only if there is a pit in an adjacent square), together with the percept $\neg B_{1,1}$ (no breeze was actually felt at [1,1]). Derive what the agent can validly conclude about $P_{1,2}$ and $P_{2,1}$.

- **Step 1 — Apply biconditional elimination.** $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$ is logically equivalent to the conjunction of two implications: $(B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$.
- **Step 2 — Take the contrapositive of the second implication.** $(P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1}$ is logically equivalent to its contrapositive, $\neg B_{1,1} \Rightarrow \neg(P_{1,2} \vee P_{2,1})$.
- **Step 3 — Apply Modus Ponens.** We are given $\neg B_{1,1}$ as a fact (the percept). Combined with the contrapositive from Step 2, Modus Ponens yields $\neg(P_{1,2} \vee P_{2,1})$.
- **Step 4 — Apply De Morgan's law.** $\neg(P_{1,2} \vee P_{2,1})$ is logically equivalent to $\neg P_{1,2} \wedge \neg P_{2,1}$.

**Answer:** $\neg P_{1,2} \wedge \neg P_{2,1}$ — the agent can conclude, with logical certainty, that *neither* [1,2] nor [2,1] contains a Pit, even though it has never directly observed either square. This is precisely the kind of useful, non-obvious inference that motivates representing an agent's knowledge logically rather than relying purely on reflexive, percept-only rules, exactly as introduced conceptually in Section 4.2.

**Advantages of propositional logic:** semantics are completely precise and mechanically checkable (via truth tables, for small numbers of symbols); inference procedures are relatively simple to implement correctly; forms the essential conceptual foundation for the more expressive first-order logic introduced next.

**Disadvantages and limitations:** propositional logic can only express facts about a fixed, pre-enumerated set of specific individual propositions (such as "there is a pit in square [2,2]" as one single, indivisible symbol), and cannot directly express general facts about objects, their properties, and their relationships (such as "every square adjacent to a pit is breezy," stated once and for all, for every square) without writing out one separate proposition instance for literally every relevant square a serious practical representational limitation that directly motivates the considerably more expressive first-order logic introduced in Section 4.5.

## 4.5 First-Order Logic

### 4.5.1 Representation, Syntax, and Semantics

**First-order logic (FOL)** extends propositional logic with the ability to talk about **objects**, the **relations** (also called *predicates*) that hold among those objects, and **functions** that map objects onto other objects, and, critically, with **quantifiers** that let a single sentence express a fact about *all* objects in the domain, or about the mere *existence* of some object, rather than requiring one separate sentence per individual object as propositional logic would.

The building blocks of FOL syntax are:

> **Definition: Building Blocks of First-Order Logic**
> **Constant symbols** name specific objects (e.g., John, Block_A). **Predicate symbols** name relations or properties (e.g., Brother(x,y), Red(x)). **Function symbols** name a mapping from objects to a single object (e.g., FatherOf(x)). **Terms** refer to objects: a constant, a variable, or a function applied to terms. **Atomic sentences** apply a predicate to one or more terms. **Quantifiers** are the universal ∀ ("for all") and existential ∃ ("there exists").

For example, ∀x *King(x)* ⇒ *Person(x)* states "every king is a person," and ∃x *Crown(x)* ∧ *OnHead(x, John)* states "something is a crown, and it is on John's head" that is, "John is wearing a crown." The semantics of FOL define a **model** as a set of objects (the *domain*) together with an interpretation that maps each constant symbol to a specific object, each predicate symbol to a relation over the domain, and each function symbol to a function over the domain; a universally quantified sentence ∀x P(x) is true in a model exactly when P(x) is true under every possible assignment of a domain object to the variable x, and an existentially quantified sentence ∃x P(x) is true exactly when there exists at least one such assignment making P(x) true.

### 4.5.2 Using First-Order Logic: The Kinship Domain

A frequently used illustrative domain is **kinship**, encoding facts and general rules about family relationships using predicates such as *Parent(x, y)*, *Male(x)*, and *Female(x)*. General rules can be stated once and then apply automatically to every object in the domain, for example:

**Key Formula: Sample Kinship Axioms**

$\forall x, y$ Parent $(x, y) \Rightarrow$ Child $(y, x)$

$\forall x$, Male$(x) \vee$ Female$(x)$

$\forall x, y$ Sibling $(x, y) \Leftrightarrow [x \neq y \wedge \exists p$ Parent $(p, x) \wedge$ Parent $(p, y)]$

Once such general axioms are asserted (TELLed) to the knowledge base, together with specific ground facts (for instance, *Parent(Elizabeth, Charles)*), the inference engine can correctly ASK and answer queries such as whether two specific named individuals are siblings for *any* pair of individuals in the entire domain, without requiring a separate hand-written rule for that specific pair, directly illustrating the substantial gain in expressive and representational efficiency that FOL provides over propositional logic (Section 4.4.2).

**Worked Example: Worked Example 4.2 Translating English into First-Order Logic**

**Problem:** Translate the following sentences into first-order logic, using predicates Student(x), Course(y), Passed(x,y), and CoreCourse(y): (a) "Every student has passed at least one course." (b) "No student has passed all core courses without being eligible for graduation" is represented instead as its positive form: "Every student who has passed all core courses is eligible for graduation."

- **Step 1 — Sentence (a).** "Every student" suggests a universally quantified x ranging over students; "has passed at least one course" suggests an existentially quantified y. Combine: $\forall x$ [Student$(x) \Rightarrow \exists y$ (Course$(y) \wedge$ Passed$(x,y))$].
- **Step 2 — Sentence (b), identify the structure.** This is a universally quantified conditional: for every student x, *if* x has passed every core course, *then* x is eligible for graduation. "Passed every core course" is itself a nested universal quantifier over courses y, restricted to core courses.
- **Step 3 — Formalize the nested quantifier.** "x has passed every core course" becomes $\forall y$ [CoreCourse$(y) \Rightarrow$ Passed$(x,y)$].
- **Step 4 — Assemble the full sentence.** $\forall x$ [Student$(x) \wedge \forall y$ (CoreCourse$(y) \Rightarrow$ Passed$(x,y)) \Rightarrow$ Eligible$(x)$].

**Answer:** (a) $\forall x$ [Student$(x) \Rightarrow \exists y$ (Course$(y) \wedge$ Passed$(x,y))$]. (b) $\forall x$ [Student$(x) \wedge \forall y$ (CoreCourse$(y) \Rightarrow$ Passed$(x,y)) \Rightarrow$ Eligible$(x)$]. Note how the nested quantifier structure in (b) directly mirrors the nested structure of the original English sentence — a useful general translation strategy.

### 4.5.3 Knowledge Engineering in First-Order Logic

**Knowledge engineering** is the general, systematic process of building a knowledge base for a particular real application domain, and it typically follows a repeatable sequence of steps:

1. **Identify the task** — determine what range of questions the knowledge base will actually need to answer, and what facts will realistically be available to tell it.
2. **Assemble the relevant knowledge** — become sufficiently familiar with the domain (through independent study, or by directly consulting domain experts) to know what genuinely needs representing, without yet committing to any particular formal representation.
3. **Decide on a vocabulary** of predicates, functions, and constants that is, translate the important domain-level concepts identified in Step 2 into a precise formal logical notation.
4. **Encode general knowledge about the domain** — write down the axioms that capture the domain's general rules, of exactly the kind illustrated above for the kinship domain.
5. **Encode a description of the specific problem instance** currently at hand, as further ground-fact sentences TELLed to the knowledge base.
6. **Pose queries to the inference procedure and obtain answers**, by ASKing the knowledge base.
7. **Debug the knowledge base**, since early versions of the vocabulary or the axioms are very frequently discovered, through this iterative process, to be subtly incomplete or outright incorrect, and must therefore be revised often more than once.

**Case Study: Case Study: A Knowledge Base for Electronic Circuit Verification**

A commonly used illustrative domain in AI education is a knowledge base describing the structure and functional behavior of a simple digital logic circuit (built from AND-gates, OR-gates, and inverters/NOT-gates). The vocabulary might include predicates such as Type(g, AndGate), and functions such as Out(1,g) (denoting the signal present on output terminal 1 of gate g), together with general functional axioms such as: ∀g [Type(g, AndGate) ⇒ Signal(Out(1,g)) = Min(Signal(In(1,g)), Signal(In(2,g)))], which encodes the functional behavior of

*any* AND-gate exactly once, applying automatically to every AND-gate subsequently described within any specific circuit instance. Genuine hardware-verification tools used in the semiconductor industry are built on essentially this same knowledge-engineering principle, though scaled up to circuits containing many millions of individual gates and using considerably more efficient specialized inference procedures than the general-purpose ones introduced pedagogically in Section 4.6 below.

## 4.6 Inference in First-Order Logic

### 4.6.1 Propositional vs. First-Order Inference

Because a universally quantified sentence such as ∀x *King(x)* ⇒ *Person(x)* is, in effect, logically equivalent to the (typically infinite) conjunction of its instances for every possible object *King(John)* ⇒ *Person(John)*, *King(Richard)* ⇒ *Person(Richard)*, and so on for every object in the domain one possible route to first-order inference is **propositionalization**: replacing each quantified sentence with a large set of instantiated propositional sentences, using **Universal Instantiation** (substituting a specific ground term for a universally quantified variable) and **Existential Instantiation** (substituting a single brand-new constant symbol, called a *Skolem constant*, for an existentially quantified variable), and then applying the propositional inference techniques of Section 4.4. Although this reduction is theoretically important, and can be proven complete for knowledge bases containing no function symbols, it is generally impractical as an engineering approach, since blindly instantiating every quantified sentence against every possible ground term generates enormous numbers of largely irrelevant sentences; practical FOL inference engines therefore instead reason directly with quantified sentences and variables, using **unification** to determine, entirely on demand, exactly which specific instantiations are actually relevant to the query at hand.

**Advantages of propositionalization:** conceptually simple; directly reuses the well-understood machinery of propositional logic.

**Disadvantages and limitations:** generates a very large number of irrelevant ground instances in practice; does not scale to knowledge bases of realistic size, motivating the unification-based methods below.

### 4.6.2 Unification

**Unification** is the algorithmic process of finding a substitution of terms for variables that makes two (or more) logical expressions syntactically identical to one another. The substitution found is called a **unifier**; among all possible unifiers of two given expressions, a **most general unifier (MGU)** is one that is at least as general as every other valid unifier (that is, every other valid unifier can itself be obtained from the MGU by applying some further, additional substitution on top of it).

**Definition: Unification and Most General Unifier**

UNIFY (p, q) returns a substitution θ such that SUBST (θ, p) = SUBST (θ, q), if such a substitution exists, or failure otherwise. The most general unifier (MGU) is the unique (up to variable renaming) unifier that imposes the fewest possible commitments on the variables involved.

**Worked Example: Worked Example 4.3 — Computing a Most General Unifier by Hand**

**Problem:** Find the most general unifier of Knows(John, x) and Knows(y, Mother(y)).

- **Step 1 — Compare the predicate symbols.** Both are Knows(·,·) with two arguments — predicate symbols match, so unification may proceed argument by argument.
- **Step 2 — Unify the first arguments.** John (a constant) versus y (a variable). Since one side is a variable, unify by substitution: {y / John}.
- **Step 3 — Apply this substitution and unify the second arguments.** After substituting y = John throughout, the second arguments become x (a variable) versus Mother(John) (a compound term, since y was just replaced by John). Since x is an unbound variable, unify by substitution: {x / Mother(John)}.
- **Step 4 — Combine the substitutions and check for consistency.** Combined substitution: {y/John, x/Mother(John)}. Check: does x appear inside Mother(y) itself, which would indicate a circular ("occurs check") failure? No — x and y are distinct variables, so no circularity arises.
- **Step 5 — Verify by direct substitution.** Applying {y/John, x/Mother(John)} to Knows(John, x) gives Knows(John, Mother(John)). Applying the same substitution to Knows(y, Mother(y)) gives Knows(John, Mother(John)). Both expressions are now syntactically identical.

**Answer:** The most general unifier is θ = {y/John, x/Mother(John)}, and both expressions unify to Knows(John, Mother(John)).

Unification is the essential computational mechanism that allows the first-order inference rules introduced next to determine, entirely automatically, exactly which specific instantiations of a general axiom are relevant to a given fact or query, without ever needing to blindly enumerate every possible ground instantiation as propositionalization would require.

### 4.6.3 Forward Chaining

**Forward chaining** is a data-driven inference strategy well suited to knowledge bases composed largely of rules in a restricted form called **definite clauses** (a disjunction of literals with exactly one positive literal, commonly written as an implication with a conjunction of positive literals as the premise and a single positive literal as the conclusion). Starting from the facts currently known to be true, forward chaining repeatedly finds every rule whose premises are fully satisfied by the currently known facts (using unification to match variables), adds each such rule's conclusion as a new fact, and iterates — continuing until either the specific queried fact is derived, or no further new facts can be added at all (a **fixed point** has been reached).

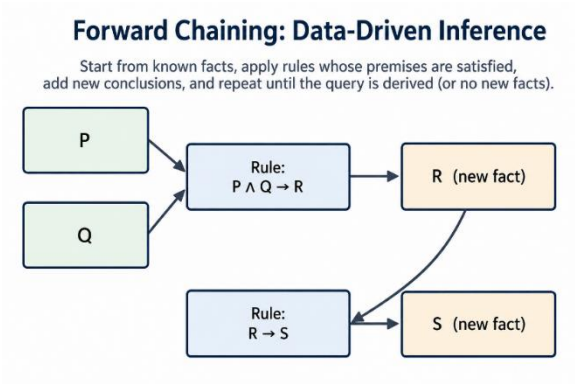


*Figure 4.2: Forward chaining is data-driven — new facts are derived by repeatedly applying rules whose premises are already satisfied, until the query is reached or no further facts can be added.*

**Worked Example: Worked Example 4.4 — Tracing Forward Chaining**

**Problem:** Given the knowledge base of facts {American (West), Weapon(M1), Sells (West, M1, Nono), Hostile (Nono)} and the rule {American(x) ∧ Weapon(y) ∧

Sells(x,y,z) ∧ Hostile(z) ⇒ Criminal(x)}, use forward chaining to determine whether Criminal (West) can be derived.

- **Step 1 — Check the rule's premises against known facts.** American(x): matches American (West), giving x=West. Weapon(y): matches Weapon(M1), giving y=M1. Sells(x,y,z) with x=West, y=M1: matches Sells (West, M1, Nono), giving z=Nono. Hostile(z) with z=Nono: matches Hostile (Nono) — **all four premises are simultaneously satisfied** under the single consistent substitution {x/West, y/M1, z/Nono}.
- **Step 2 — Apply the rule.** Since every premise is satisfied, add the conclusion Criminal(x) with x=West, i.e., add the new fact Criminal (West) to the knowledge base.
- **Step 3 — Check for a fixed point.** No further rules remain to apply (this toy knowledge base has only one rule), so forward chaining halts here.

**Answer:** Criminal (West) is successfully derived by forward chaining after a single rule application, using the unifying substitution {x/West, y/M1, z/Nono}.

Forward chaining is **sound** (every fact it derives is genuinely entailed) and **complete** for knowledge bases restricted to definite clauses, and, because it works forward from all currently known facts rather than being directed by any single particular query, it is especially natural for reactive or continuously updated systems (such as production-rule systems) where new facts arrive incrementally and the system should proactively keep all of their logical consequences up to date at all times.

**Advantages:** naturally incremental — new facts can be TELLed at any time and their consequences derived immediately; well suited to systems that must keep a complete, continuously updated picture of all current consequences.

**Disadvantages and limitations:** can derive many facts that are never actually needed to answer any specific query the user cares about, wasting computation on irrelevant conclusions; less efficient than backward chaining specifically when only a single, narrow query needs answering.

### 4.6.4 Backward Chaining

**Backward chaining** is a goal-driven inference strategy: rather than deriving all possible consequences of the known facts up front, it starts from the specific query (goal) to be proven and works backward, recursively, finding rules whose

conclusion matches the goal and then attempting to prove each of that rule's premises in turn as a new sub-goal, continuing this recursive process until every remaining sub-goal is a fact already known to be true directly.

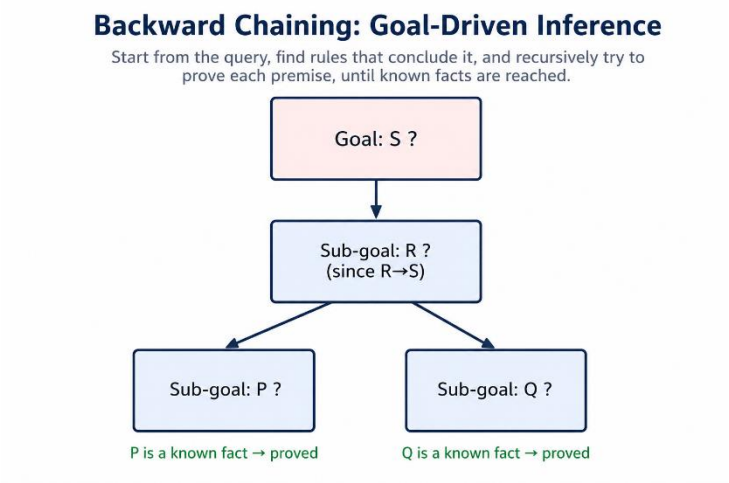


*Figure 4.3: Backward chaining is goal-driven starting from the query, the algorithm recursively reduces it to sub-goals until each is matched against a known fact.*

**Worked Example: Worked Example 4.5 — Tracing Backward Chaining**

**Problem:** Using the same knowledge base and rule as Worked Example 4.4, use backward chaining to prove the query Criminal(West).

- **Step 1 — Identify the goal.** Prove Criminal(West).
- **Step 2 — Find a rule whose conclusion matches the goal.** Criminal(x) ⇐ American(x) ∧ Weapon(y) ∧ Sells(x,y,z) ∧ Hostile(z) matches with x=West.
- **Step 3 — Recursively generate and attempt to prove each sub-goal, left to right.** Sub-goal 1: American(West) — matches a known fact directly, proved immediately. Sub-goal 2: Weapon(y) — matches Weapon(M1), giving y=M1, proved. Sub-goal 3: Sells(West, M1, z) — matches Sells(West, M1, Nono), giving z=Nono, proved. Sub-goal 4: Hostile(Nono) — matches a known fact directly, proved.
- **Step 4 — Conclude.** All four sub-goals are proved under the single consistent substitution {x/West, y/M1, z/Nono}, so the original goal Criminal (West) is proved.

**Answer:** Criminal (West) is proved by backward chaining, exploring exactly the sub-goals relevant to this specific query notice that, unlike forward chaining, no

> irrelevant facts or rules outside this single reasoning chain were ever even considered.

Backward chaining, like forward chaining, is sound and complete for definite-clause knowledge bases, but because it only ever explores facts and rules that are actually relevant to answering the one specific query posed, it is typically far more computationally efficient than forward chaining whenever only a single, specific query needs to be answered, rather than the complete set of consequences of the entire knowledge base. This important distinction (data-driven versus goal-driven reasoning) directly parallels the inference strategy used by Prolog, a widely used programming language built almost entirely around backward-chained first-order definite-clause resolution.

**Advantages:** explores only facts and rules genuinely relevant to the query, avoiding wasted computation on irrelevant consequences; naturally supports interactive, query-driven systems.

**Disadvantages and limitations:** can be considerably less efficient than forward chaining when the *same* few queries are asked repeatedly against a largely static knowledge base, since each query re-derives its answer from scratch rather than reusing previously derived consequences; can loop indefinitely on certain classes of recursively defined rules unless additional loop-checking machinery is added.

### 4.6.5 Resolution

**Resolution**, briefly introduced for propositional logic in Section 4.4.2, generalizes to a single, sound, and (when combined with an appropriate search strategy) **refutation-complete** inference rule for the *full* first-order logic — not merely the restricted definite-clause fragment handled by forward and backward chaining — meaning it can, in principle, prove the entailment of *any* sentence that genuinely follows from the knowledge base, given unbounded computation time.

To use resolution, every sentence in the knowledge base (together with the negation of the query to be proved) must first be converted into **Conjunctive Normal Form (CNF)** — a conjunction of clauses, where each clause is itself a disjunction of literals.

**Definition: Conjunctive Normal Form (CNF) Conversion Steps**

**Worked Example: Worked Example 4.6 — Converting a Sentence to CNF Step by Step**

**Problem:** Convert ∀x [American(x) ∧ Weapon(y) ∧ Sells(x,y,z) ∧ Hostile(z)] ⇒ Criminal(x) to CNF. (Treat y and z here as implicitly universally quantified at the front, as is standard convention for definite-clause-style rules.)

- **Step 1 — Eliminate the implication.** ∀x,y,z ¬[American(x) ∧ Weapon(y) ∧ Sells(x,y,z) ∧ Hostile(z)] ∨ Criminal(x).
- **Step 2 — Move negation inward using De Morgan's law.** ∀x,y,z [¬American(x) ∨ ¬Weapon(y) ∨ ¬Sells(x,y,z) ∨ ¬Hostile(z)] ∨ Criminal(x).
- **Step 3 — Variables are already standardized apart (only one quantifier scope here).**
- **Step 4 — Quantifiers are already at the front (prenex form).**
- **Step 5 — No existential quantifiers are present, so no Skolemization is required.**
- **Step 6 — Drop the universal quantifiers (implicit) and note the sentence is already a single disjunction of literals — a single clause.** ¬American(x) ∨ ¬Weapon(y) ∨ ¬Sells(x,y,z) ∨ ¬Hostile(z) ∨ Criminal(x).

**Answer:** The CNF form is the single clause: ¬American(x) ∨ ¬Weapon(y) ∨ ¬Sells(x,y,z) ∨ ¬Hostile(z) ∨ Criminal(x), which is exactly the clausal form of the original implication rule used in Worked Examples 4.4 and 4.5.

The **resolution inference rule** itself states that from two clauses containing complementary literals (a literal $l$ in one clause, and its negation $\neg l$ — after unification, in the first-order case — in the other clause), a new clause can be derived consisting of the disjunction of all the literals of both original clauses, *except* for the complementary pair itself, which is removed. **Resolution refutation** proves that a knowledge base KB entails a sentence α by: (1) negating α and adding it to KB; (2) converting the resulting complete set of sentences to CNF using the six-step procedure above; and (3) repeatedly applying the resolution rule to pairs of clauses until either the **empty clause** (conventionally written NIL) is derived representing a direct logical contradiction, and thereby proving that KB ∧ ¬α is unsatisfiable, and hence that KB does indeed entail α or until no further resolutions are possible at all, in which case KB does not entail α.

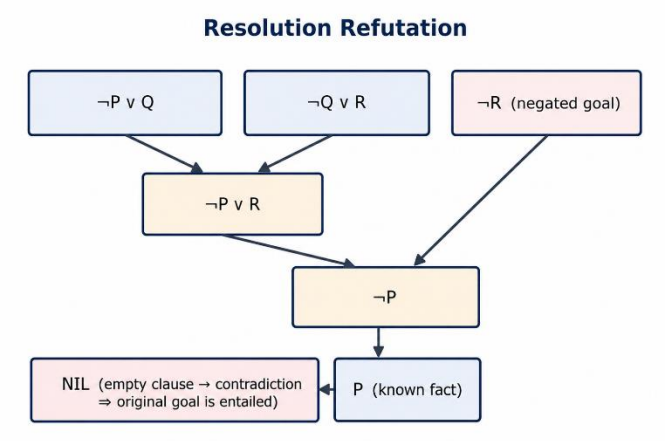


*Figure 4.4: Resolution refutation: clauses are repeatedly resolved against each other; deriving the empty clause (NIL) proves the negated goal leads to a contradiction, and therefore that the original goal is entailed.*

**Worked Example: Worked Example 4.7 — A Complete Resolution Refutation**

**Problem:** Given the clauses {¬P ∨ Q, ¬Q ∨ R, P} (three known facts/rules in CNF), prove R by resolution refutation.

- **Step 1 — Negate the goal and add it to the clause set.** Add ¬R. Full clause set: {¬P∨Q, ¬Q∨R, P, ¬R}.
- **Step 2 — Resolve ¬Q∨R with ¬R.** The complementary pair is R and ¬R. Resolving removes both, leaving: ¬Q. Add ¬Q to the clause set.
- **Step 3 — Resolve ¬P∨Q with ¬Q.** The complementary pair is Q and ¬Q. Resolving removes both, leaving: ¬P. Add ¬P to the clause set.
- **Step 4 — Resolve P with ¬P.** The complementary pair is P and ¬P. Resolving removes both, leaving: the empty clause, NIL.
- **Step 5 — Conclude.** Deriving NIL proves that the clause set {¬P∨Q, ¬Q∨R, P, ¬R} is unsatisfiable, meaning the original three clauses {¬P∨Q, ¬Q∨R, P} together logically entail R.

**Answer:** R is proved by resolution refutation in exactly three resolution steps, deriving the empty clause and thereby confirming the entailment. This exact derivation is illustrated graphically in Figure 4.4.

Resolution's completeness (specifically for the goal of proving unsatisfiability, hence "refutation-completeness") together with unification's ability to find exactly the substitutions needed to make literals complementary, together make

resolution one of the most important and general-purpose inference procedures studied in classical AI, and it underlies both classical automated theorem-proving systems and, in its restricted Horn-clause form, the logic-programming paradigm exemplified by Prolog.

**Advantages of resolution:** works on the *full* first-order logic, not merely the restricted definite-clause fragment; refutation-complete, meaning it will eventually find a proof if one exists.

**Disadvantages and limitations:** the CNF conversion process can substantially obscure the original, more readable structure of the knowledge base; the search for which pairs of clauses to resolve next can itself be very large and requires good heuristic guidance (analogous to the search heuristics of Module 3) to remain practical on realistic-sized knowledge bases; termination is not guaranteed within any fixed time bound when the original sentence is *not*, in fact, entailed (the procedure may simply run forever without ever concluding failure, a direct practical consequence of the underlying undecidability results discussed in Module 1, Section 1.2.2).

### 4.6.6 Comparative Summary of the Three Inference Methods

| Method | Direction | Completeness | Best suited for |
|---|---|---|---|
| Forward chaining | Data-driven (facts → conclusions) | Complete for definite clauses | Systems needing all consequences kept continuously up to date |
| Backward chaining | Goal-driven (query → sub-goals) | Complete for definite clauses | Interactive systems answering one specific query at a time |
| Resolution | Refutation-based (negate goal, derive contradiction) | Refutation-complete for full FOL | General-purpose automated theorem proving beyond definite clauses |

**Case Study: Case Study: Prolog and Backward-Chaining Logic Programming**

The Prolog programming language, still widely used in AI education and in certain specialized industrial applications (natural-language processing, symbolic

mathematics, and expert-system shells), is built almost entirely around backward-chained resolution restricted to Horn clauses (a generalization of the definite clauses discussed above). A Prolog program is, quite literally, a first-order knowledge base; running a Prolog query is, quite literally, invoking backward chaining (Section 4.6.4) to prove that query against the program's clauses, with unification (Section 4.6.2) automatically binding variables along the way. This makes Prolog a direct, practical, and still-used embodiment of the theoretical inference machinery developed in this section of the module.

## Key Terms Introduced in This Module

Knowledge-based agent · knowledge base (KB) · TELL/ASK · inference engine · Wumpus World · syntax · semantics · model · entailment · sound / complete inference · propositional logic · atomic sentence · connective · truth table · Modus Ponens · first-order logic · constant/predicate/function symbol · term · quantifier (∀, ∃) · kinship domain · knowledge engineering · propositionalization · Universal/Existential Instantiation · Skolem constant · unification · most general unifier (MGU) · forward chaining · definite clause · backward chaining · Prolog · resolution · Conjunctive Normal Form (CNF) · Skolemization · resolution refutation · empty clause (NIL)

## Summary of Module 4

- A knowledge-based agent stores sentences in a knowledge base and manipulates it via TELL and ASK, cleanly separating declarative knowledge from the inference mechanism.
- The Wumpus World is a small, partially observable grid environment used to motivate and illustrate logical inference from percept sequences (Breeze, Stench, Glitter).
- A logic has a syntax (well-formed sentences) and a semantics (truth in models); KB ⊨ α means α is true in every model where KB is true; sound inference derives only entailed sentences, complete inference derives all entailed sentences.
- Propositional logic represents facts using atomic propositions and connectives (¬, ∧, ∨, ⇒, ⇔), and supports inference rules such as Modus Ponens and resolution, but cannot compactly express general facts about all objects.

- First-order logic adds objects, predicates, functions, and quantifiers (∀, ∃), giving vastly greater expressive power; knowledge engineering is the systematic seven-step process of building an FOL knowledge base for a domain.
- FOL inference is enabled by unification (finding substitutions that make expressions syntactically identical, computed here by hand in Worked Example 4.3); forward chaining is data-driven and complete for definite clauses; backward chaining is goal-driven and typically more efficient for single queries; resolution is sound and refutation-complete for the full first-order logic, operating on sentences converted to Conjunctive Normal Form via the six-step procedure derived in this module.

## Review Questions

1. Explain the roles of TELL and ASK in a knowledge-based agent, and describe the overall percept–inference–action cycle such an agent follows, referencing Figure 4.2's cycle diagram.
2. Give the PEAS description of the Wumpus World and explain why it is only partially observable.
3. Define entailment (KB ⊨ α) and distinguish between a sound and a complete inference procedure.
4. Reproduce the derivation in Worked Example 4.1 for a different room configuration of your own choosing, showing every biconditional-elimination and De Morgan step explicitly.
5. Explain, with an example, why first-order logic is more expressive than propositional logic.
6. Using Worked Example 4.2 as a model, translate "Every student who has failed a core course is not eligible for graduation" into first-order logic.
7. List and briefly describe the seven steps of the knowledge engineering process, using the electronic-circuit case study as your running example.
8. Reproduce the unification derivation of Worked Example 4.3 for a different pair of atomic sentences of your own choosing, and check your answer by direct substitution.
9. Using Worked Examples 4.4 and 4.5 as models, trace both forward and backward chaining on a knowledge base of your own design with at least two rules, and compare the number of facts/sub-goals each method actually examines.
10. Reproduce the six-step CNF conversion procedure on a first-order sentence of your own choosing that includes at least one existential quantifier, showing the Skolemization step explicitly.
11. Reproduce the resolution refutation of Worked Example 4.7 for a knowledge base of your own design, and draw the corresponding refutation diagram in the style of Figure 4.4. Compare forward chaining, backward chaining, and resolution using the summary table in Section 4.6.6, and justify which you would choose for a chatbot that answers one user question at a time against a large, mostly static knowledge base.

# MODULE 5:
# PLANNING

## Learning Objectives

By the end of this module, the student should be able to:

- Represent a real-world task using STRIPS/PDDL action schemas, including preconditions, add-lists, and delete-lists.
- Trace forward and backward state-space planning search by hand on a small blocks-world problem.
- Construct a planning graph level by level for a small problem, and explain the role of mutex relations.
- Compute a planning heuristic from a relaxed problem, and apply the Critical Path Method to a small project schedule.
- Compare conditional planning against execution monitoring with replanning, and justify a choice between them for a given real-world scenario.

## 5.1 Definition of Classical Planning

The search techniques studied in Module 3 treat states, actions, and goals as atomic, opaque, "black-box" entities, described only by problem-specific functions (ACTIONS(s), RESULT(s,a), and so on) supplied individually by the problem designer. This works well for small or specially structured problems, but it forces the designer to write a substantial amount of problem-specific code for every new problem, and it prevents a single general-purpose reasoning system from reusing knowledge about actions across genuinely different problems. **Automated planning** addresses this limitation directly by adopting a **factored** representation: states, goals, and actions are all described using a structured logical language (closely related to the propositional and first-order logics of Module 4), so that a *single, domain-independent* planning algorithm can be applied across a huge variety of specific problems, simply by changing the description of the available actions and the goal.

**Definition: Classical Planning**
Classical planning is the problem of finding a sequence of actions that transforms a fully observable, deterministic, static, finite initial state into a state satisfying a given goal, where states, actions, and goals are all represented in a factored (STRIPS/PDDL) logical language rather than as opaque black-box objects.

### 5.1.1 The STRIPS/PDDL Representation

**Classical planning** makes several simplifying assumptions: the environment is fully observable, deterministic, static, and finite, with a known initial state and actions that have fully deterministic effects. Under these assumptions, a planning problem is typically described using a language descended from **STRIPS** (Stanford Research Institute Problem Solver) and standardized, in its modern form, as **PDDL** (Planning Domain Definition Language). Its key elements are:

- **States** are represented as a conjunction of function-free, ground (variable-free) positive literals for example, *At (Truck1, Depot) ∧ Empty (Truck1)*. Anything not explicitly stated to be true in a given state is assumed false by default (the **closed-world assumption**).
- **Goals** are also represented as a conjunction of literals (which may themselves include variables, treated as implicitly existentially quantified), and a state *satisfies* a goal if that state's literals form a superset of the goal's literals that is, the goal is, in general, only a partial description of an acceptable final state, not a complete one.
- **Actions** are described schematically using action **schemas**, each consisting of an action name with formal parameters, a **precondition** (a conjunction of literals that must hold in a state for the action to be legally applicable there), and an **effect**, itself split into an **add-list** (literals that become true as a result) and a **delete-list** (literals that become false as a result) after the action is executed.

**Key Formula: A STRIPS Action Schema**
Action (Stack (b, c),

PRECOND: Clear(c) ∧ Holding(b)

EFFECT: On (b, c) ∧ Clear(b) ∧ ¬Clear(c) ∧ ¬Holding(b))

This single schema, with variables b and c, stands in for an unbounded number of concretes "stack block X on block Y" actions one for every pair of blocks that might

appear in a particular problem instance exactly the kind of compact, reusable, variable-based representation a factored planning language is designed to provide.

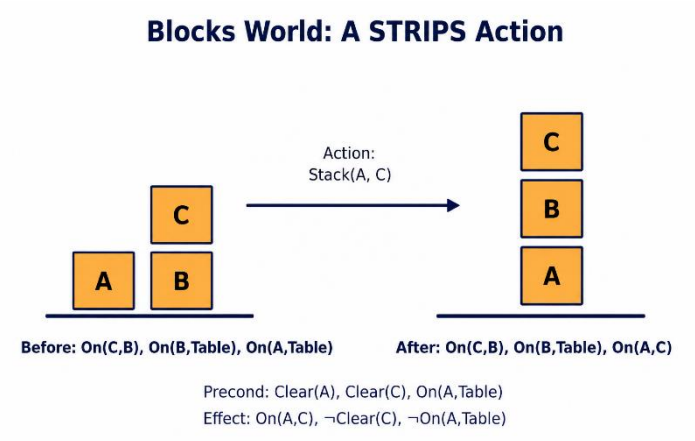


*Figure 5.1: A single STRIPS action, Stack (A, C), shown with its precondition and effect applied to a blocks-world state.*

**Worked Example: Worked Example 5.1 — Applying a STRIPS Action Step by Step**

**Problem:** Given the state {On(B,Table), On(C,B), Clear(C), On(A,Table), Clear(A), Holding(nothing)} — actually, to keep this concrete, assume the agent has just picked up block A, so the state is {On(B,Table), On(C,B), Clear(C), Clear(B) is false since C is on B, Holding(A)} — apply the action Stack(A, C) using the schema above.

- **Step 1 — Check the precondition.** Precondition requires Clear(C) ∧ Holding(A). Clear(C) holds (nothing is on C). Holding(A) holds (given). Precondition is satisfied.
- **Step 2 — Apply the add-list.** Add On(A,C) and Clear(A) to the state.
- **Step 3 — Apply the delete-list.** Remove Clear(C) and Holding(A) from the state (block C is no longer clear since A now sits on it, and the agent's hand is no longer holding anything).
- **Step 4 — Assemble the resulting state.** {On(B,Table), On(C,B), On(A,C), Clear(A), Holding(nothing)}.

**Answer:** After Stack(A,C), block A sits on block C, which sits on block B, which sits on the table — a three-block tower — exactly matching the "After" state shown in Figure 5.1

## 5.2 Algorithms for Classical Planning

### 5.2.1 Forward (Progression) State-Space Search

The most direct way to solve a planning problem is to treat it as an ordinary search problem, exactly as in Module 3: states are sets of ground literals, the initial state is given directly by the problem, actions are the ground instances of action schemas whose preconditions are satisfied by the current state, RESULT(s, a) is computed by removing the delete-list literals from *s* and adding the add-list literals, and the goal test checks whether the goal's literals are a subset of the current state. This is called **forward search** or **progression planning**, because it moves forward, step by step, from the initial state toward the goal. Its principal weakness is that the branching factor at each state can be very large in realistic domains, since *any* action whose preconditions happen to be satisfied is a candidate, most of which are entirely irrelevant to the actual goal so forward search performs poorly in practice without a good heuristic to focus it, motivating the planning-specific heuristics of Section 5.3.

**Advantages:** conceptually simple; directly reuses the search algorithms and heuristic-search machinery of Module 3.

**Disadvantages and limitations:** large branching factor without a good heuristic, since every applicable action is a candidate regardless of relevance to the goal.

### 5.2.2 Backward (Regression) Search

**Backward search**, or **regression planning**, instead starts from a description of the goal and works backward, at each step choosing an action whose effect is *relevant* (its add-list intersects the current sub-goal, and its delete-list does not undo any literal still needed) and computing the predecessor sub-goal that must have held immediately before that action was applied, continuing until a sub-goal is reached that is fully satisfied by the problem's initial state. Because backward search only ever considers actions that are *relevant* to the goal (rather than every action applicable in the current state, as forward search does), it can in principle be considerably more efficient in domains where the total number of applicable actions in a typical state vastly exceeds the number that are actually useful though it requires additional care with variable substitution, since sub-goals generated during regression can themselves contain unbound variables.

**Advantages:** considers only actions relevant to the goal, avoiding forward search's large irrelevant branching factor.

**Disadvantages and limitations:** requires more careful bookkeeping of variable substitutions than forward search; can still generate a large number of irrelevant sub-goals in domains with many ways to achieve the same literal.

**Worked Example: Worked Example 5.2 — Tracing Forward Search on a Small Blocks-World Problem**

**Problem:** Initial state {On(A,Table), On(B,Table), Clear(A), Clear(B), Holding(nothing)}. Goal: On(A,B). Available action schemas: Pickup(x) [PRECOND: Clear(x) ∧ Holding(nothing) ∧ On(x,Table); EFFECT: Holding(x) ∧ ¬Clear(x) ∧ ¬On(x,Table) ∧ ¬Holding(nothing)] and Stack(x,y) [PRECOND: Clear(y) ∧ Holding(x); EFFECT: On(x,y) ∧ Clear(x) ∧ ¬Clear(y) ∧ ¬Holding(x) ∧ Holding(nothing)]. Trace forward search to the goal.

- **Step 1 — Check the goal against the initial state.** On(A,B) is not present initially — search must proceed.
- **Step 2 — Enumerate applicable actions in the initial state.** Pickup(A): precondition Clear(A) ∧ Holding(nothing) ∧ On(A,Table) all satisfied. Pickup(B): similarly satisfied. Stack(x,y) requires Holding(x), which does not hold for any x yet not applicable. So the only candidates are Pickup(A) and Pickup(B); since the goal On(A,B) requires holding A, choose Pickup(A).
- **Step 3 — Apply Pickup(A).** New state: {On(B,Table), Clear(B), Holding(A)} (Clear(A) and On(A,Table) removed; Holding(nothing) removed).
- **Step 4 — Enumerate applicable actions.** Stack(A,B): precondition Clear(B) ∧ Holding(A) both satisfied. Apply it.
- **Step 5 — Apply Stack(A,B).** New state: {On(B,Table), On(A,B), Clear(A), Holding(nothing)} (Clear(B) and Holding(A) removed; On(A,B), Clear(A), Holding(nothing) added).
- **Step 6 — Test the goal.** On(A,B) is present goal satisfied.

**Answer:** The plan Pickup(A), Stack(A,B) achieves the goal in two steps, found directly by forward search since the branching factor in this tiny two-block domain is small enough not to require heuristic guidance.

### 5.2.3 Planning Graphs and the GRAPHPLAN Algorithm

A **planning graph** is a specialized, compact data structure distinct from, and typically far cheaper to construct than, a full state-space search tree that records, level by level, a *superset* of the literals and actions that could *possibly* be true or applicable at each successive time step. The graph alternates **literal levels** ($S_0$, $S_1$, $S_2$, ...), each holding every literal that *might* be achievable by that time step, and **action levels** ($A_0$, $A_1$, ...), each holding every action whose preconditions might be satisfiable at that step, together with explicit **mutual exclusion (mutex)** relations that record pairs of literals or actions that provably *cannot* co-occur at a given level (for instance, two actions that interfere because one deletes a precondition or effect the other needs).

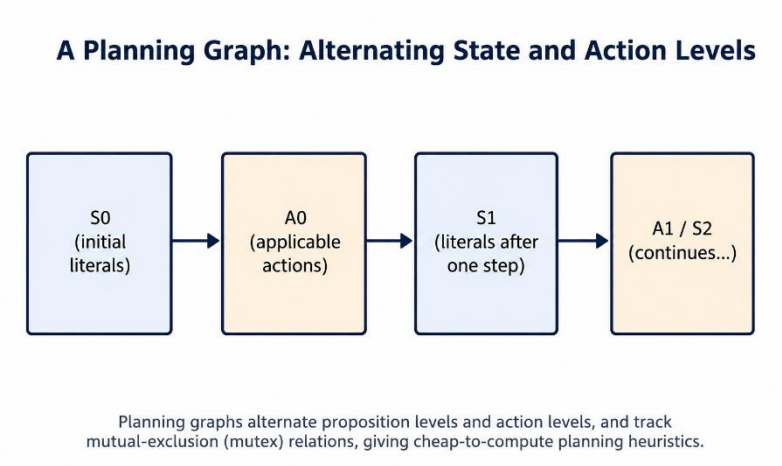


*Figure 5.2: A planning graph alternates literal levels and action levels; mutex relations (not shown) additionally record which literal or action pairs cannot co-occur at a given level.*

**Worked Example: Worked Example 5.3 — Constructing the First Two Levels of a Planning Graph**

**Problem:** Using the same initial state and action schemas as Worked Example 5.2, construct levels $S_0$, $A_0$, and $S_1$ of the planning graph.

- **Step 1 — $S_0$ (literal level 0).** Simply the initial state's literals: {On(A,Table), On(B,Table), Clear(A), Clear(B), Holding(nothing)}.
- **Step 2 — $A_0$ (action level 0).** Every ground action whose preconditions are a subset of $S_0$: Pickup(A) (precondition satisfied), Pickup(B) (precondition satisfied). Stack(x,y) is not included, since Holding(x) is not

in $S_0$ for any x. Also include a "no-op" persistence action for every literal in $S_0$, allowing it to optionally persist unchanged into $S_1$.

- **Step 3 — Compute mutex relations in $A_0$.** Pickup(A) and Pickup(B) are mutex with each other, because both require Holding(nothing) as a precondition, but only one can result in that hand actually holding a single object — more precisely, they compete for the same precondition resource in a way the STRIPS Holding(nothing)/Holding(x) representation renders mutually exclusive (an "inconsistent support" mutex).
- **Step 4 — $S_1$ (literal level 1).** The union of all add-list literals reachable via $A_0$'s actions, together with every persisted literal from $S_0$: {On(A,Table), On(B,Table), Clear(A), Clear(B), Holding(nothing), Holding(A), Holding(B), ¬Clear(A) [via Pickup(A)], ¬Clear(B) [via Pickup(B)]} represented properly as the *presence* of Holding(A) and Holding(B) as newly reachable literals at level 1, each still marked mutex with each other and with Holding(nothing), since no single real state can actually contain both.

**Answer:** Even after just one level of expansion, the planning graph already records that Holding(A) and Holding(B) can never co-occur (mutex), and that On (A, B) is still not reachable until level 2 (since Stack (A, B) first requires Holding(A), only newly available starting at level 1) this level-by-level reachability information is exactly what the planning heuristics of Section 5.3 exploit.

The **GRAPHPLAN** algorithm alternately extends the planning graph by one further level and then searches backward through the resulting (much smaller, more tightly constrained) graph for a valid plan; because the planning graph never omits a literal or action that could genuinely be needed, but does aggressively prune those provably mutually exclusive, GRAPHPLAN is both sound and complete, and in practice its use of the mutex-annotated graph as a powerful pruning and heuristic-estimation structure makes it substantially more efficient than naive forward or backward search on many benchmark planning domains.

**Advantages:** compact; sound and complete; the mutex structure gives a strong basis for the planning heuristics of Section 5.3.

**Disadvantages and limitations:** graph construction cost grows with the number of levels (and hence with plan length); mutex computation itself adds overhead per level.

### 5.2.4 Planning as Boolean Satisfiability (SAT Plan)

A further, quite different approach translates the entire planning problem for a fixed plan length *n* into a single large **propositional satisfiability (SAT)** problem: the initial state, the goal, and every possible action instance at every time step from 0 to *n* are all encoded as propositional sentences (in a manner reminiscent of the propositionalization approach discussed in Module 4, Section 4.6.1), and a general-purpose, highly optimized SAT solver is then used to find a satisfying assignment, from which a valid plan can be read off directly. If no satisfying assignment exists for length *n*, the horizon *n* is increased and the process is repeated. This **SATPlan** approach benefits directly from decades of engineering progress in industrial-strength SAT solvers, and for many planning domains it is competitive with, or even superior to, specialized planning-graph- or search-based planners.

**Advantages:** leverages highly optimized, general-purpose SAT solvers built by a separate, very mature research community.

**Disadvantages and limitations:** requires guessing and then iterating over the plan-length horizon *n*, since the correct horizon is not known in advance.

**Case Study: Case Study: SATPlan in Practice — Bounded-Length Planning Competitions**

In the biennial International Planning Competition, SATPlan-based systems have repeatedly proven highly competitive against dedicated heuristic-search planners on several benchmark domain families, particularly domains with dense negative interactions between actions (where many actions compete for the same resources), because modern industrial SAT solvers incorporate decades of highly optimized conflict-driven clause-learning engineering that a bespoke planner would otherwise need to reimplement from scratch. This is a concrete, competition-verified illustration of a broader engineering principle worth remembering: it is very often more effective to translate a new problem into a well-studied, heavily optimized existing solver's input format (here, Boolean satisfiability) than to write a new specialized algorithm from first principles.

## 5.3 Heuristics for Planning

Because planning problems are, in the worst case, formulated over enormous state spaces, forward or backward state-space search (Sections 5.2.1–5.2.2) is only practical when guided by an effective heuristic function *h(s)*, estimating the cost of

reaching the goal from state's precisely the role played by heuristics in the A\ *search algorithm of Module 3, Section 3.7.2. Planning heuristics are almost always derived automatically from a relaxed version** of the actual problem, rather than being hand-crafted separately for each new domain, which is one of the chief practical advantages of the factored STRIPS/PDDL representation.

**Definition: Planning Heuristics via Relaxation**

The **ignore-preconditions heuristic** removes all preconditions from every action (so any action can be applied at any time), reducing the problem to one of simply counting (or optimally combining) the actions whose add-lists are needed to cover the goal literals — this relaxation is easy to compute and is provably admissible, since the original, more constrained problem can never be solved in fewer actions than this unconstrained relaxation. The **ignore-delete-lists heuristic** keeps action preconditions but drops all delete-list effects, so that facts, once made true, are never subsequently undone; the resulting relaxed problem, although still NP-hard to solve *optimally* in general, can be solved *approximately* very efficiently (for example, greedily), giving a very effective and widely used heuristic in practice.

**Worked Example: Worked Example 5.4 — Computing the Ignore-Delete-Lists Heuristic by Hand**

**Problem:** Using the state and action schemas of Worked Example 5.2, estimate h(s) for the initial state {On(A,Table), On(B,Table), Clear(A), Clear(B), Holding(nothing)} with goal On(A,B), using the ignore-delete-lists relaxation.

- **Step 1 — Apply the relaxation.** Remove all delete-lists from Pickup and Stack, so applying an action only ever *adds* literals, never removes them.
- **Step 2 — Search for the shortest relaxed plan.** Apply Pickup(A) (precondition Clear(A) ∧ Holding(nothing) ∧ On(A,Table), all satisfied): adds Holding(A) (Clear(A), On(A,Table), Holding(nothing) are no longer removed, since delete-lists are ignored). State now includes both the old literals and Holding(A).
- **Step 3 — Continue the relaxed search.** Apply Stack(A,B) (precondition Clear(B) ∧ Holding(A), both now satisfied): adds On(A,B). Goal reached.
- **Step 4 — Count the relaxed plan length.** Two actions were used: Pickup(A), Stack(A,B).

**Answer:** h(s) = 2 under the ignore-delete-lists heuristic — and in this small example the relaxed plan length happens to exactly equal the true optimal plan

length found in Worked Example 5.2, illustrating that this heuristic, while not always exact, is frequently very close to the true cost in practice, which is precisely why it is so widely used in real planning systems.

The **planning-graph-based heuristics** use the level at which each individual goal literal first appears in a planning graph (Section 5.2.3), or the level at which the entire goal set first appears jointly free of mutex conflicts, as a heuristic estimate, since this level number is, by construction, never greater than the true number of steps required, and is frequently a close approximation to it in practice.

**Advantages:** automatically derived from the problem's own action schemas, requiring no hand-crafting for each new domain, unlike the search heuristics of Module 3 which are often somewhat domain-specific.

**Disadvantages and limitations:** the ignore-delete-lists relaxed problem, while efficiently *approximable*, is itself still NP-hard to solve *exactly* in general, so practical implementations use a greedy approximation rather than the true optimal relaxed-plan length.

**Worked Example: Worked Example 5.4b — Reading a Heuristic Value Directly from a Planning Graph**

**Problem:** A planning graph for a delivery-robot problem shows the goal literal At(Package, Destination) first appearing at literal level $S_3$, with no mutex conflict against the rest of the goal set at that level. What planning-graph-based heuristic estimate does this give for the number of remaining actions needed?

- **Step 1 — Recall the planning-graph heuristic definition (Section 5.2.3, Section 5.3).** The heuristic estimate is the level at which a goal literal (or the full goal set, mutex-free) first appears in the graph.
- **Step 2 — Read off the level.** At (Package, Destination) first appears at $S_3$, meaning at least 3 action levels ($A_0$, $A_1$, $A_2$) were needed to first make it reachable.
- **Step 3 — Apply admissibility reasoning.** Because the planning graph only ever includes literals that are genuinely reachable (or optimistically assumed reachable, ignoring mutex-driven false positives at lower levels), this level number can never exceed the true minimum number of steps required.

**Answer:** h(s) = 3 for this state, and because planning-graph levels never overestimate true reachability depth, this heuristic is admissible making it

directly usable to guide an A*-style forward search exactly as described in Section 5.3.

## 5.4 Planning and Acting in Nondeterministic Domains

The classical planning framework of Sections 5.1–5.3 assumes a fully observable, deterministic, known environment, but real-world environments frequently violate one or more of these assumptions, requiring extensions of the basic framework.

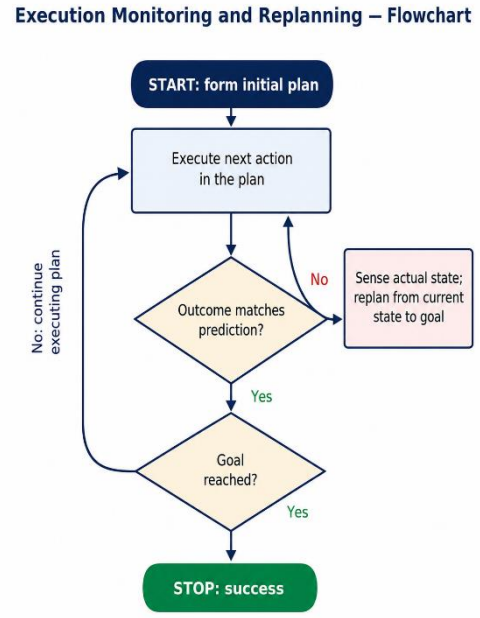


*Figure: Execution monitoring and replanning shown as a flowchart.*

### 5.4.1 Conditional (Contingency) Planning

When an environment is **nondeterministic** (a given action may have more than one possible outcome) or only **partially observable**, a single, fixed sequence of actions computed entirely in advance may not remain executable, because the agent may not know, at plan time, exactly which specific situation it will find itself in once a given step is actually reached. A **conditional plan** (or *contingency plan*) addresses this by including explicit branches, of the form "if condition C holds (as determined by a sensing action), do plan-branch A; otherwise do plan-branch B," so that the single plan produced in advance can correctly handle every contingency that sensing might reveal during execution  at the cost of the plan's total size

potentially growing very large if many independent contingencies must all be anticipated up front.

**Advantages:** the complete plan is available before execution begins, useful when replanning during execution is expensive or impossible (for example, a probe with a long communication delay to mission control).

**Disadvantages and limitations:** plan size can grow explosively with the number of independent contingencies that must be anticipated; wasted effort is spent planning for contingencies that may never actually occur in a given run.

### 5.4.2 Execution Monitoring and Replanning

An alternative, and often more practical, strategy for nondeterministic or partially known environments is to abandon the attempt to anticipate every contingency in advance, and instead to interleave planning with execution: the agent executes a classically computed plan step by step while continuously monitoring whether the actual, observed outcome of each action matches the outcome predicted by its model. If a discrepancy is detected an action fails, or the environment turns out to differ from the model's assumptions the agent invokes **replanning**: it reformulates a new planning problem starting from the actual current state (as revealed by its sensors) and re-invokes the planning algorithm to compute a revised plan for reaching the goal from this updated starting point.

**Advantages:** avoids the combinatorial blow-up of anticipating every contingency in advance; only spends planning effort on contingencies that actually occur.

**Disadvantages and limitations:** provides no advance guarantee that every future situation will be successfully handled; replanning itself takes time, which may be unacceptable in fast-moving or safety-critical situations, or impossible where communication delays prevent timely replanning support.

### 5.4.3 Continuous Planning

Many realistic agents a household robot, a factory-control system, an autonomous vehicle fleet manager never reach a final "done" state at all: new goals arrive continuously over the agent's entire operational lifetime, sometimes even before earlier goals have been fully completed. A **continuous planning agent** is designed to accept and interleave new goals as they arrive, maintaining and updating an

ongoing plan (or a small set of concurrent plans) rather than treating each planning episode as an isolated, self-contained problem solved once and then discarded, typically combining the execution-monitoring and replanning ideas of Section 5.4.2 with additional mechanisms for prioritizing and merging multiple simultaneously active goals.

**Case Study: Case Study: Daily Activity Planning for a Mars Rover**

NASA's Mars rovers (Spirit, Opportunity, and Curiosity) rely on automated planning systems closely related to the classical planning framework of this module. Each Martian day (sol), mission controllers specify a set of high-level science goals  photograph a specific rock formation, drill a sample at a specific site, transmit collected data during a specific communication window  and an onboard or ground-based planner produces a detailed, STRIPS-like sequence of actions (Move, Photograph, Drill, Transmit) that respects tight resource constraints on battery power and available daylight hours. Because round-trip radio communication with Earth can take many minutes, the rover cannot simply request new instructions the moment an unexpected obstacle is encountered; it must instead perform its own on-board execution monitoring (Section 5.4.2), and in some cases replan locally and immediately, rather than waiting for a new plan from Earth — directly illustrating why the choice between conditional planning and execution monitoring/replanning discussed in this section is a genuine, consequential engineering decision, not merely a theoretical distinction.

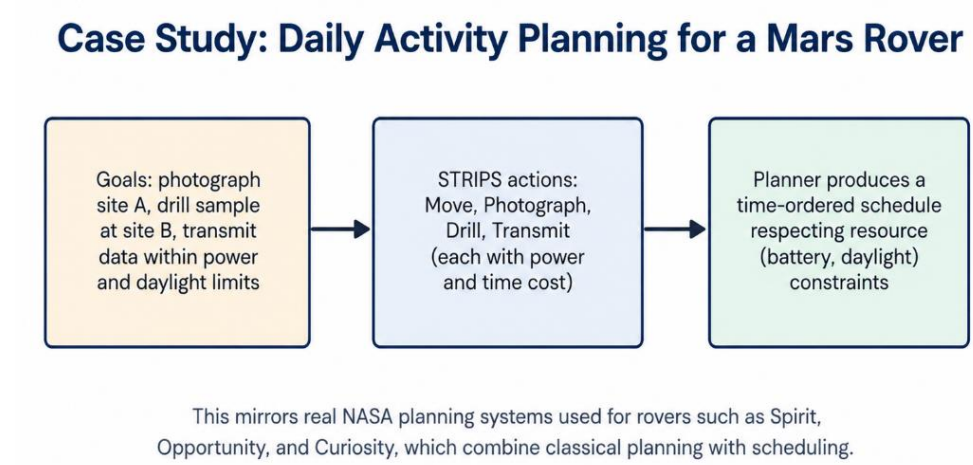


*Figure: Case study — daily activity planning for a Mars rover.*

## 5.5 Time, Schedules, and Resources

Classical planning, as presented in Sections 5.1–5.2, is concerned only with finding a correct *sequence* (or partial order) of actions; it says nothing about *when*, in real (wall-clock) time, each action should actually be performed, nor about competition for shared, limited **resources** (machines, workers, budget, raw materials) needed to carry those actions out. **Scheduling** extends planning to address exactly these additional concerns.

### 5.5.1 Job-Shop Scheduling

In a **job-shop scheduling problem**, a fixed set of **jobs** must each be processed through a sequence of operations, each operation requiring a specific **resource** (a particular machine, worker, or tool) for a specific **duration**; the objective is typically to find a schedule  an assignment of start times to every operation, respecting both the required ordering of operations within each job and the constraint that a given resource can process only one operation at a time  that minimizes the total completion time (the **makespan**) or some other cost measure.

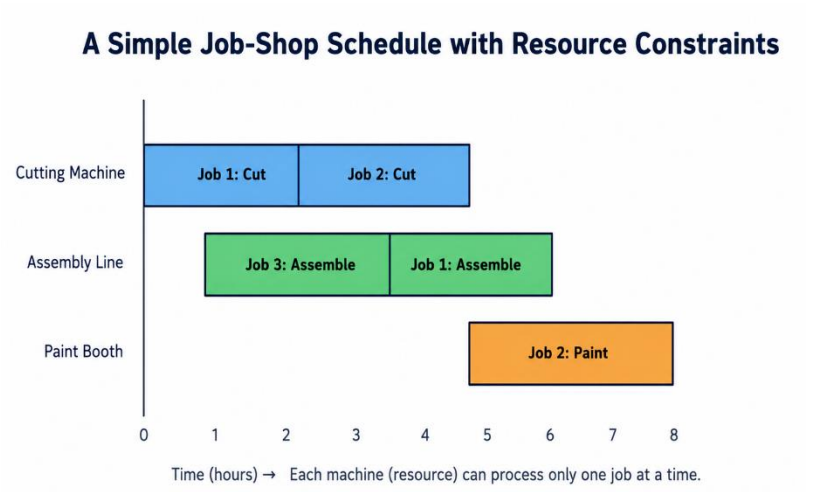


*Figure 5.3: A simple job-shop schedule for three jobs sharing three resources (machines). Each resource can process only one operation at a time, forcing some operations to wait.*

Job-shop scheduling is naturally formulated as a constraint satisfaction problem (Module 3, Section 3.9), where variables are the start times of each operation, with

**precedence constraints** (an operation cannot start until the previous operation in its job has finished) and **resource (capacity) constraints** (no two operations requiring the same resource may overlap in time) and, given optimization objectives such as minimizing makespan, is more precisely classified as a **constraint optimization problem**.

### 5.5.2 Critical Path Method and Resource Constraints

When only precedence constraints (and not resource contention) are considered, the minimum possible makespan, and the specific operations that determine it, can be found efficiently using the classical **Critical Path Method (CPM)**.

**Definition: Critical Path Method**

The critical path is the longest chain of precedence-linked operations from project start to finish; it determines the minimum possible completion time of the whole project, since delaying any operation on the critical path delays the entire project by exactly the same amount. Operations not on the critical path have positive slack (float) they can be delayed, up to a limit, without affecting the overall project completion time

**Worked Example: Worked Example 5.5 — Computing the Critical Path by Hand**

**Problem:** A small project has four tasks: T1 (duration 3 days, no dependencies), T2 (duration 2 days, depends on T1), T3 (duration 4 days, depends on T1), T4 (duration 1 day, depends on both T2 and T3). Find the critical path and the project's minimum completion time.

- **Step 1 — Compute the earliest start (ES) and earliest finish (EF) for each task, in dependency order.** T1: ES=0, EF=0+3=3. T2 (depends on T1): ES=3 (T1's EF), EF=3+2=5. T3 (depends on T1): ES=3, EF=3+4=7. T4 (depends on both T2 and T3): ES = max (T2's EF, T3's EF) = max (5,7) = 7, EF = 7+1=8.
- **Step 2 — Identify the project's overall completion time.** The maximum EF among all tasks with no successors is T4's EF = 8 days.
- **Step 3 — Compute the latest finish (LF) and latest start (LS) for each task, working backward from the project deadline of 8 days.** T4: LF=8, LS=8-1=7. T3 (only predecessor of T4): LF = T4's LS = 7, LS = 7-4=3. T2 (only predecessor of T4): LF = T4's LS = 7, LS = 7-2=5. T1 (predecessor of both T2 and T3): LF = min (T2's LS, T3's LS) = min (5,3) = 3, LS = 3-3=0.

- **Step 4 — Compute slack for each task (LS − ES, or equivalently LF − EF).** T1: slack = 0−0 = 0. T2: slack = 5−3 = 2. T3: slack = 3−3 = 0. T4: slack = 7−7 = 0.
- **Step 5 — Identify the critical path as the chain of zero-slack tasks.** T1 → T3 → T4 all have zero slack, forming the critical path; T2 has 2 days of slack and is not on the critical path.

**Answer:** The critical path is T1 → T3 → T4, with a minimum project completion time of 8 days. T2 can be delayed by up to 2 days without affecting the overall 8-day project completion time.

Once resource constraints are added back into the problem, scheduling generally becomes substantially harder (NP-hard in general), and practical solvers typically combine CPM-style critical-path analysis with CSP-style backtracking or local-search heuristics (such as greedily resolving the resource conflict with the least available slack first) to find good, though not always provably optimal, schedules within a practical amount of computation time.

**Advantages of CPM:** efficient (polynomial-time) to compute; directly identifies which specific tasks genuinely determine the project deadline, focusing management attention precisely where it matters most.

**Disadvantages and limitations:** ignores resource contention entirely in its basic form; real schedules with genuine resource constraints require the more expensive constraint-optimization techniques discussed above.

## 5.6 Analysis of Planning Approaches

Each of the planning approaches introduced in this module carries distinct strengths and weaknesses, summarized below:

| Approach | Key idea | Strength | Limitation |
|---|---|---|---|
| Forward (progression) search | Ordinary state-space search from the initial state | Simple; straightforward to implement and to combine with A*-style heuristics | Large branching factor without a good heuristic |

| Approach | Key idea | Strength | Limitation |
|---|---|---|---|
| Backward (regression) search | State-space search from the goal | Considers only relevant actions | Complicated by variable substitution in sub-goals |
| Planning graph / GRAPHPLAN | Level-by-level graph with mutex pruning | Compact; sound and complete; strong basis for heuristics | Graph construction cost grows with plan length |
| SATPlan | Encode as a Boolean satisfiability problem | Leverages highly optimized, general SAT solvers | Requires guessing/iterating over the plan-length horizon |
| Conditional planning | Branching plans for nondeterministic/partial observability | Plan handles every contingency in advance | Plan size can explode with the number of contingencies |
| Execution monitoring + replanning | Interleave execution with re-planning on failure | Efficient; avoids planning for contingencies that never occur | No guarantee in advance that every future situation is handled |

No single approach dominates in every domain: forward search with strong heuristics (Section 5.3) is often the most practical general-purpose method, GRAPHPLAN and SATPlan are frequently competitive or superior on problems with rich negative interactions between actions, and the choice between committing to conditional plans versus monitoring-and-replanning depends heavily on how expensive replanning is relative to the cost of anticipating contingencies in advance, and on how observable and how deterministic the actual deployment environment turns out to be  exactly the considerations weighed in the Mars rover case study of Section 5.4.3.

## Key Terms Introduced in This Module

Classical planning · STRIPS · PDDL · closed-world assumption · action schema · precondition · add-list · delete-list · forward (progression) search · backward (regression) search · planning graph · mutex (mutual exclusion) · GRAPHPLAN · SATPlan · ignore-preconditions heuristic · ignore-delete-lists heuristic · conditional (contingency) planning · execution monitoring · replanning · continuous planning ·

job-shop scheduling · makespan · precedence constraint · resource constraint · Critical Path Method (CPM) · earliest/latest start and finish · slack (float) · critical path

## Summary of Module 5

- Classical planning uses a factored (STRIPS/PDDL) representation of states, goals, and actions (with preconditions, add-lists, and delete-lists), enabling a single domain-independent algorithm to solve many different problems.
- Forward (progression) search and backward (regression) search both reduce planning to state-space search but differ in which actions they consider at each step, as traced by hand in Worked Example 5.2.
- Planning graphs (and the GRAPHPLAN algorithm) and SATPlan offer alternative, often more efficient, sound-and-complete solution methods based on level-by-level graph construction and Boolean satisfiability, respectively, as constructed by hand in Worked Example 5.3.
- Effective heuristics for state-space planning search (ignore-preconditions, ignore-delete-lists, planning-graph-level heuristics) are generally derived automatically from relaxed versions of the planning problem, as computed by hand in Worked Example 5.4.
- Nondeterministic or partially observable domains require either explicit conditional (contingency) plans or, often more practically, execution monitoring combined with replanning, as illustrated by the Mars rover case study; continuous planning further extends this to agents that must handle a continuous stream of arriving goals.
- Scheduling extends planning with real time and shared, limited resources; job-shop scheduling is naturally posed as a constraint (optimization) problem, and the Critical Path Method, derived and computed by hand in Worked Example 5.5, identifies the operations that bound the minimum achievable completion time when resource contention is ignored.

## Review Questions

1. Explain the STRIPS representation of an action schema, using a precondition, add-list, and delete-list, with an example other than Stack (b, c).
2. Using Worked Example 5.1 as a model, apply a Pickup(x) action to a state of your own choosing and show the resulting state.
3. Compare forward (progression) and backward (regression) state-space planning search in terms of which actions each considers at every step.
4. Using Worked Example 5.2 as a model, trace forward search on a three-block stacking problem of your own design.
5. What is a planning graph, and what role do mutex relations play within it? Extend Worked Example 5.3 by constructing level $A_1$ and $S_2$.
6. Briefly describe how SATPlan converts a planning problem into a Boolean satisfiability problem, and explain why the plan-length horizon must be iterated.
7. Using Worked Example 5.4 as a model, compute the ignore-delete-lists heuristic for a different initial state of the same blocks-world problem, and compare it to the true optimal plan length.
8. Distinguish between conditional (contingency) planning and execution-monitoring-with-replanning, using the Mars rover case study to justify which is preferable when communication delay is large.
9. What is meant by a "continuous planning" agent, and why is it needed for many real-world deployments?
10. Formulate a small job-shop scheduling example as a constraint satisfaction problem, identifying the variables, domains, and constraints.
11. Reproduce the Critical Path Method derivation of Worked Example 5.5 for a project of your own design with at least five tasks, and identify which tasks have positive slack.

1. Discuss, with reference to the comparison table in Section 5.6, the circumstances under which you would prefer GRAPHPLAN over plain forward-search planning.

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*Key terms are listed alphabetically with the Module and Section in which they are principally introduced or discussed.*

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Manhattan-distance heuristic — Module 3, §3.8

McCulloch–Pitts neuron — Module 1, §1.3.1

Means-ends analysis — Module 1, §1.3.3

Model-based reflex agent — Module 2, §2.4.2

Moore's Law — Module 1, §1.2.6

Most general unifier (MGU) — Module 4, §4.6.2

Mutex (mutual exclusion) — Module 5, §5.2.3

MYCIN — Module 1, §1.3.5

## P

Path cost — Module 3, §3.2

PDDL — Module 5, §5.1.1

PEAS description — Module 2, §2.3.1

Percept / percept sequence — Module 2, §2.1

Perceptron — Module 1, §1.3.4

Performance measure — Module 2, §2.2.1

**R**



**S**



**T**



**U**

## V



## W



## X